

Information and Entertainment System

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware or equivalent diagnostic scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3104-A	Back-probe Pins POM6411 or equivalent
 ST3051-A	Universal Serial Bus (USB) Male-A To Male-A Cable CCMUSB2-AM-AM-10
 ST3086-A	Multi-Media Interface Tester 105-00120

DTC Charts

NOTE: For SYNC® related concerns, REFER to the SYNC® Symptom Chart before connecting the scan tool and beginning DTC-based diagnostics.

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Audio Front Control Module (ACM) DTC Chart

DTC	Description	Action
B1A56:21	Antenna: Signal Amplitude < Minimum	If DTC B1A56:21 sets after running the bezel diagnostic tests, disregard this DTC. This test is used for end of line or repair bay testing at the Ford plant. Each plant has its own assigned frequency and signal strength requirements. The running and passing of this test depends on local radio stations. If no local station is broadcasting on the required frequency, the test will fail. Therefore, tests ran from the bezel diagnostic screen and the DTC generated from the test can be disregarded.
B1A89:11	Satellite Antenna: Circuit Short To Ground	- For a satellite radio concern, GO to Pinpoint Test B - For a SIRIUS Data Services concern, GO to Pinpoint Test C
B1A89:13	Satellite Antenna: Circuit Open	- For a satellite radio concern, GO to Pinpoint Test B - For a SIRIUS Data Services concern, GO to Pinpoint Test C
B1D19:49	Compact Disc Unit: Internal Electronic Failure	- CLEAR the DTC. - WAIT at least 30 seconds. - REPEAT the self-test. - If DTC B1D19:49 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
B1D19:4B	Compact Disc Unit: Over Temperature	The ACM was over temperature. This may occur during extended use. Audio operation can resume after the ACM cools. CLEAR the DTC. This is normal operation.
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AF.
U0238:00	Lost Communication With Digital Audio Control Module "D" : No Sub Type Information	GO to Pinpoint Test AI.
U0253:00	Lost Communication With Accessory Protocol Interface Module: No Sub Type Information	GO to Pinpoint Test AJ.
U0256:00	Lost Communication With Front Controls Interface Module "A": No Sub Type Information	GO to Pinpoint Test AK.
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	GO to Pinpoint Test AL.
U201A:51	Control Module Main Calibration Data: Not Programmed	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.

DTC	Description	Action
		If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM).
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM).
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new ACM to correct the failure to retain configuration data. REFER to Audio Control Module (ACM).
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
U3000:96	Control Module: Component Internal Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:96 is retrieved again, INSTALL a new ACM . REFER to Audio Control Module (ACM).
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AM.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AN.

Accessory Protocol Interface Module (APIM) DTC Chart

DTC	Description	Action
C1001:01	Vision System Camera: General Electrical Failure	REFER to The rear video is inoperative in Section 413-13.

DTC	Description	Action
B108E:01	Display: General Electrical Failure	GO to Pinpoint Test XA.
B116A:01	Handset Microphone: General Electrical Failure	GO to Pinpoint Test R.
B1215:00	External Memory Card: No Sub Type Information	GO to Pinpoint Test AA.
B1252:11	USB Port: Circuit Short to Ground	- For an inoperative USB port, GO to Pinpoint Test T - For an inoperative SD card slot, GO to Pinpoint Test U
B1252:13	USB Port: Circuit Open	- For an inoperative USB port, GO to Pinpoint Test T - For an inoperative SD card slot, GO to Pinpoint Test U
B12B8:11	USB Port #2: Circuit Short To Ground	NOTE: <i>This DTC refers to USB port #2 on the APIM itself, which is not used.</i> - INSPECT the APIM USB port #2 for damage and REMOVE any debris. - CLEAR the DTCs. - WAIT at least 10 seconds. - REPEAT the self-test. - If DTC B12B8:11 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U0140:00	Lost Communication With Body Control Module: No Sub Type Information	GO to Pinpoint Test AD.
U0151:00	Lost Communication With Restraints Control Module: No Sub Type Information	GO to Pinpoint Test AE.
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AF.
U0162:00	Lost Communication With Navigation Display Module: No Sub Type Information	GO to Pinpoint Test XA.
U016A:00	Lost Communication With Global Positioning System Module: No Sub Type Information	GO to Pinpoint Test AG.
U0184:00	Lost Communication With Radio: No Sub Type Information	GO to Pinpoint Test AH.
U0238:00	Lost Communication With Digital Audio Control Module "D" : No Sub Type Information	GO to Pinpoint Test AI.
U0256:00	Lost Communication With Front Controls Interface	GO to Pinpoint Test AK.

DTC	Description	Action
	Module "A": No Sub Type Information	
U0423:00	Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the IPC . REFER to Section 413-01.
U1A00:87	Private Communication Network: Missing Message	GO to Pinpoint Test C.
U2017:45	Control Module Software #2: Program Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U2017:45 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U2017:51	Control Module Software #2: Not Programmed	This DTC sets due to incomplete or incorrect APIM programming/provisioning. CHECK vehicle service history for recent service actions related to the APIM . If there have been recent service actions with the APIM , REPEAT/PERFORM the APIM programming as directed by the scan tool. REFER to APIM Programming in the Description and Operation, Section 415-00E. If there have been no recent service actions, the APIM has failed to retain configuration data. PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U2017:52	Control Module Software #2: Not Activated	This DTC sets due to incomplete or incorrect APIM programming/provisioning. CHECK vehicle service history for recent service actions related to the APIM . If there have been recent service actions with the APIM , REPEAT/PERFORM the APIM programming as directed by the scan tool. REFER to APIM Programming in the Description and Operation, Section 415-00E. If there have been no recent service actions, the APIM has failed to retain configuration data. PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U2017:54	Control Module Software #2: Missing Calibration	This DTC sets due to incomplete or incorrect APIM programming/provisioning. CHECK vehicle service history for recent service actions related to the APIM . If there have been recent service actions with the APIM , REPEAT/PERFORM the APIM programming as directed by the scan tool. REFER to APIM Programming in the Description and Operation, Section 415-00E. If there have been no recent service actions, the APIM has failed to retain configuration data. PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	This DTC sets due to incomplete or incorrect APIM programming/provisioning. CHECK vehicle service history for recent service actions related to the APIM . If there have been recent service actions with the APIM , REPEAT/PERFORM the APIM programming as directed by the scan tool. REFER to APIM Programming in the Description and Operation, Section 415-00E. If there have been no recent service actions, the APIM has failed to retain configuration data. PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	This DTC sets due to incomplete or incorrect APIM programming/provisioning. CHECK vehicle service history for recent service actions related to the APIM . If there have been recent service actions with the APIM , REPEAT/PERFORM the

DTC	Description	Action
		APIM programming as directed by the scan tool. REFER to APIM Programming in the Description and Operation, Section 415-00E. If there have been no recent service actions, the APIM has failed to retain configuration data. PERFORM the APIM Hardware Test. INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U3000:04	Control Module: System Internal Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:04 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
U3000:88	Control Module: Bus Off	NOTE: <i>Network communication/message DTCs can result from intermittent concerns such as damaged wiring or low battery voltage occurrences. Additionally, vehicle repair procedures such as module reprogramming often set these DTCs. Replacing a module to resolve a network DTC is unlikely to resolve the concern. The module was unable to communicate on the network at a point in time. The fault is not currently present.</i> To avoid repeat network concerns: • INSPECT for damaged or corroded wiring at the APIM . • TEST the vehicle battery. REFER to Charging Systems in Section 414-00. • CLEAR the DTC. • REPEAT the network test with the diagnostic scan tool.
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AM.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AN.

Audio DSP Module DTC Chart

DTC	Description	Action
B1A01:01	Speaker #1: General Electrical Failure	GO to Pinpoint Test I.

DTC	Description	Action
B1A01:11	Speaker #1: Circuit Short to Ground	GO to Pinpoint Test I.
B1A01:12	Speaker #1: Circuit Short to Battery	GO to Pinpoint Test I.
B1A01:13	Speaker #1: Circuit Open	GO to Pinpoint Test I.
B1A02:01	Speaker #2: General Electrical Failure	GO to Pinpoint Test G.
B1A02:11	Speaker #2: Circuit Short to Ground	GO to Pinpoint Test G.
B1A02:12	Speaker #2: Circuit Short to Battery	GO to Pinpoint Test G.
B1A02:13	Speaker #2: Circuit Open	GO to Pinpoint Test G.
B1A03:01	Speaker #3: General Electrical Failure	GO to Pinpoint Test H.
B1A03:11	Speaker #3: Circuit Short to Ground	GO to Pinpoint Test H.
B1A03:12	Speaker #3: Circuit Short to Battery	GO to Pinpoint Test H.
B1A03:13	Speaker #3: Circuit Open	GO to Pinpoint Test H.
B1A04:01	Speaker #4: General Electrical Failure	GO to Pinpoint Test N.
B1A04:11	Speaker #4: Circuit Short to Ground	GO to Pinpoint Test N.
B1A04:12	Speaker #4: Circuit Short to Battery	GO to Pinpoint Test N.
B1A04:13	Speaker #4: Circuit Open	GO to Pinpoint Test N.
B1A06:01	Speaker #6: General Electrical Failure	GO to Pinpoint Test J.
B1A06:11	Speaker #6: Circuit Short to Ground	GO to Pinpoint Test J.
B1A06:12	Speaker #6: Circuit Short to Battery	GO to Pinpoint Test J.
B1A06:13	Speaker #6: Circuit Open	GO to Pinpoint Test J.
B1A07:01	Speaker #7: General Electrical Failure	GO to Pinpoint Test K.
B1A07:11	Speaker #7: Circuit Short to Ground	GO to Pinpoint Test K.
B1A07:12	Speaker #7: Circuit Short to Battery	GO to Pinpoint Test K.
B1A07:13	Speaker #7: Circuit Open	GO to Pinpoint Test K.
B1A08:01	Speaker #8: General Electrical Failure	GO to Pinpoint Test L.
B1A08:11	Speaker #8: Circuit Short to Ground	GO to Pinpoint Test L.

DTC	Description	Action
B1A08:12	Speaker #8: Circuit Short to Battery	GO to Pinpoint Test L.
B1A08:13	Speaker #8: Circuit Open	GO to Pinpoint Test L.
B1A09:11	Speaker #9: Circuit Short to Ground	GO to Pinpoint Test M.
B1A09:12	Speaker #9: Circuit Short to Battery	GO to Pinpoint Test M.
B1A09:13	Speaker #9: Circuit Open	GO to Pinpoint Test M.
B1A10:11	Speaker #10: Circuit Short to Ground	GO to Pinpoint Test M.
B1A10:12	Speaker #10: Circuit Short to Battery	GO to Pinpoint Test M.
B1A10:13	Speaker #10: Circuit Open	GO to Pinpoint Test M.
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AF.
U0184:00	Lost Communication With Radio: No Sub Type Information	GO to Pinpoint Test AH.
U0253:00	Lost Communication With Accessory Protocol Interface Module: No Sub Type Information	GO to Pinpoint Test AJ.
U0256:00	Lost Communication With Front Controls Interface Module "A": No Sub Type Information	GO to Pinpoint Test AK.
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	GO to Pinpoint Test AL.
U0423:00	Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the IPC. REFER to Section 413-01.
U2014:01	Control Module Hardware: General Electrical Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U2014:01 is retrieved again, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
U2014:41	Control Module Hardware: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U2014:41 is retrieved again, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front

DTC	Description	Action
		Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
U201A:51	Control Module Main Calibration Data: Not Programmed	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new audio DSP module to correct the failure to retain configuration data. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new audio DSP module to correct the failure to retain configuration data. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new audio DSP module to correct the failure to retain configuration data. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AM.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AN.

FCIM DTC Chart

DTC	Description	Action
B00D5:01	Restraint System Passenger Disable Indicator: General Electrical Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B00D5:01 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
B113C:11	Hazard switch illumination: Circuit Short To Ground	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B113C:11 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
B1202:01	Restraint System Passenger Enable Indicator: General Electrical Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC B1202:01 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AF.
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	GO to Pinpoint Test AL.
U0423:00	Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the IPC . REFER to Section 413-01.
U2013:63	Switch Pack: Circuit / Component Protection Time-Out	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U2013:63 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test.

DTC	Description	Action
		If DTC U3000:42 is retrieved again, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AM.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AN.

Global Positioning System Module (GPSM) DTC Chart

DTC	Description	Action
U0100:00	Lost Communication with ECM/PCM "A": No Sub Type Information	GO to Pinpoint Test AC.
U0140:00	Lost Communication With Body Control Module: No Sub Type Information: No Sub Type Information	GO to Pinpoint Test AD.
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	GO to Pinpoint Test AF.
U0401:00	Invalid Data Received from ECM/PCM A: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the PCM . REFER to Section 303-14.
U0422:00	Invalid Data Received From Body Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the BCM . REFER to Section 419-10.
U0423:00	Invalid Data Received from Instrument Panel Cluster Control Module: No Sub Type Information	RETRIEVE and FOLLOW DTCs present in the IPC . REFER to Section 413-01.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01. If there have been no recent service actions, INSTALL a new Global Positioning System Module (GPSM) to correct the failure to retain configuration data. REFER to Global Positioning System Module (GPSM).
U2101:00	Control Module Configuration Incompatible: No Sub Type Information	CHECK vehicle service history for recent service actions related to this module. This DTC sets due to incomplete or incorrect PMI procedures. If there have been recent service actions with this module, REPEAT/PERFORM the PMI procedure as directed by the diagnostic scan tool. For more information, REFER to Programmable Module Installation (PMI) in Section 418-01.

DTC	Description	Action
		If there have been no recent service actions, INSTALL a new Global Positioning System Module (GPSM) to correct the failure to retain configuration data. REFER to Global Positioning System Module (GPSM).
U3000:09	Control Module: Component Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:09 is retrieved again, PERFORM a battery reset by disconnecting the battery for 1 minute, then reconnecting it. • CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:09 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM).
U3000:41	Control Module: General Checksum Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:41 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM).
U3000:42	Control Module: General Memory Failure	• CLEAR the DTC. • WAIT at least 10 seconds. • REPEAT the self-test. • If DTC U3000:42 is retrieved again, INSTALL a new Global Positioning System Module (GPSM) . REFER to Global Positioning System Module (GPSM).
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test AM.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test AN.

SCCM DTC Chart

DTC	Description	Action
B1380:09	Steering Wheel Right Switch Pack: Component Failure	GO to Pinpoint Test X.
B1380:11	Steering Wheel Right Switch Pack: Circuit Short To Ground	GO to Pinpoint Test X.
B1380:17	Steering Wheel Right Switch Pack: Circuit Voltage Above Threshold	GO to Pinpoint Test X.
All other DTCs	—	REFER to Section 211-05.

Symptom Charts

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Symptom Chart — General Audio System

Condition	Possible Causes	Action
A module does not respond to the diagnostic scan tool	- Fuse - Wiring, terminals, or connectors - Module	REFER to Section 418-00.
The RH steering wheel controls are inoperative or do not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test X.
The CD player is inoperative or does not operate correctly	- CD - ACM	INSPECT the CD for scratches, fingerprints, a loose paper label, incorrect format, or damage. INSERT a known good CD and TEST the system. If the system operates correctly, the concern was caused by a damaged CD. If the system does not operate correctly, INSTALL a new ACM . REFER to Audio Control Module (ACM).
The speed compensated volume does not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test D.
The compass is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test AB.
The FCIM illumination is inoperative or does not dim	- Illumination system concern - FCIM	REFER to Section 413-00.
The audio system does not operate correctly from the FCIM	Refer to the Pinpoint Test	GO to Pinpoint Test Y.

Symptom Chart — Touchscreen Display

Condition	Possible Causes	Action
The FDIS is blank or black	Refer to the Pinpoint Test	GO to Pinpoint Test XA.
The FDIS is not responsive to touch	Refer to the Pinpoint Test	GO to Pinpoint Test XA.
The FDIS is slow to respond to touch	Refer to the Pinpoint Test	GO to Pinpoint Test XA.

The image displayed on the FDIM is incorrect, the image color is incorrect, or the image is distorted, flickering, or unclear	Refer to the Pinpoint Test	GO to Pinpoint Test XA.
The image on the FDIM is locked or frozen	Refer to the Pinpoint Test	GO to Pinpoint Test XA.
The FDIM changes brightness	Refer to the Pinpoint Test	GO to Pinpoint Test XB.
The FDIM displays “SD CARD FAULT”	Refer to the Pinpoint Test	GO to Pinpoint Test AA.
The FDIM displays a battery saver message	- Communication network concern - Battery voltage concern - Charging system concern	- CARRY OUT the APIM self-test. - If any Diagnostic Trouble Codes (DTCs) are present, REFER to the APIM DTC Chart in this section. - If no Diagnostic Trouble Codes (DTCs) are present, DIAGNOSE the charging system concern. REFER to: Section 414-00.

Symptom Chart — **HDD** Radio® System (If Equipped)

Condition	Possible Causes	Action
Data errors for text display	Data error from radio broadcaster	This condition is due to the data from the radio broadcaster. ADVISE the customer that the HDD Radio® is functioning correctly.
Echo, stutter, skip, repeat or volume fluctuation in HDD	Signal alignment error from radio broadcaster	This condition is due to the signal alignment from the radio broadcaster. ADVISE the customer that the HDD Radio® is functioning correctly.
Sound fading in and out or sound delay	Transition between HDD and analog	This condition is due to the transition between HDD and analog. ADVISE the customer that the HDD Radio® must have an available channel in order to output HDD content.
Sound dropout or channel unavailable in HDD 2 or HDD 3	HDD 2 and HDD 3 do not have analog equivalents causing complete signal loss when HDD signal fades	This condition is due to the HDD 2 and HDD 3 do not have analog equivalents causing complete signal loss when HDD signal fades. ADVISE the customer that the HDD Radio® must have an available channel in order to output HDD content.

Symptom Chart — Navigation (If Equipped)

Condition	Possible Causes	Action
Navigation application is inoperative or unresponsive	Refer to the Pinpoint Test	GO to Pinpoint Test AA.

Map does not load or show what is expected	Refer to the Pinpoint Test	GO to Pinpoint Test AA.
Navigation does not track vehicle correctly on map	Refer to the Pinpoint Test	GO to Pinpoint Test AA.
GPS icon displayed over red X with FDM navigation tab selected	Refer to the Pinpoint Test	GO to Pinpoint Test AA.
Voice recognition does not operate correctly to voice navigate to a destination	Refer to the Pinpoint Test	GO to Pinpoint Test AO.
The SIRIUS data services are not available (if equipped)	Refer to the Pinpoint Test	GO to Pinpoint Test C.

Symptom Chart — Satellite Radio (If Equipped)

Condition	Possible Causes	Action
Poor reception — satellite radio	Refer to the Pinpoint Test	GO to Pinpoint Test B.
Poor sound quality or no sound while in satellite radio mode - all other functions operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test B.
The satellite radio is inoperative or does not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test B.

Symptom Chart — Sound Quality

Condition	Possible Causes	Action
Poor reception — AM/FM	Refer to the Pinpoint Test	GO to Pinpoint Test A.
Continuous seek or scan — AM/FM	Refer to the Pinpoint Test	GO to Pinpoint Test A.
Poor sound quality or distorted sound from one or more speakers (not all speakers)	Refer to the Pinpoint Test	GO to Pinpoint Test E.
No sound from all speakers	Refer to the Pinpoint Test	GO to Pinpoint Test F.
The LF door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test G.
The LF tweeter speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test H.
The instrument panel speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test I.
The RF door speaker is inoperative	Refer to the Pinpoint Test	GO to Pinpoint Test J.

• The RF tweeter speaker is inoperative	• Refer to the Pinpoint Test	• GO to Pinpoint Test K.
• The RR door speaker is inoperative	• Refer to the Pinpoint Test	• GO to Pinpoint Test L.
• The subwoofer speaker is inoperative	• Refer to the Pinpoint Test	• GO to Pinpoint Test M.
• The LR door speaker is inoperative	• Refer to the Pinpoint Test	• GO to Pinpoint Test N.

SYNC System

Condition	Possible Causes	Action
• All SYNC® Services features are inoperative or inaccurate (traffic, directions, and information)	• Unregistered SYNC® owner account • Unregistered or expired SYNC® Services subscription • Unregistered mobile phone number • Mobile phone being used not selected as "Active" • SYNC® Services server • Accessory Protocol Interface Module (APIM) • GPSM	• PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test Q .
• The SYNC® system voice or tone prompts, Text-To-Speech (TTS) feature, or ringtones are inoperative or do not operate correctly	• Customer's setting • Wiring, terminals, or connectors • APIM • Audio Front Control Module (ACM) (for vehicles without the Audio DSP Module) or Audio DSP Module (If equipped)	• PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test P.

During a phone call, no incoming audio is heard in the vehicle	- Customer's setting - Wiring, terminals, or connectors - APIM - Audio Front Control Module (ACM) (for vehicles without the Audio DSP Module) or Audio DSP Module (If equipped)	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test P.
Voice recognition is inoperative or does not operate correctly	- Wiring, terminals, or connectors - Microphone - APIM	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test R.
During a phone call, poor quality or no outgoing audio is heard on the outside device	- Wiring, terminals, or connectors - Microphone - APIM	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test R.
No sound from all SYNC® audio sources (Bluetooth, USB, audio input jack)	- Customer's device - Wiring, terminals, or connectors - APIM - Audio Front Control Module (ACM)	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test O.
The SYNC® steering wheel switches are inoperative or do not operate correctly	- Steering wheel switch - Wiring, terminals, or connectors - Clockspring - SCCM	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test X.
All media hub functions are inoperative or do not operate correctly (USB ports, Secure Digital (SD) card, and RCA jacks)	- Customer's device - Media hub - Wiring, terminals, or connectors - APIM	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test S.

The USB port(s) are/is inoperative or do/does not operate correctly	- USB cable/port - Customer's device - APIM	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test T.
The Secure Digital (SD) card slot is inoperative or does not operate correctly	- Customer's device - Media hub USB cable - APIM	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test U.
The RCA jacks are inoperative or do not operate correctly	- Customer's device - Wiring, terminals, or connectors - Media hub - APIM	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test V.
Unable to pair Bluetooth device	- Incompatible Bluetooth device - APIM	PERFORM the APIM Reset., REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, GO to Pinpoint Test W.
An individual Bluetooth device feature is inoperative (text messaging, media playback or controls, phone book download, call history, or call waiting)	Customer's device compatibility	PERFORM the APIM Reset, REFER to Accessory Protocol Interface Module (APIM) Reset. If the concern is still present, INSTRUCT the customer to review the device compatibility list. REFER to www.syncmyride.com. It is normal operation for unsupported or incompatible Bluetooth features to be unavailable through the SYNC® system. If the inoperative feature is compatible, ADVISE the customer to check their device for firmware updates, and if required, reset their device following the manufacturer's instructions.
911 Assist™ or Vehicle Health Report (VHR) is inoperative or does not operate correctly	- Unregistered SYNC® owner account - RCM misconfiguration - IPC misconfiguration	Verify that the customer has a SYNC® Owner Account on www.syncmyride.com. This error message appears due to incomplete or incorrect PMI procedures. CHECK the vehicle service history for recent service actions related to the IPC or RCM . If there have been recent service actions with the IPC or RCM , REPEAT/PERFORM the Programmable Module Installation (PMI) procedure as directed by the diagnostic scan tool. REFER to: Section 418-01. If there have been no recent service actions, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.

Pinpoint Tests

Pinpoint Test A: Poor Reception Or Continuous Seek Or Scan — AM/FM

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to AM/FM Radio, ~~ACM~~ , and Antenna in Information and Entertainment System.

The ~~RDS~~ function allows the vehicle operator to choose which genre (category) of music to listen to. When the SCAN button is pressed, the ~~ACM~~ only stops at channels falling within the selected category. If no channel in the selected category is found, the ~~ACM~~ continues to scan without stopping. This can appear to be a reception issue, but it is actually due to the customer's setting.

Possible Sources

- AM/FM antenna
- AM/FM antenna coaxial cable
- Charging system
- Ignition system
- No channel found in the selected category
- Noise suppression equipment
- Wiring, terminals, or connectors
- ~~ACM~~

PINPOINT TEST A : POOR RECEPTION OR CONTINUOUS SEEK OR SCAN — AM/FM					
A1 CHECK THE AUDIO SYSTEM RECEPTION					
• Operate the audio system in radio tuner AM/FM mode. • Check the reception with the engine running, and with the engine off. Does the poor reception only occur with the engine running? <table><tr><td>Yes</td><td>GO to A2.</td></tr><tr><td>No</td><td>GO to A4.</td></tr></table>		Yes	GO to A2 .	No	GO to A4 .
Yes	GO to A2 .				
No	GO to A4 .				
A2 CHECK THE GENERATOR					
• Ignition OFF. • Disconnect: Wiring harness to the generator voltage regulator. • Start the engine. • Operate the audio system in radio tuner AM/FM mode. Is the reception OK?					

Yes	INSTALL a new generator. REFER to Section 414-00.
No	GO to A3 .

A3 CHECK THE IGNITION CIRCUITS

- Ignition OFF.
- Connect: Wiring harness to the generator voltage regulator.
- Visually inspect the engine compartment and make sure all ignition coils are correctly and securely connected, and that there are no visible cracks in the coil housings.
- Inspect all the wiring harnesses and connectors for damaged insulation and loose or broken conditions.

Are the ignition components OK?

Yes	USE a jumper cable to ground various parts of the vehicle to the frame (for example: engine, fenders, quarter panels, stone deflectors, air cleaner, body sheet metal). When the noise is eliminated, PROVIDE a permanent ground where necessary.
No	REPAIR the ignition system as necessary.

A4 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: AM/FM antenna coaxial cable at the ~~ACM~~ . Refer to Audio Control Module (ACM).
- Disconnect: AM/FM antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground
AM/FM antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.
No	GO to A5 .

A5 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to A6 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A6 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM coaxial cable core at the ACM	—	Lower AM/FM antenna coaxial cable core at the inline connection	—
AM/FM antenna coaxial cable shield at the ACM	—	Lower AM/FM antenna coaxial cable shield at the inline connection	—

Is the resistance less than 1 ohm?

Yes	GO to A7 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A7 CHECK THE LOWER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to A8 .
No	REPAIR or INSTALL a new lower AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A8 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: AM/FM antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: AM/FM antenna coaxial cable at the antenna unit. Refer to Antenna — AM/FM.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>	—	—	Ground
AM/FM antenna coaxial cable shield at the <u>ACM</u>	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.
No	GO to A9 .

A9 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the <u>ACM</u>	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to A10 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A10 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable core at the antenna unit (harness side)	—
AM/FM antenna coaxial cable shield at the ACM	—	AM/FM antenna coaxial cable shield at the antenna unit (harness side)	—

Is the resistance less than 1 ohm?

Yes	GO to A11 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A11 CHECK THE UPPER AM/FM ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to A12 .
No	REPAIR or INSTALL a new upper AM/FM antenna coaxial cable. REFER to Antenna Cable — AM/FM.

A12 CHECK THE AM/FM ANTENNA UNIT FOR A SHORT TO VOLTAGE

- Ignition OFF.

- Connect: AM/FM antenna coaxial cable at the antenna unit.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground
AM/FM antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.
No	GO to A13 .

A13 CHECK THE AM/FM ANTENNA UNIT CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to A14 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

A14 CHECK THE AM/FM ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna coaxial cable core at the ACM	—	AM/FM antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to A15 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

A15 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND

- Disconnect: AM/FM antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna unit shield at the antenna unit (component side)		—	Ground

Is the resistance less than 1 ohm?

Yes	GO to A17 .
No	GO to A16 .

A16 CHECK THE AM/FM ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE

- Remove the AM/FM antenna. Refer to Antenna — AM/FM.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
AM/FM antenna unit shield at the antenna unit (component side)		—	AM/FM antenna base (unpainted surface)

Is the resistance less than 1 ohm?

Yes	GO to A17 .
No	INSTALL a new AM/FM antenna. REFER to Antenna — AM/FM.

A17 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.

- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test B: The Satellite Radio Is Inoperative, Has Poor Reception, Or Has No Sound

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to MyKey® and Satellite Radio in Information and Entertainment System.

When a satellite radio subscription is activated, the satellite receiver **ESN** is associated with the **VIN** .

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1A89:11	Satellite Antenna: Circuit Short To Ground	Set by the ACM when a short to ground is detected in the satellite antenna circuit for more than 250 milliseconds. This DTC can be either continuous or on-demand.
B1A89:13	Satellite Antenna: Circuit Open	Set by the ACM when an open is detected in the satellite antenna circuit. This DTC can be either continuous or on-demand.

Possible Sources

- Expired subscription
- Incorrectly matched **ESN**
- Wiring, terminals, or connectors
- Obstructions in the antenna's line of sight

- Satellite radio antenna
- Satellite radio antenna coaxial cable
- ACM

PINPOINT TEST B : THE SATELLITE RADIO IS INOPERATIVE, HAS POOR RECEPTION, OR HAS NO SOUND

B1 VERIFY AN ACTIVE SUBSCRIPTION

- Operate the infotainment system in satellite radio mode and observe the infotainment display.

Does the display indicate that the subscription has expired?

Yes	The subscription has expired. INFORM the customer to contact SIRIUS to reactivate the subscription.
No	GO to B2 .

B2 CHECK THE RECEPTION

- Drive the vehicle to an open location that is free from obstacles such as tall buildings or large trees.
- Using a diagnostic scan tool, view the ACM PIDs.
- Monitor the satellite signal strength (SAT_SIG_STR) PID.
- If the PID reads "No Signal" or "Poor," move the vehicle to another area and retest until "Good," "Very Good," or "Excellent" PID values are obtained.

Can a "Good," "Very Good," or "Excellent" PID value be obtained?

Yes	GO to B3 .
No	GO to B6 .

B3 CHECK THE SATELLITE RADIO SYSTEM FOR CORRECT OPERATION

- Operate multiple channels with the infotainment system in satellite radio mode.

Does the satellite radio system receive multiple channels with normal sound quality?

Yes	The system is operating correctly at this time. The concern was a result of obstructions in the antenna's line of sight.
No	GO to B4 .

B4 CHECK FOR PROMOTIONAL CHANNEL 184

- Select channel 184.

Does the satellite radio system receive channel 184 with normal sound quality?

Yes	GO to B5 .

No	GO to B6 .
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B5 VERIFY THAT THE VIN AND ESN ON FILE MATCH THE VIN AND ESN FROM THE VEHICLE

- Retrieve the satellite radio **ESN** . Refer to Bezel Diagnostics in this section.
 - Contact the SIRIUS Dealer Support Line (888-465-8528).
 - Ensure that the **VIN** and **ESN** on the customer's file match the **VIN** and **ESN** retrieved from the vehicle.
 - Follow the actions from step 2 to obtain "Good," "Very Good," or "Excellent" reception.
 - Operate the infotainment system in satellite radio mode.
 - **NOTE:** *It may take up to 30 minutes for the activation signal to be received and satellite radio functions to operate.*
- Request for SIRIUS to send the activation signal to the vehicle 2-3 times.
- Operate multiple channels with the infotainment system in satellite radio mode.

Does the satellite radio system receive multiple channels with normal sound quality?

Yes	The system is operating correctly at this time. The concern was a result of the activation signal needing to be resent or the VIN and ESN SIRIUS had on file not matching the VIN and ESN from the vehicle.
No	GO to B6 .

B6 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Satellite radio antenna coaxial cable at the **ACM** . Refer to Audio Control Module (ACM).
- Disconnect: Satellite radio antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground
Satellite radio antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.
No	GO to B7 .

B7 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B8 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B8 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio coaxial cable core at the ACM	—	Lower satellite radio antenna coaxial cable core at the inline connection	—
Satellite radio antenna coaxial cable shield at the ACM	—	Lower satellite radio antenna coaxial cable shield at the inline connection	—

Is the resistance less than 1 ohm?

Yes	GO to B9 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B9 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B10 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B10 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: Satellite radio antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: Satellite radio antenna coaxial cable at the antenna unit. Refer to Antenna — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground
Satellite radio antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.
No	GO to B11 .

B11 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B12 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B12 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable core at the antenna unit (harness side)	—
Satellite radio antenna coaxial cable shield at the ACM	—	Satellite radio antenna coaxial cable shield at the antenna unit (harness side)	—

Is the resistance less than 1 ohm?

Yes	GO to B13 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B13 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B14 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

B14 CHECK THE SATELLITE RADIO ANTENNA UNIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: Satellite radio antenna coaxial cable at the antenna unit.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground
Satellite radio antenna coaxial cable shield at the ACM	—	—	Ground

Is any voltage present?

Yes	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.
No	GO to B15 .

B15 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to B16 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

B16 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to B17 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

B17 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND

- Disconnect: Satellite radio antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

Positive Lead			Negative Lead	
Pin		Circuit	Pin	Circuit
Satellite radio antenna unit shield at the antenna unit (component side)			—	Ground

Is the resistance less than 1 ohm?

Yes	GO to B19 .
No	GO to B18 .

B18 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE

- Remove the satellite radio antenna. Refer to Antenna — Satellite Radio.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna unit shield at the antenna unit (component side)	—	Satellite radio antenna base (unpainted surface)	—

Is the resistance less than 1 ohm?

Yes	GO to B19 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

B19 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test C: The SIRIUS Data Services Are Not Available (If Equipped)

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to SIRIUS Data Services in Information and Entertainment System.

When a SIRIUS Data Services subscription is activated, the satellite receiver **ESN** is associated with the **VIN** .

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1A89:11	Satellite Antenna: Circuit Short To Ground	Set by the ACM when a short to ground is detected in the satellite antenna circuit for more than 250 milliseconds. This DTC can be either continuous or on-demand.
B1A89:13	Satellite Antenna: Circuit Open	Set by the ACM when an open is detected in the satellite antenna circuit. This DTC can be either continuous or on-demand.
U1A00:87	Private Communication Network: Missing Message	Set by the APIM when messages are missing from the ACM on the satellite digital radio data link (SDL) circuits.

Removal of the navigation map secure digital (SD) card results in loss of SIRIUS Data Services while the card is removed. This is normal operation.

Possible Sources

- Navigation map secure digital (SD) card not present, not fully seated, or damaged
- Media hub
- Expired subscription
- Obstructions in the antenna's line of sight
- Incorrectly matched ~~ESN~~
- Satellite radio antenna
- Satellite radio antenna coaxial cable
- Wiring, terminals, or connectors
- ~~APIM~~
- ~~ACM~~

PINPOINT TEST C : THE SIRIUS DATA SERVICES ARE NOT AVAILABLE (IF EQUIPPED)	
C1 VERIFY THE NAVIGATION MAP SECURE DIGITAL (SD) CARD IS PRESENT	
• Check for the presence of a navigation map secure digital (SD) card in the media hub secure digital (SD) slot.	
Is the navigation map secure digital (SD) card present and fully seated?	
Yes	GO to C2 .
No	INSTALL the navigation map secure digital (SD) card. TEST the system for normal operation.
C2 INSPECT THE NAVIGATION MAP DATA SECURE DIGITAL (SD) CARD	
• Remove the navigation map data secure digital (SD) card. • Inspect the navigation map data secure digital (SD) card case and pins for damage.	
Is any damage present?	
Yes	INSTALL a new navigation map data secure digital (SD) card.
No	GO to C3 .
C3 CHECK THE USB PORTS FOR POWER OUTPUT	
• Connect Multi-Media Interface Tester 105-00120.	

Does the Multi-Media Interface Tester 105-00120 power on from both USB ports?

Yes	GO to C4 .
No	If the Multi-Media Interface Tester 105-00120 powers on from only one USB port, INSTALL a new media hub. REFER to Media Hub If the Multi-Media Interface Tester 105-00120 does not power on from either USB port, GO to Pinpoint Test S .

C4 RESET THE APIM AND VERIFY OPERATION OF THE USB CABLE

- Remove any devices connected to the media hub.
- Carry out an APIM power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Wait at least 2 minutes for the APIM to re-initialize.
- Connect Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using each USB port.

Is the audio output OK for each USB port?

Yes	GO to C5 .
No	If the audio output is not OK from one USB port, INSTALL a new media hub. REFER to Media Hub If the audio output is not OK from both USB ports, GO to Pinpoint Test T .

C5 VERIFY AN ACTIVE SUBSCRIPTION

NOTE: *The status of the satellite radio subscription should be disregarded in this step since SIRIUS data services does not require a satellite radio subscription to operate.*

- Access the Subscription Info selection in the touchscreen Travel Link menu.

Does the display indicate "Unsubscribed" for traffic, weather, fuel prices, movie times, sports, weather map, or ski conditions?

Yes	If the display indicates "Unsubscribed" for one or more, but not all categories mentioned above, GO to C28 . If the display indicates "Unsubscribed" for all categories mentioned above, INFORM the customer to contact SIRIUS to reactivate the subscription.
No	GO to C6 .

C6 CHECK THE RECEPTION

- Drive the vehicle to an open location that is free from obstacles such as tall buildings or large trees.
- Using a diagnostic scan tool, view the ACM PIDs.
- Monitor the satellite signal strength (SAT_SIG_STR) PID.

- If the PID reads "No Signal" or "Poor," move the vehicle to another area and retest until "Good," "Very Good," or "Excellent" PID values are obtained.

Can a "Good," "Very Good," or "Excellent" PID value be obtained?

Yes	GO to C7 .
No	GO to C10 .

C7 CHECK THE SIRIUS DATA SERVICES FOR CORRECT OPERATION

- Operate the infotainment system in SIRIUS data services mode and observe the infotainment display.

Do SIRIUS data services operate correctly?

Yes	The system is operating correctly at this time. The concern was a result of obstructions in the antenna's line of sight.
No	GO to C8 .

C8 CHECK FOR PROMOTIONAL CHANNEL 184

- Operate the infotainment system in satellite radio mode.
- Select channel 184.

Does the satellite radio system receive promotional channel 184 with normal sound quality?

Yes	GO to C9 .
No	GO to C10 .

C9 VERIFY THAT THE VIN AND ESN ON FILE MATCH THE VIN AND ESN FROM THE VEHICLE

- Retrieve the satellite radio ESN. Refer to Bezel Diagnostics in this section.
- Contact the SIRIUS Dealer Support Line (888-465-8528).
- Ensure that the VIN and ESN on the customer's file match the VIN and ESN retrieved from the vehicle.
- Follow the actions from step 4 to obtain "Good," "Very Good," or "Excellent" reception.
- Operate the infotainment system in SIRIUS data services mode.
- **NOTE:** *It may take up to 30 minutes for the activation signal to be received and satellite radio functions to operate.*

Request for SIRIUS to send the activation signal to the vehicle 2-3 times.
- Operate the infotainment system in SIRIUS data services mode and observe the infotainment display.

Do SIRIUS data services operate correctly?

Yes	The system is operating correctly at this time. The concern was a result of the activation signal needing to be resent or the VIN and ESN SIRIUS had on file not matching the VIN and ESN from the vehicle.
No	GO to C10 .

C10 CHECK FOR ACM DTCS

- Ignition ON.
- Using a diagnostic scan tool, clear the ACM DTCS.
- Using a diagnostic scan tool, perform the ACM self-test.

Are any DTCS recorded in the ACM ?

Yes	For DTC B1A89:11 or DTC B1A89:13, GO to C12 . For all other ACM DTCS, REFER to the ACM DTC chart in Information and Entertainment System.
No	GO to C11 .

C11 CHECK FOR APIM DTCS

- Ignition ON.
- Using a diagnostic scan tool, clear the APIM DTCS.
- Using a diagnostic scan tool, perform the APIM self-test.

Are any DTCS recorded in the APIM ?

Yes	For DTC U1A00:87, GO to C25 . For all other APIM DTCS, REFER to the APIM DTC chart in Information and Entertainment System.
No	GO to C29 .

C12 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Satellite radio antenna coaxial cable at the ACM . Refer to Audio Control Module (ACM).
- Disconnect: Satellite radio antenna inline coaxial cable connection behind the glove compartment. Refer to Antenna Cable — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the <u>ACM</u>	—	—	Ground
Satellite radio antenna coaxial cable shield at the <u>ACM</u>	—	—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.
No	GO to C13 .

C13 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the <u>ACM</u>	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to C14 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

C14 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio coaxial cable core at the <u>ACM</u>	—	Lower satellite radio antenna coaxial cable core at the inline connection	—
Satellite radio antenna coaxial cable shield at the <u>ACM</u>	—	Lower satellite radio antenna coaxial cable shield at the inline connection	—

Is the resistance less than 1 ohm?

Yes	GO to C15 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

C15 CHECK THE LOWER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to C16 .
No	REPAIR or INSTALL a new lower satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

C16 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: Satellite radio antenna inline coaxial cable connection behind the glove compartment.
- Disconnect: Satellite radio antenna coaxial cable at the antenna unit. Refer to Antenna — Satellite Radio.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead			Negative Lead	
Pin	Circuit		Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—		—	Ground
Satellite radio antenna coaxial cable shield at the ACM	—		—	Ground

Is any voltage present?

Yes	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.
No	GO to C17 .

C17 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	—	Ground

Is the resistance greater than 10,000 ohms?

Yes	GO to C18 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

C18 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable core at the antenna unit (harness side)	—
Satellite radio antenna coaxial cable shield at the ACM	—	Satellite radio antenna coaxial cable shield at the antenna unit (harness side)	—

Is the resistance less than 1 ohm?

Yes	GO to C19 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

C19 CHECK THE UPPER SATELLITE RADIO ANTENNA COAXIAL CABLE CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to C20 .
No	REPAIR or INSTALL a new upper satellite radio antenna coaxial cable. REFER to Antenna Cable — Satellite Radio.

C20 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE AND SHIELD FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Connect: Satellite radio antenna coaxial cable at the antenna unit.
- Ignition ON.
- Measure the **voltage** between:

Positive Lead			Negative Lead	
Pin		Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM		—	—	Ground
Satellite radio antenna coaxial cable shield at the ACM		—	—	Ground

Is any voltage present?

Yes	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.
No	GO to C21 .

C21 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead			Negative Lead	
Pin		Circuit	Pin	Circuit

Satellite radio antenna coaxial cable core at the ACM	—	—	Ground
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Is the resistance greater than 10,000 ohms?

Yes	GO to C22 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

C22 CHECK THE SATELLITE RADIO ANTENNA UNIT CORE AND SHIELD FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna coaxial cable core at the ACM	—	Satellite radio antenna coaxial cable shield at the ACM	—

Is the resistance greater than 10,000 ohms?

Yes	GO to C23 .
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

C23 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO GROUND

- Disconnect: Satellite radio antenna coaxial cable at the antenna unit.
- Measure the **resistance** between:

Positive Lead			Negative Lead	
Pin		Circuit	Pin	Circuit
Satellite radio antenna unit shield at the antenna unit (component side)		—	—	Ground

Is the resistance less than 1 ohm?

Yes	GO to C29 .
No	GO to C24 .

C24 CHECK THE SATELLITE RADIO ANTENNA UNIT SHIELD FOR CONTINUITY TO ANTENNA UNIT BASE

- Remove the satellite radio antenna. Refer to Antenna — Satellite Radio.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Satellite radio antenna unit shield at the antenna unit (component side)	—	Satellite radio antenna base (unpainted surface)	—

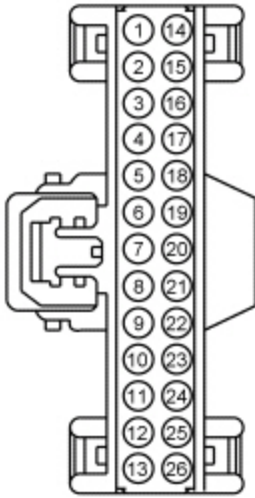
Is the resistance less than 1 ohm?

Yes	INSPECT and REPAIR any corrosion on the antenna unit base and/or vehicle mounting surface. REINSTALL the antenna and ensure the proper torque is achieved. REFER to Antenna — Satellite Radio.
No	INSTALL a new satellite radio antenna. REFER to Antenna — Satellite Radio.

C25 CHECK THE SATELLITE DATA CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~ACM~~ [C240B](#).
- Disconnect: ~~APIM~~ [C2383](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 6	VME90 (GY/BU)	—	Ground
C240B Pin 19	VME91 (WH/BN)	—	Ground



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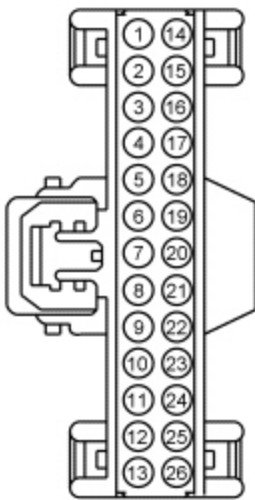
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to C26 .

C26 CHECK THE SATELLITE DATA CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 6	VME90 (GY/BU)	—	Ground
C240B Pin 19	VME91 (WH/BN)	—	Ground



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Are the resistances greater than 10,000 ohms?

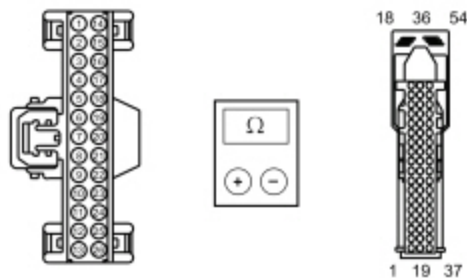
Yes	GO to C27 .
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No	REPAIR the circuit in question.
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C27 CHECK THE SATELLITE DATA CIRCUITS FOR AN OPEN

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 6	VME90 (GY/BU)	C2383 Pin 51	VME90 (GY/BU)
C240B Pin 19	VME91 (WH/BN)	C2383 Pin 52	VME91 (WH/BN)



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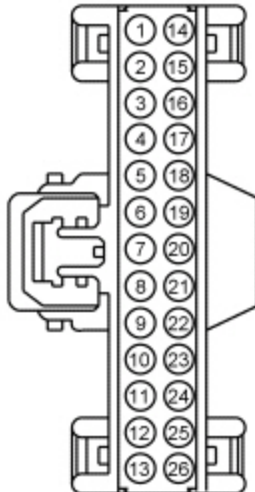
Are the resistances less than 3 ohms?

Yes	GO to C28 .
No	REPAIR the circuit in question.

C28 CHECK THE SATELLITE DATA CIRCUITS FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 6	VME90 (GY/BU)	C240B Pin 19	VME91 (WH/BN)



•

Is the resistance greater than 10,000 ohms?

Yes	GO to C30 .
No	REPAIR the circuit in question.

C29 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

C30 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)

- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test D: The Speed Compensated Volume Does Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to Speed Compensated Volume in Information and Entertainment System.

Possible Sources

- Speed compensated volume setting
- **YSS** signal concern
- Network communication concern
- Audio **DSP** module

PINPOINT TEST D : THE SPEED COMPENSATED VOLUME DOES NOT OPERATE CORRECTLY

D1 CHECK THE SPEEDOMETER OPERATION

- Drive the vehicle and observe the speedometer.

Does the speedometer operate correctly?

Yes	GO to D2 .
No	REFER to Section 413-01.

D2 CHECK THE SPEED COMPENSATED VOLUME SETTING

- Turn the speed compensated volume off. Refer to the Owner's Literature.
- Operate the audio system in radio tuner AM/FM mode.
- Drive the vehicle at various speeds and observe the speaker volume.
- Set the speed compensated volume to maximum compensation. Refer to the Owner's Literature.

- Operate the audio system in radio tuner AM/FM mode.
- Drive the vehicle at various speeds and observe the speaker volume.

Does the volume remain constant with the speed compensated volume turned off, and increase and decrease with vehicle speed with the speed compensated volume set to maximum?

Yes	The system is operating correctly at this time. INSTRUCT the customer in the correct usage of the speed compensated volume feature.
No	GO to D3 .

D3 CHECK FOR DTC U0155:00

- Using a diagnostic scan tool, retrieve all **CMDTCs** .

Is DTC U0155:00 or U0238:00 retrieved in any audio system module?

Yes	For DTC U0155:00, GO to Pinpoint Test AF . For DTC U0238:00, GO to Pinpoint Test AI .
No	GO to D4 .

D4 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test E: Poor Sound Quality Or Distorted Sound (Buzz/Crackle/Rattle) From One Or More Speakers (Not All Speakers)

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

The audio DSP module sends audio signals to the speakers in the form of AC voltage, resulting in clear audio output. REFER to Information and Entertainment System. See USB Audio Mode.

Possible Sources

- Loose component fasteners
- Loose door handle, lock, and other trim panel components and fasteners
- Loose items in storage areas
- Loose speaker grilles
- Loose wire harnesses or wire harness fasteners
- Speaker(s)
- Water shield bonding and placement

Visual Inspection and Diagnostic Pre-checks

Visually inspect around the suspect area for any possible rattle conditions such as:

- Any storage compartments including cup holders (if equipped)
- Door map pockets for contents that can rattle (if equipped)
- Door pull cup attachments (if equipped)
- Door switch bezels (if equipped)
- Package tray (if equipped)
- Speaker grilles
- Trim panels and fasteners

PINPOINT TEST E : POOR SOUND QUALITY OR DISTORTED SOUND (BUZZ/CRACKLE/RATTLE) FROM ONE OR MORE SPEAKERS (NOT ALL SPEAKERS)

E1 ISOLATE THE ZONE	
• Download the diagnostic sound tracks to a suitable USB drive then install the USB drive into the Media Hub. Refer to Speaker Audio Test. • On the audio system select USB input. • Using the audio system fade and balance feature, adjust the audio sound to each of the four zones (LF , RF , LR , and RR) of the vehicle and use the 6 diagnostic sound tracks to isolate the poor sound quality. Allow each track to run for 10 seconds, then use the Seek buttons to change the test track. • Locate and apply pressure to the trim panel(s) around the poor sound quality area in question.	
Does applying pressure to a trim panel reduce or eliminate the audible noise?	
Yes	REPAIR or REPLACE the trim panel as needed.
No	GO to E2 .

E2 REMOVE AND INSPECT BEHIND/UNDERNEATH THE SUSPECT TRIM PANEL(S)	
• Remove the trim panel to access the suspect speaker. • Operate the audio system using digital media (CD, MP3, etc.).	

- Validate where the poor sound quality area is located.
- Check:
 - Trim panel for loose components around the speaker area such as the speaker grille
 - Trim panel joining components for missing or broken pieces
 - Lock and handle mechanical parts for correct attachment
 - Wire harnesses for correct routing
 - Wire harness fasteners for correct attachment
 - Water shield for correct placement (speaker air path not blocked)
 - Water shield for correct bonding to sheet metal
 - Storage areas for loose items
 - All child safety belt anchors (if equipped)
 - All safety belt retractors
 - All speaker bracket fasteners and other fasteners are secured and tightened to specified torque

Was the source of the noise located?

Yes	REPAIR or REPLACE any loose or broken component or fastener as needed.
No	GO to E3 .

E3 CHECK THE SUSPECT SPEAKER FOR WATER INTRUSION

- Inspect for watermarks.
- Check the:
 - Cone
 - Magnet
 - Basket

Are any watermarks present on the speaker?

Yes	VERIFY the water shield is in the correct location, REPAIR or REPLACE any trim, door, or speaker seal as required. DRY the speaker in question and TEST the system for normal operation.
No	GO to E4 .

E4 ISOLATE SPEAKER TO VERIFY NOISE

- Remove the suspect speaker from its location and leave the speaker connected to the harness.
- Hold the speaker away from any trim panel(s) and ensure the speaker is isolated from contact.
- Operate the audio system using digital media (CD, MP3, etc.).

Is the noise still present in the suspect speaker?

Yes	INSTALL a new speaker for the one in question.
No	LOCATE the source of the noise and REPAIR as needed.

Pinpoint Test F: No Sound From All Speakers

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** in Information and Entertainment System.

When the ignition is in the START position, the **ACM** receives voltage on the START signal circuit and mutes the speaker output. This prevents voltage spikes from producing popping noises through the speakers. A short to voltage on this circuit causes the **ACM** output to be muted continuously.

The audio **DSP** module is enabled through software. Additionally, the **ACM** can mute the audio **DSP** module through the mute/unmute circuit. If the audio **DSP** module detects less than 3.5 volts on the mute/unmute circuit, it mutes the speaker output. As a result, an open or short to ground in the mute/unmute circuit causes the audio system to produce no sound, because the enable signal does not reach the audio **DSP** module. In the event of an overload, the audio **DSP** module clips the audio output signal to the speakers and subwoofer (heard as distortion) in order to prevent damage to the speakers.

A short to ground or short to voltage in the circuitry to one of the speakers may cause multiple speakers to lose sound due to the built-in overload protection feature of the **ACM** . In this case, a speaker fault DTC sets.

Possible Sources

- Wiring, terminals, or connectors
- **ACM**
- Audio **DSP** module

PINPOINT TEST F : NO SOUND FROM ALL SPEAKERS

F1 CHECK FOR SPEAKER DTCS

- Ignition ON.
- Using a diagnostic scan tool, perform the audio **DSP** module self-test.

Are any speaker fault Diagnostic Trouble Codes (DTCs) present?

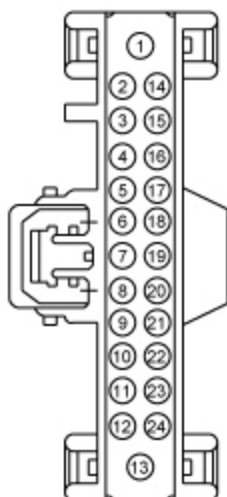
Yes	REFER to the audio DSP module DTC chart in Information and Entertainment System.
No	GO to F2 .

F2 CHECK THE START SENSE CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: **ACM** [C240A](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead	Negative Lead
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Pin	Circuit	Pin	Circuit
C240A Pin 3	CDC54 (WH/GN)	—	Ground



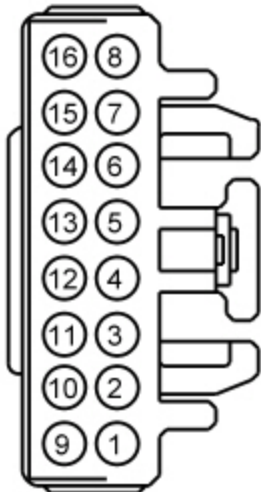
Is the voltage 5V or greater?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to F3 .

F3 CHECK THE AUDIO DSP MODULE VOLTAGE SUPPLY

- Ignition OFF.
- Disconnect: Audio DSP, [C3154C](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154C Pin 5	SBP19 (BU/RD)	—	Ground
C3154C Pin 6	SBP19 (BU/RD)	—	Ground



•

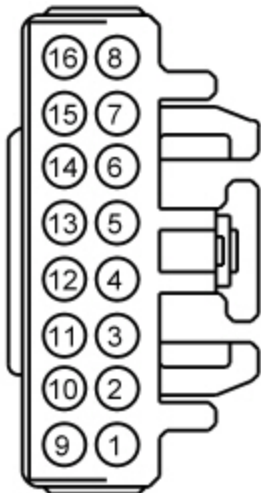
Are the voltages greater than 11 volts?

Yes	GO to F4 .
No	REPAIR the circuit.

F4 CHECK THE AUDIO DSP MODULE GROUND CIRCUITS

- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154C Pin 5	SBP19 (BU/RD)	C3154C Pin 15	GD148 (BK/YE)
C3154C Pin 6	SBP19 (BU/RD)	C3154C Pin 16	GD148 (BK/YE)



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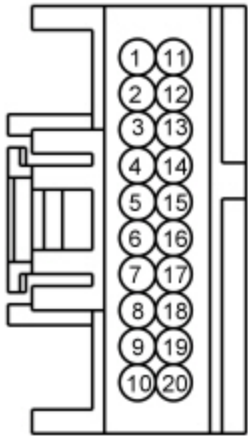
Are the voltages greater than 11 volts?

Yes	GO to F5 .
No	REPAIR the circuit.

F5 CHECK THE AUDIO DSP MODULE ENABLE CIRCUIT FOR CORRECT VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP. [C3154A](#).
- Ignition ON.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 3	SME23 (VT/RD)	—	Ground



•

Is the voltage between 3.5 and 8.5 volts?

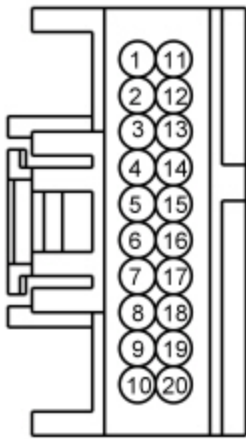
Yes	GO to F10 .
No	GO to F6 .

F6 CHECK THE AUDIO DSP MODULE ENABLE CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM. [C240B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit

C3154A Pin 3	SME23 (VT/RD)	—	Ground
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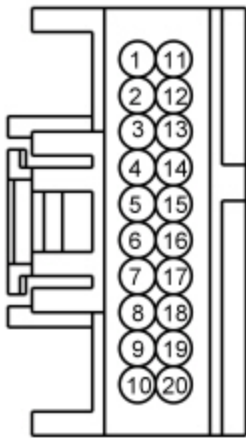
Is any voltage present?

Yes	REPAIR the circuit.
No	GO to F7 .

F7 CHECK THE AUDIO DSP MODULE ENABLE CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 3	SME23 (VT/RD)	—	Ground



•

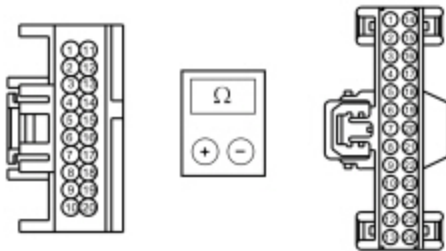
Is the resistance greater than 10,000 ohms?

Yes	GO to F8 .
No	REPAIR the circuit in question.

F8 CHECK THE AUDIO DSP MODULE ENABLE CIRCUIT FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 3	SME23 (VT/RD)	C240B Pin 18	SME23 (VT/RD)



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Is the resistance less than 3 ohms?

Yes	GO to F9 .
No	REPAIR the circuit in question.

F9 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
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No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.
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F10 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio DSP module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio DSP module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test G: The LF Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM , Audio DSP Module, and DSP in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A02:01	Speaker #2: General Electrical Failure	Set by the audio DSP module when a failure is detected on the LF door speaker circuits.
B1A02:11	Speaker #2: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the LF door speaker circuits. When DTC B1A02:11 is set, the speaker output is disabled.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A02:12	Speaker #2: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the LF door speaker circuits. When DTC B1A02:12 is set, the speaker output is disabled.
B1A02:13	Speaker #2: Circuit Open	Set by the audio DSP module when an open is detected on the LF door speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- ACM
- Audio DSP module

PINPOINT TEST G : THE LF DOOR SPEAKER IS INOPERATIVE

G1 CHECK THE AUDIO SIGNAL TO THE LF DOOR SPEAKER			
• Ignition OFF. • Disconnect: LF Door Speaker C523. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C523 Pin 2	RME07 (WH/BN)
			
•			
Is a fluctuating AC voltage present?			
Yes	INSTALL a new LF door speaker. REFER to Speaker — Door.		
No	GO to G2 .		

G2 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	—	Ground
C523 Pin 2	RME07 (WH/BN)	—	Ground



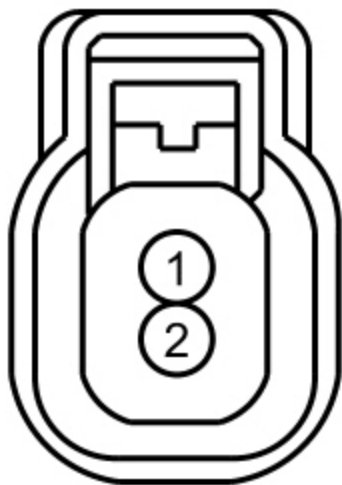
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to G3 .

G3 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	—	Ground
C523 Pin 2	RME07 (WH/BN)	—	Ground



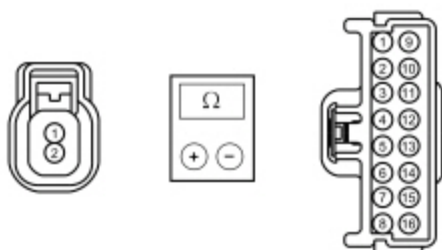
Are the resistances greater than 10,000 ohms?

Yes	GO to G4 .
No	REPAIR the circuit in question.

G4 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C3154B Pin 2	VME07 (WH)
C523 Pin 2	RME07 (WH/BN)	C3154B Pin 10	RME07 (WH/BN)



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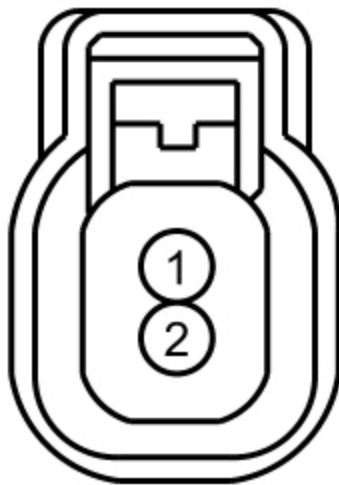
Are the resistances less than 3 ohms?

Yes	GO to G5 .
No	REPAIR the circuit in question.

G5 CHECK THE AUDIO CIRCUITS TO THE LF DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C523 Pin 1	VME07 (WH)	C523 Pin 2	RME07 (WH/BN)



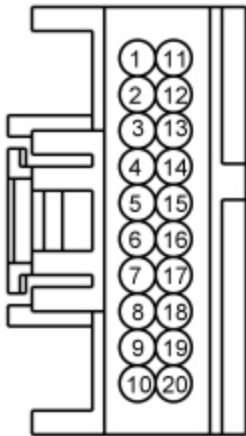
Is the resistance greater than 10,000 ohms?

Yes	GO to G6 .
No	REPAIR the circuit in question.

G6 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154A](#).
- Ignition ON.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C3154A Pin 17	RME17 (GY)



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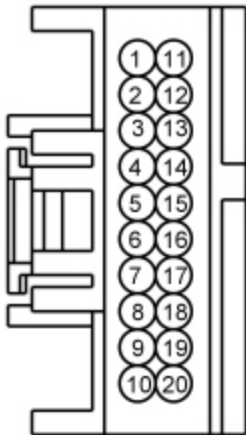
Is a fluctuating AC voltage present?

Yes	GO to G12 .
No	GO to G7 .

G7 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	—	Ground
C3154A Pin 17	RME17 (GY)	—	Ground



•

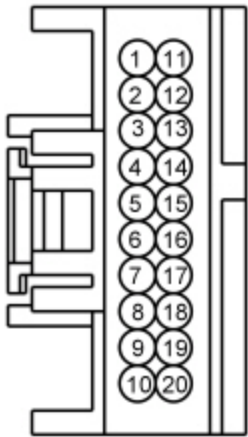
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to G8 .

G8 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	—	Ground
C3154A Pin 17	RME17 (GY)	—	Ground



-

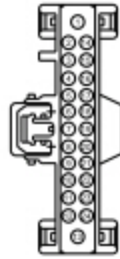
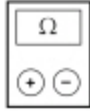
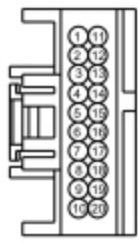
Are the resistances greater than 10,000 ohms?

Yes	GO to G9 .
No	REPAIR the circuit in question.

G9 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C240A Pin 22	VME17 (GN)
C3154A Pin 17	RME17 (GY)	C240A Pin 10	RME17 (GY)



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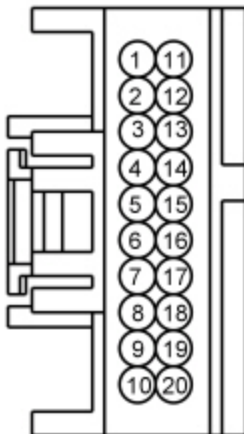
Are the resistances less than 3 ohms?

Yes	GO to G10 .
No	REPAIR the circuit in question.

G10 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C3154A Pin 17	RME17 (GY)



Is the resistance greater than 10,000 ohms?

Yes	GO to G11 .
No	REPAIR the circuit in question.

G11 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

G12 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test H: The LLF Tweeter Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM , Audio DSP Module, and DSP in Information and Entertainment System.

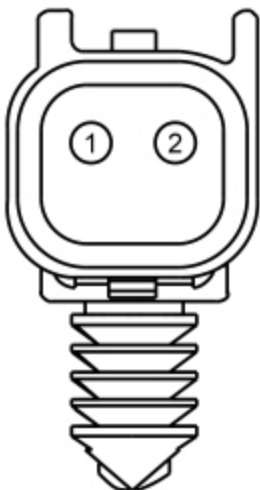
Root DTC Description	Failure Types	Fault Trigger Conditions
B1A03:01	Speaker #3: General Electrical Failure	Set by the audio DSP module when a failure is detected on the LF tweeter speaker circuits.
B1A03:11	Speaker #3: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the LF tweeter speaker circuits. When DTC B1A03:11 is set, the speaker output is disabled.
B1A03:12	Speaker #3: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the LF tweeter speaker circuits. When DTC B1A03:12 is set, the speaker output is disabled.
B1A03:13	Speaker #3: Circuit Open	Set by the audio DSP module when an open is detected on the LF tweeter speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- ACM
- Audio DSP module

PINPOINT TEST H : THE LF TWEETER SPEAKER IS INOPERATIVE

H1 CHECK THE AUDIO SIGNAL TO THE LF TWEETER SPEAKER			
• Ignition OFF. • Disconnect: LF Tweeter Speaker C230. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME08 (GN/OG)	C230 Pin 2	RME08 (GY/OG)



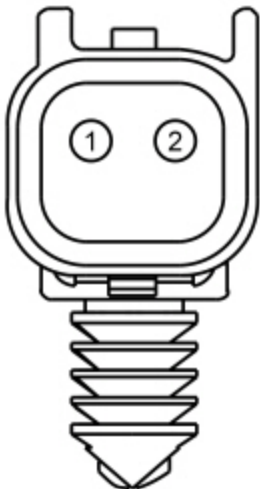
Is a fluctuating AC voltage present?

Yes	INSTALL a new LF tweeter speaker. REFER to Speaker — A-Pillar.
No	GO to H2 .

H2 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME08 (GN/OG)	—	Ground
C230 Pin 2	RME08 (GY/OG)	—	Ground



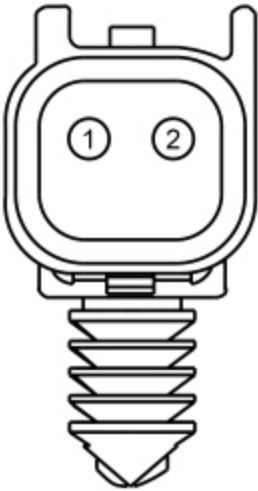
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to H3 .

H3 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME08 (GN/OG)	—	Ground
C230 Pin 2	RME08 (GY/OG)	—	Ground



Are the resistances greater than 10,000 ohms?

Yes	GO to H4 .
No	REPAIR the circuit in question.

H4 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME08 (GN/OG)	C3154B Pin 3	VME08 (GN/OG)
C230 Pin 2	RME08 (GY/OG)	C3154B Pin 11	RME08 (GY/OG)



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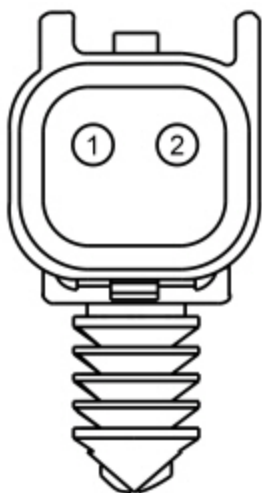
Are the resistances less than 3 ohms?

Yes	GO to H5 .
No	REPAIR the circuit in question.

H5 CHECK THE AUDIO CIRCUITS TO THE LF TWEETER SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C230 Pin 1	VME08 (GN/OG)	C230 Pin 2	RME08 (GY/OG)



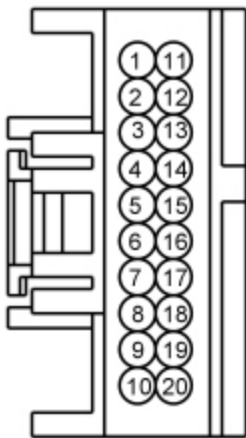
Is the resistance greater than 10,000 ohms?

Yes	GO to H6 .
No	REPAIR the circuit in question.

H6 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154A](#).
- Ignition ON.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C3154A Pin 17	RME17 (GY)



-

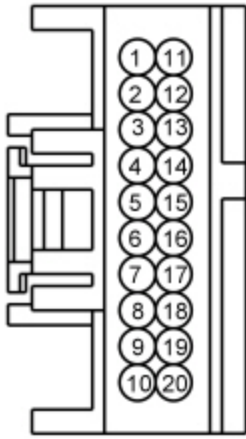
Is a fluctuating AC voltage present?

Yes	GO to H12 .
No	GO to H7 .

H7 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	—	Ground
C3154A Pin 17	RME17 (GY)	—	Ground



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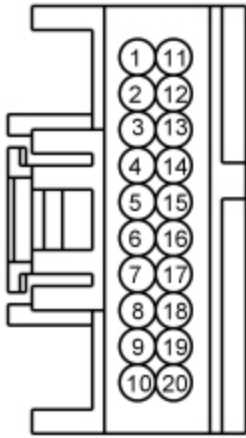
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to H8 .

H8 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	—	Ground
C3154A Pin 17	RME17 (GY)	—	Ground



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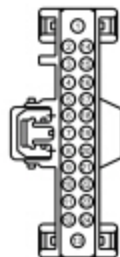
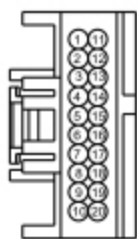
Are the resistances greater than 10,000 ohms?

Yes	GO to H9 .
No	REPAIR the circuit in question.

H9 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C240A Pin 22	VME17 (GN)
C3154A Pin 17	RME17 (GY)	C240A Pin 10	RME17 (GY)



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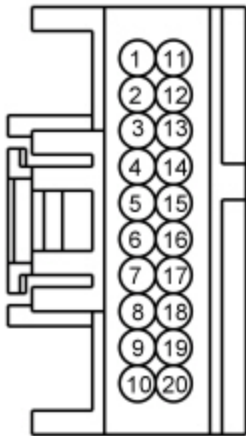
Are the resistances less than 3 ohms?

Yes	GO to H10 .
No	REPAIR the circuit in question.

H10 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C3154A Pin 17	RME17 (GY)



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Is the resistance greater than 10,000 ohms?

Yes	GO to H11 .
No	REPAIR the circuit in question.

H11 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

H12 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)

- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test I: The Instrument Panel Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** , Audio **DSP** Module, and **DSP** in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A01:01	Speaker #1: General Electrical Failure	Set by the audio DSP module when a failure is detected on the instrument panel speaker circuits.
B1A01:11	Speaker #1: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the instrument panel speaker circuits. When DTC B1A01:11 is set, the speaker output is disabled.
B1A01:12	Speaker #1: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the instrument panel speaker circuits. When DTC B1A01:12 is set, the speaker output is disabled.
B1A01:13	Speaker #1: Circuit Open	Set by the audio DSP module when an open is detected on the instrument panel speaker circuits.

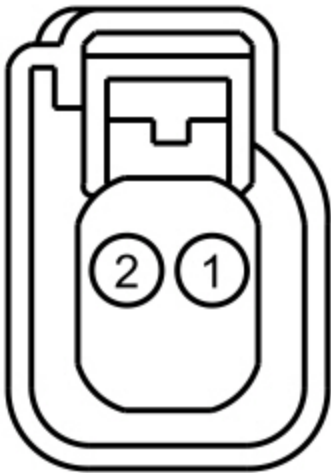
Possible Sources

- Wiring, terminals, or connectors
- Speaker
- Audio **DSP** module

I1 CHECK THE AUDIO SIGNAL TO THE INSTRUMENT PANEL SPEAKER

- Ignition OFF.
- Disconnect: Instrument Panel Speaker [C2356](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2356 Pin 1	VME06 (GN)	C2356 Pin 2	RME06 (GY/YE)



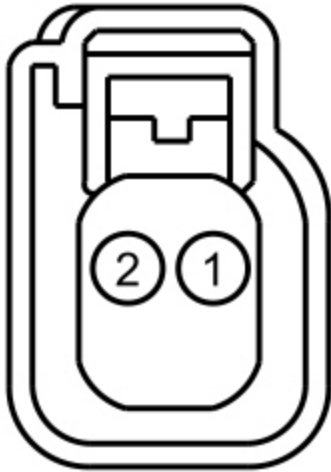
Is a fluctuating AC voltage present?

Yes	INSTALL a new instrument panel speaker. REFER to Speaker — Instrument Panel.
No	GO to I2 .

I2 CHECK THE AUDIO CIRCUITS TO THE INSTRUMENT PANEL SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP Module [C3154B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2356 Pin 1	VME06 (GN)	—	Ground
C2356 Pin 2	RME06 (GY/YE)	—	Ground



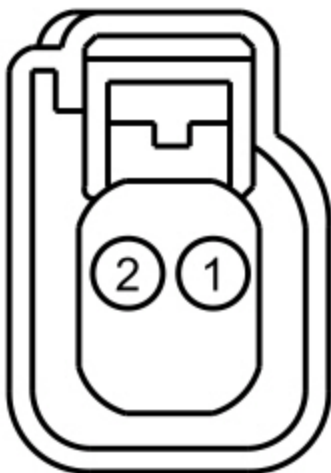
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to I3 .

I3 CHECK THE AUDIO CIRCUITS TO THE INSTRUMENT PANEL SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2356 Pin 1	VME06 (GN)	—	Ground
C2356 Pin 2	RME06 (GY/YE)	—	Ground



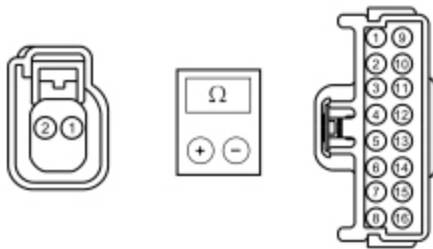
Are the resistances greater than 10,000 ohms?

Yes	GO to I4 .
No	REPAIR the circuit in question.

I4 CHECK THE AUDIO CIRCUITS TO THE INSTRUMENT PANEL SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2356 Pin 1	VME06 (GN)	C3154B Pin 1	VME06 (GN)
C2356 Pin 2	RME06 (GY/YE)	C3154B Pin 9	RME06 (GY/YE)



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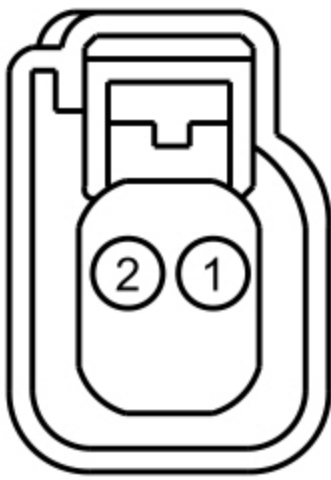
Are the resistances less than 3 ohms?

Yes	GO to I5 .
No	REPAIR the circuit in question.

I5 CHECK THE AUDIO CIRCUITS TO THE INSTRUMENT PANEL SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2356 Pin 1	VME06 (GN)	C2356 Pin 2	RME06 (GY/YE)



•

Is the resistance greater than 10,000 ohms?

Yes	GO to I6 .
No	REPAIR the circuit in question.

I6 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio DSP module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio DSP module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test J: The RF Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to [ACM](#) , Audio [DSP](#) Module, and [DSP](#) in Information and Entertainment System.

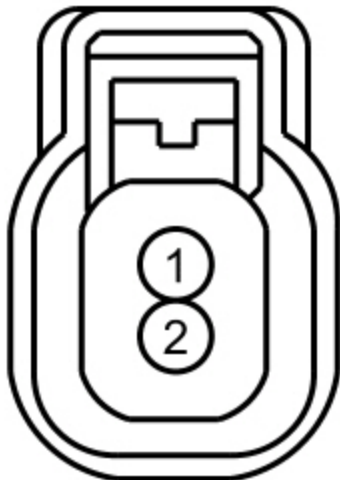
Root DTC Description	Failure Types	Fault Trigger Conditions
B1A06:01	Speaker #6: General Electrical Failure	Set by the audio DSP module when a failure is detected on the RF door speaker circuits.
B1A06:11	Speaker #6: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the RF door speaker circuits. When DTC B1A06:11 is set, the speaker output is disabled.
B1A06:12	Speaker #6: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the RF door speaker circuits. When DTC B1A06:12 is set, the speaker output is disabled.
B1A06:13	Speaker #6: Circuit Open	Set by the audio DSP module when an open is detected on the RF door speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- [ACM](#)
- Audio [DSP](#) module

PINPOINT TEST J : THE RF DOOR SPEAKER IS INOPERATIVE

J1 CHECK THE AUDIO SIGNAL TO THE RF DOOR SPEAKER			
• Ignition OFF. • Disconnect: RF Door Speaker C612. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C612 Pin 2	RME10 (WH/OG)



Is a fluctuating AC voltage present?

Yes	INSTALL a new RF door speaker. REFER to Speaker — Door.
No	GO to J2 .

J2 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	—	Ground
C612 Pin 2	RME10 (WH/OG)	—	Ground



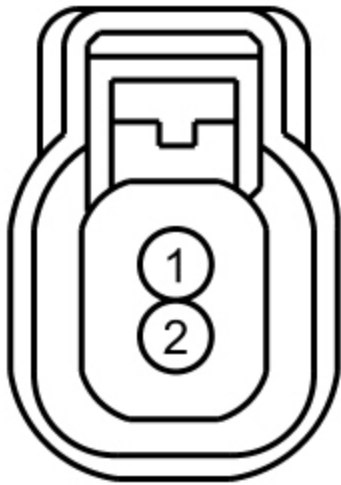
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to J3 .

J3 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	—	Ground
C612 Pin 2	RME10 (WH/OG)	—	Ground



Are the resistances greater than 10,000 ohms?

Yes	GO to J4 .
No	REPAIR the circuit in question.

J4 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C3154B Pin 6	VME10 (WH/VT)
C612 Pin 2	RME10 (WH/OG)	C3154B Pin 14	RME10 (WH/OG)



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Are the resistances less than 3 ohms?

Yes	GO to J5 .
No	REPAIR the circuit in question.

J5 CHECK THE AUDIO CIRCUITS TO THE RF DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C612 Pin 1	VME10 (WH/VT)	C612 Pin 2	RME10 (WH/OG)



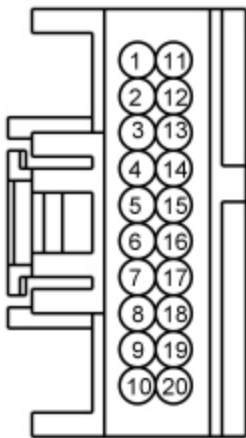
Is the resistance greater than 10,000 ohms?

Yes	GO to J6 .
No	REPAIR the circuit in question.

J6 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154A](#).
- Ignition ON.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C3154A Pin 19	RME18 (YE)



•

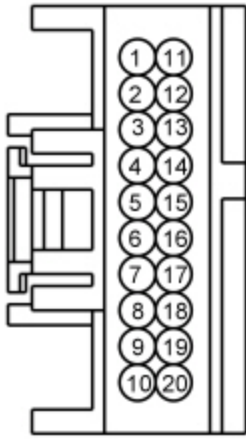
Is a fluctuating AC voltage present?

Yes	GO to J12 .
No	GO to J7 .

J7 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	—	Ground
C3154A Pin 19	RME18 (YE)	—	Ground



•

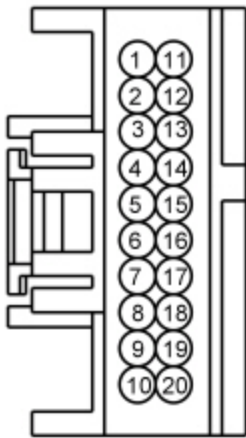
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to J8 .

J8 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	—	Ground
C3154A Pin 19	RME18 (YE)	—	Ground



•

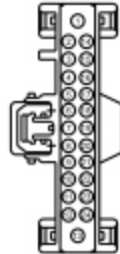
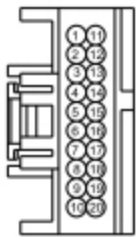
Are the resistances greater than 10,000 ohms?

Yes	GO to J9 .
No	REPAIR the circuit in question.

J9 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C240A Pin 23	VME18 (VT)
C3154A Pin 19	RME18 (YE)	C240A Pin 11	RME18 (YE)



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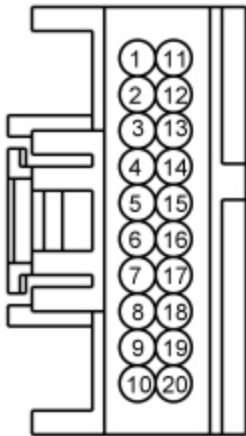
Are the resistances less than 3 ohms?

Yes	GO to J10 .
No	REPAIR the circuit in question.

J10 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C3154A Pin 19	RME18 (YE)



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Is the resistance greater than 10,000 ohms?

Yes	GO to J11 .
No	REPAIR the circuit in question.

J11 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

J12 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)

- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test K: The **RF Tweeter Speaker Is Inoperative**

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to **ACM** , Audio **DSP** Module, and **DSP** in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A07:01	Speaker #7: General Electrical Failure	Set by the audio DSP module when a failure is detected on the RF tweeter speaker circuits.
B1A07:11	Speaker #7: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the RF tweeter speaker circuits. When DTC B1A07:11 is set, the speaker output is disabled.
B1A07:12	Speaker #7: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the RF tweeter speaker circuits. When DTC B1A07:12 is set, the speaker output is disabled.
B1A07:13	Speaker #7: Circuit Open	Set by the audio DSP module when an open is detected on the RF tweeter speaker circuits.

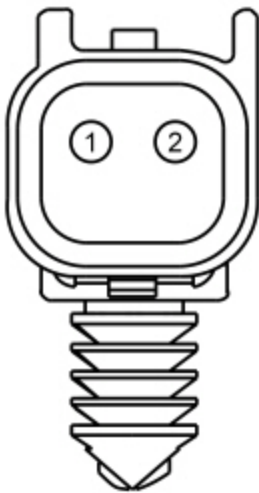
Possible Sources

- Wiring, terminals, or connectors
- Speaker
- **ACM**
- Audio **DSP** module

PINPOINT TEST K : THE RF TWEETER SPEAKER IS INOPERATIVE**K1 CHECK THE AUDIO SIGNAL TO THE RF TWEETER SPEAKER**

- Ignition OFF.
- Disconnect: RF Tweeter Speaker [C231](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME11 (VT/OG)	C231 Pin 2	RME11 (YE/OG)



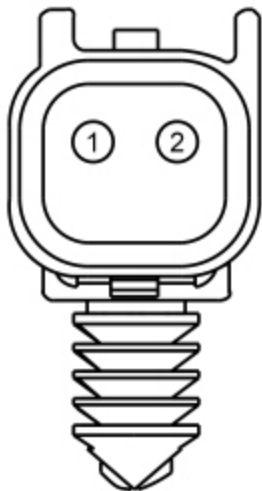
Is a fluctuating AC voltage present?

Yes	INSTALL a new RF tweeter speaker. REFER to Speaker — A-Pillar.
No	GO to K2 .

K2 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME11 (VT/OG)	—	Ground
C231 Pin 2	RME11 (YE/OG)	—	Ground



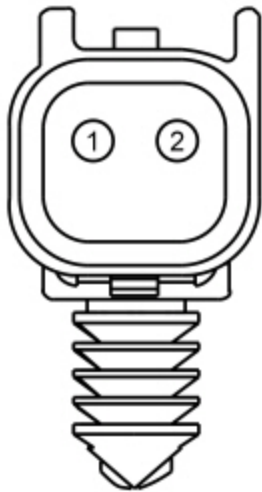
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to K3 .

K3 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME11 (VT/OG)	—	Ground
C231 Pin 2	RME11 (YE/OG)	—	Ground



Are the resistances greater than 10,000 ohms?

Yes	GO to K4 .
No	REPAIR the circuit in question.

K4 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME11 (VT/OG)	C3154B Pin 7	VME11 (VT/OG)
C231 Pin 2	RME11 (YE/OG)	C3154B Pin 15	RME11 (YE/OG)



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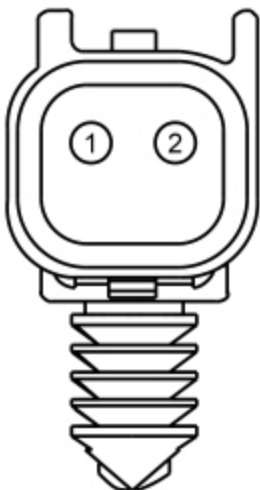
Are the resistances less than 3 ohms?

Yes	GO to K5 .
No	REPAIR the circuit in question.

K5 CHECK THE AUDIO CIRCUITS TO THE RF TWEETER SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C231 Pin 1	VME11 (VT/OG)	C231 Pin 2	RME11 (YE/OG)



•

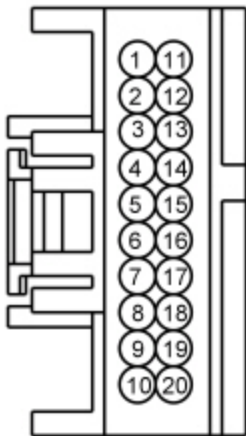
Is the resistance greater than 10,000 ohms?

Yes	GO to K6 .
No	REPAIR the circuit in question.

K6 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154A](#).
- Ignition ON.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C3154A Pin 19	RME18 (YE)



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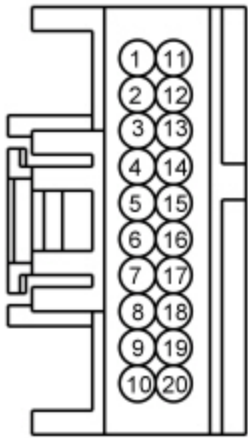
Is a fluctuating AC voltage present?

Yes	GO to K12 .
No	GO to K7 .

K7 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~ACM~~ [C240A](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	—	Ground
C3154A Pin 19	RME18 (YE)	—	Ground



-

Is any voltage present?

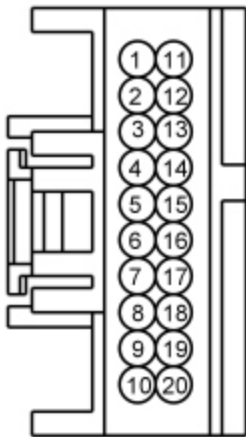
Yes	REPAIR the circuit in question.
No	GO to K8 .

K8 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
---------------	--	---------------	--

Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	—	Ground
C3154A Pin 19	RME18 (YE)	—	Ground



•

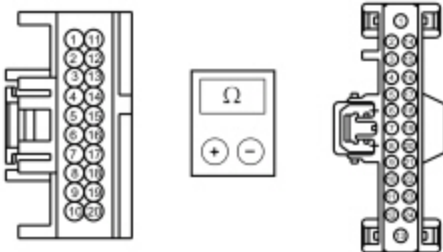
Are the resistances greater than 10,000 ohms?

Yes	GO to K9 .
No	REPAIR the circuit in question.

K9 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C240A Pin 23	VME18 (VT)
C3154A Pin 19	RME18 (YE)	C240A Pin 11	RME18 (YE)



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Are the resistances less than 3 ohms?

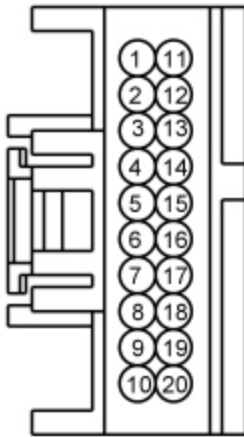
Yes	GO to K10 .
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No	REPAIR the circuit in question.
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K10 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C3154A Pin 19	RME18 (YE)



-

Is the resistance greater than 10,000 ohms?

Yes	GO to K11 .
No	REPAIR the circuit in question.

K11 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM .
------------	--

	REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

K12 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION				
• Ignition OFF. • Disconnect and inspect all the audio DSP module connectors. • Repair: ▪ Corrosion (clean module pins or install new connectors or terminals) ▪ Damaged or bent pins (install new terminals or pins) ▪ Pushed-out pins (install new pins as necessary) • Reconnect the audio DSP module connectors. Make sure they seat and latch correctly. • Operate the system and determine if the concern is still present.				
Is the concern still present?				
<table border="1"><tr><td>Yes</td><td>CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.</td></tr><tr><td>No</td><td>The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.</td></tr></table>	Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.	No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.
Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.			
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.			

Pinpoint Test L: The RR Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM , Audio DSP Module, and DSP in Information and Entertainment System.

Root DTC Description	Failure Types	Fault Trigger Conditions
B1A08:01	Speaker #8: General Electrical Failure	Set by the audio DSP module when a failure is detected on the RR door speaker circuits.
B1A08:11	Speaker #8: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the RR door speaker circuits. When DTC B1A08:11 is set, the speaker output is

Root DTC Description	Failure Types	Fault Trigger Conditions
		disabled.
B1A08:12	Speaker #8: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the RR door speaker circuits. When DTC B1A08:12 is set, the speaker output is disabled.
B1A08:13	Speaker #8: Circuit Open	Set by the audio DSP module when an open is detected on the RR door speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- ACM
- Audio DSP module

PINPOINT TEST L : THE RR DOOR SPEAKER IS INOPERATIVE

L1 CHECK THE AUDIO SIGNAL TO THE RR DOOR SPEAKER

- Ignition OFF.
- Disconnect: RR Door Speaker [C802](#).
- Operate the audio system in radio tuner AM/FM mode.
- Measure the AC voltage between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C802 Pin 2	RME12 (BN/BU)



Is a fluctuating AC voltage present?

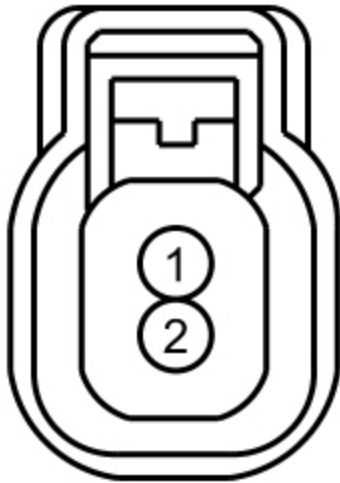
Yes	INSTALL a new RR door speaker. REFER to Speaker — Door.
------------	---

No	GO to L2 .
----	----------------------------

L2 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	—	Ground
C802 Pin 2	RME12 (BN/BU)	—	Ground



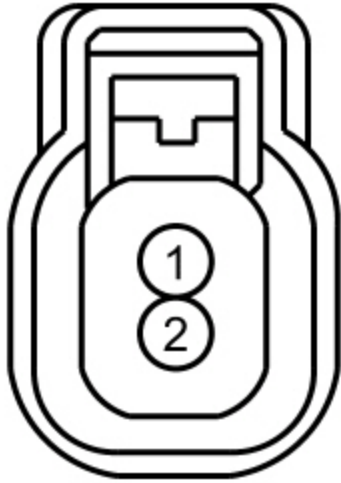
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to L3 .

L3 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	—	Ground
C802 Pin 2	RME12 (BN/BU)	—	Ground



Are the resistances greater than 10,000 ohms?

Yes	GO to L4 .
No	REPAIR the circuit in question.

L4 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C3154B Pin 8	VME12 (BN/WH)
C802 Pin 2	RME12 (BN/BU)	C3154B Pin 16	RME12 (BN/BU)



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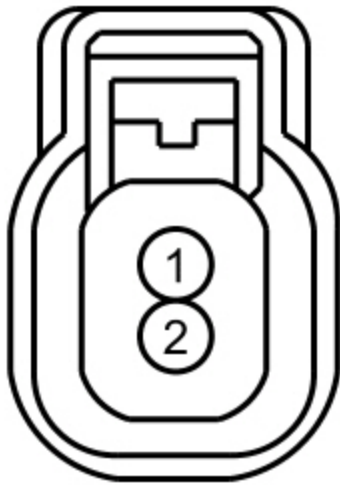
Are the resistances less than 3 ohms?

Yes	GO to L5 .
No	REPAIR the circuit in question.

L5 CHECK THE AUDIO CIRCUITS TO THE RR DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C802 Pin 1	VME12 (BN/WH)	C802 Pin 2	RME12 (BN/BU)



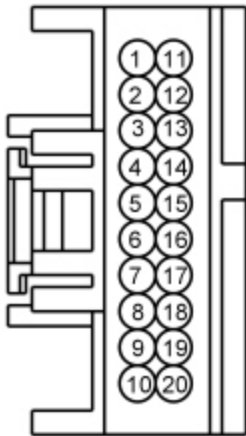
Is the resistance greater than 10,000 ohms?

Yes	GO to L6 .
No	REPAIR the circuit in question.

L6 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154A](#).
- Ignition ON.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C3154A Pin 19	RME18 (YE)



•

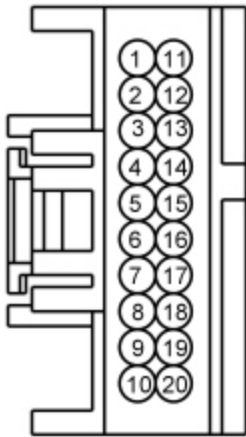
Is a fluctuating AC voltage present?

Yes	GO to L12 .
No	GO to L7 .

L7 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	—	Ground
C3154A Pin 19	RME18 (YE)	—	Ground



•

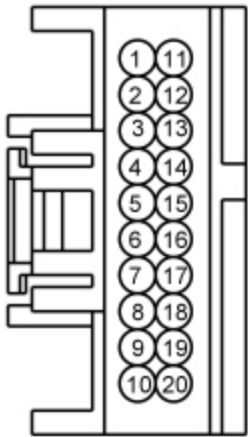
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to L8 .

L8 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	—	Ground
C3154A Pin 19	RME18 (YE)	—	Ground



-

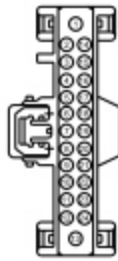
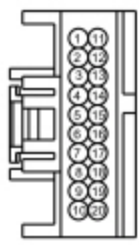
Are the resistances greater than 10,000 ohms?

Yes	GO to L9 .
No	REPAIR the circuit in question.

L9 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C240A Pin 23	VME18 (VT)
C3154A Pin 19	RME18 (YE)	C240A Pin 11	RME18 (YE)



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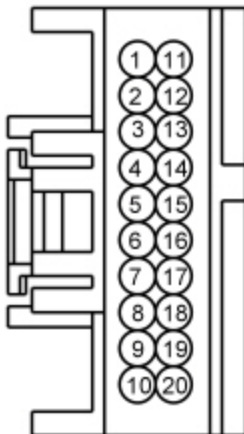
Are the resistances less than 3 ohms?

Yes	GO to L10 .
No	REPAIR the circuit in question.

L10 CHECK THE RIGHT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 9	VME18 (VT)	C3154A Pin 19	RME18 (YE)



Is the resistance greater than 10,000 ohms?

Yes	GO to L11 .
No	REPAIR the circuit in question.

L11 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

L12 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test M: The Subwoofer Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM , Audio DSP Module, DSP , and Subwoofer Speaker in Information and Entertainment System.

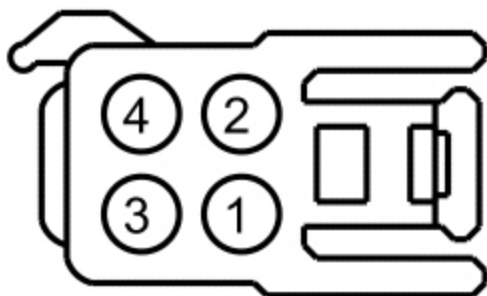
Root DTC Description	Failure Types	Fault Trigger Conditions
B1A09:11	Speaker #9: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the subwoofer speaker circuits. When DTC B1A09:11 is set, the speaker output is disabled.
B1A09:12	Speaker #9: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the subwoofer speaker circuits. When DTC B1A09:12 is set, the speaker output is disabled.
B1A09:13	Speaker #9: Circuit Open	Set by the audio DSP module when an open is detected on the subwoofer speaker circuits.
B1A10:11	Speaker #10: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the subwoofer speaker circuits. When DTC B1A10:11 is set, the speaker output is disabled.
B1A10:12	Speaker #10: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the subwoofer speaker circuits. When DTC B1A10:12 is set, the speaker output is disabled.
B1A10:13	Speaker #10: Circuit Open	Set by the audio DSP module when an open is detected on the subwoofer speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- Audio DSP module

PINPOINT TEST M : THE SUBWOOFER SPEAKER IS INOPERATIVE

M1 CHECK THE AUDIO SIGNAL TO THE SUBWOOFER SPEAKER			
• Ignition OFF. • Disconnect: Subwoofer Speaker C3020. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3020 Pin 1	VME22 (VT/GN)	C3020 Pin 2	RME22 (GN/WH)
C3020 Pin 3	VME01 (GN/VT)	C3020 Pin 4	RME01 (GY)



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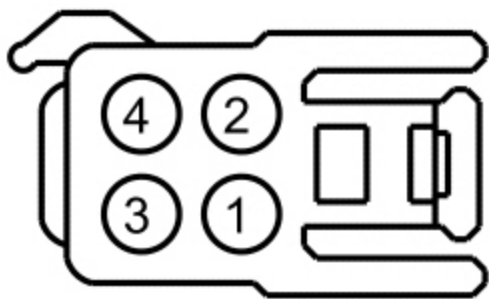
Is a fluctuating AC voltage present?

Yes	INSTALL a new subwoofer speaker. REFER to Speaker Enclosure.
No	GO to M2 .

M2 CHECK THE AUDIO CIRCUITS TO THE SUBWOOFER SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP Module [C3154C](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3020 Pin 1	VME22 (VT/GN)	—	Ground
C3020 Pin 2	RME22 (GN/WH)	—	Ground
C3020 Pin 3	VME01 (GN/VT)	—	Ground
C3020 Pin 4	RME01 (GY)	—	Ground



•

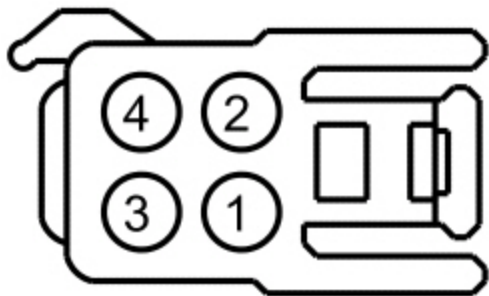
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to M3 .

M3 CHECK THE AUDIO CIRCUITS TO THE SUBWOOFER SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3020 Pin 1	VME22 (VT/GN)	—	Ground
C3020 Pin 2	RME22 (GN/WH)	—	Ground
C3020 Pin 3	VME01 (GN/VT)	—	Ground
C3020 Pin 4	RME01 (GY)	—	Ground



•

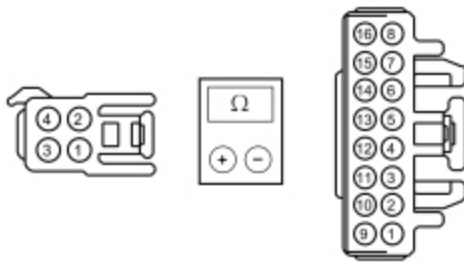
Are the resistances greater than 10,000 ohms?

Yes	GO to M4 .
No	REPAIR the circuit in question.

M4 CHECK THE AUDIO CIRCUITS TO THE SUBWOOFER SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3020 Pin 1	VME22 (VT/GN)	C3154C Pin 1	VME22 (VT/GN)
C3020 Pin 2	RME22 (GN/WH)	C3154C Pin 9	RME22 (GN/WH)
C3020 Pin 3	VME01 (GN/VT)	C3154C Pin 2	VME01 (GN/VT)
C3020 Pin 4	RME01 (GY)	C3154C Pin 10	RME01 (GY)



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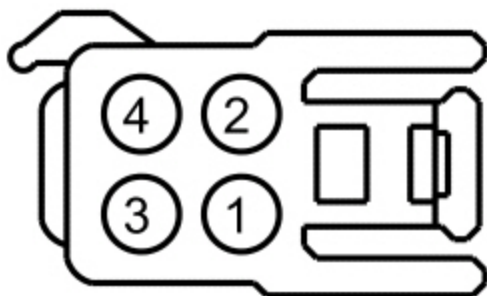
Are the resistances less than 3 ohms?

Yes	GO to M5 .
No	REPAIR the circuit in question.

M5 CHECK THE AUDIO CIRCUITS TO THE SUBWOOFER SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3020 Pin 1	VME22 (VT/GN)	C3020 Pin 2	RME22 (GN/WH)
C3020 Pin 1	VME22 (VT/GN)	C3020 Pin 3	VME01 (GN/VT)
C3020 Pin 1	VME22 (VT/GN)	C3020 Pin 4	RME01 (GY)
C3020 Pin 2	RME22 (GN/WH)	C3020 Pin 3	VME01 (GN/VT)
C3020 Pin 2	RME22 (GN/WH)	C3020 Pin 4	RME01 (GY)
C3020 Pin 3	VME01 (GN/VT)	C3020 Pin 4	RME01 (GY)



•

Are the resistances greater than 10,000 ohms?

Yes	GO to M6 .
No	REPAIR the circuit in question.

M6 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio DSP module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio DSP module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test N: The LR Door Speaker Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to ACM , Audio DSP Module, and DSP in Information and Entertainment System.

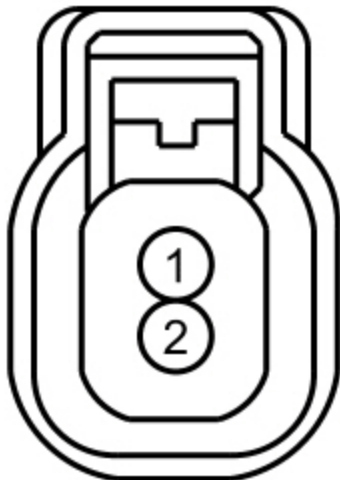
Root DTC Description	Failure Types	Fault Trigger Conditions
B1A04:01	Speaker #4: General Electrical Failure	Set by the audio DSP module when a failure is detected on the LR door speaker circuits.
B1A04:11	Speaker #4: Circuit Short to Ground	Set by the audio DSP module when a short to ground is detected on the LR door speaker circuits. When DTC B1A04:11 is set, the speaker output is disabled.
B1A04:12	Speaker #4: Circuit Short to Battery	Set by the audio DSP module when a short to battery is detected on the LR door speaker circuits. When DTC B1A04:12 is set, the speaker output is disabled.
B1A04:13	Speaker #4: Circuit Open	Set by the audio DSP module when an open is detected on the LR door speaker circuits.

Possible Sources

- Wiring, terminals, or connectors
- Speaker
- ACM
- Audio DSP module

PINPOINT TEST N : THE LR DOOR SPEAKER IS INOPERATIVE

N1 CHECK THE AUDIO SIGNAL TO THE LR DOOR SPEAKER			
• Ignition OFF. • Disconnect: LR Door Speaker C702. • Operate the audio system in radio tuner AM/FM mode. • Measure the AC voltage between:			
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C702 Pin 2	RME09 (BN/YE)



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Is a fluctuating AC voltage present?

Yes	INSTALL a new LR door speaker. REFER to Speaker — Door.
No	GO to N2 .

N2 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154B](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	—	Ground
C702 Pin 2	RME09 (BN/YE)	—	Ground



•

Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to N3 .

N3 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	—	Ground
C702 Pin 2	RME09 (BN/YE)	—	Ground



Are the resistances greater than 10,000 ohms?

Yes	GO to N4 .
No	REPAIR the circuit in question.

N4 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C3154B Pin 4	VME09 (WH/GN)
C702 Pin 2	RME09 (BN/YE)	C3154B Pin 12	RME09 (BN/YE)



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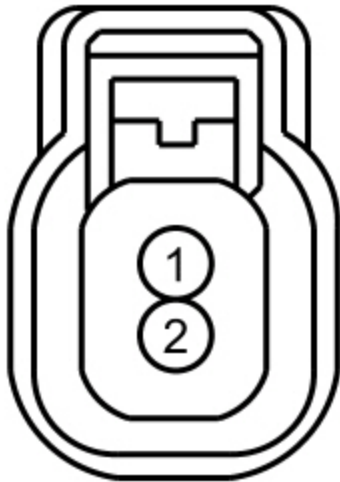
Are the resistances less than 3 ohms?

Yes	GO to N5 .
No	REPAIR the circuit in question.

N5 CHECK THE AUDIO CIRCUITS TO THE LR DOOR SPEAKER FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C702 Pin 1	VME09 (WH/GN)	C702 Pin 2	RME09 (BN/YE)



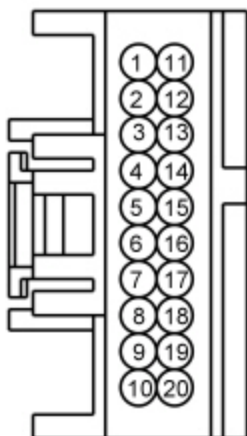
Is the resistance greater than 10,000 ohms?

Yes	GO to N6 .
No	REPAIR the circuit in question.

N6 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE

- Ignition OFF.
- Disconnect: Audio DSP module [C3154A](#).
- Ignition ON.
- Operate the audio system in radio tuner AM/FM mode.
- Measure the **AC voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C3154A Pin 17	RME17 (GY)



•

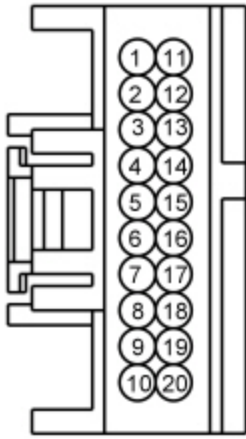
Is a fluctuating AC voltage present?

Yes	GO to N12 .
No	GO to N7 .

N7 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240A](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	—	Ground
C3154A Pin 17	RME17 (GY)	—	Ground



•

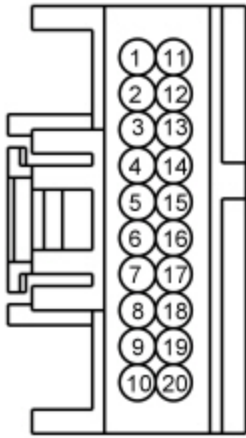
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to N8 .

N8 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	—	Ground
C3154A Pin 17	RME17 (GY)	—	Ground



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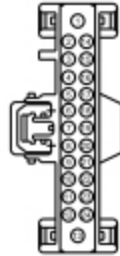
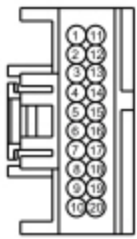
Are the resistances greater than 10,000 ohms?

Yes	GO to N9 .
No	REPAIR the circuit in question.

N9 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C240A Pin 22	VME17 (GN)
C3154A Pin 17	RME17 (GY)	C240A Pin 10	RME17 (GY)



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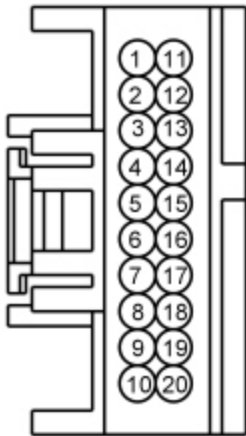
Are the resistances less than 3 ohms?

Yes	GO to N10 .
No	REPAIR the circuit in question.

N10 CHECK THE LEFT AUDIO SIGNAL TO THE AUDIO DSP MODULE FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 7	VME17 (GN)	C3154A Pin 17	RME17 (GY)



•

Is the resistance greater than 10,000 ohms?

Yes	GO to N11 .
No	REPAIR the circuit in question.

N11 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

N12 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)

- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test O: Poor Quality, Distorted, Or No Sound While In SYNC® Mode

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to SYNC® System in Information and Entertainment System.

The **APIM** transmits left and right channel analog audio signals to the **ACM** . These signals include inputs from the **USB** port, the SD card slot, the RCA jacks, and a Bluetooth media device. If a **USB** mass storage device is used to play an audio file, the SYNC® system only plays audio files that do not have **DRM** protection.

Possible Sources

- Customer's device
- Wiring, terminals, or connectors
- **APIM**
- **ACM**

PINPOINT TEST O : POOR QUALITY, DISTORTED, OR NO SOUND WHILE IN SYNC® MODE

O1 VERIFY THE OPERATION OF THE SYNC® SYSTEM AUDIO SOURCES

- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120 and RCA Adapter 105-00121, test the audio output for the RCA jacks, **USB** ports, and Bluetooth.

Is there poor quality, distorted, or no sound from all SYNC® audio sources?

Yes	GO to O2 .
No	If there is a concern with one or more, but not all audio sources, GO to Symptom Chart — SYNC® System to diagnose the observed symptom. If all audio sources operate correctly, the concern is with the customer's device.

O2 CHECK THE AUDIO SIGNALS TO THE ACM

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- Operate the audio system in SYNC® mode and play an audio file.
- Measure the **AC voltage** by back-probing ACM connector [C240B](#):

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240B Pin 12	VME52 (BU)	C240B Pin 25	RME52 (GY/OG)
C240B Pin 13	VME53 (VT/GN)	C240B Pin 26	RME53 (BN/WH)

Is a fluctuating AC voltage present?

Yes	GO to O7 .
No	GO to O3 .

O3 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ACM [C240B](#) .
- Disconnect: APIM [C2383](#) .
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 23	VME52 (BU)	—	Ground
C2383 Pin 24	RME52 (GY/OG)	—	Ground
C2383 Pin 25	VME53 (VT/GN)	—	Ground
C2383 Pin 26	RME53 (BN/WH)	—	Ground



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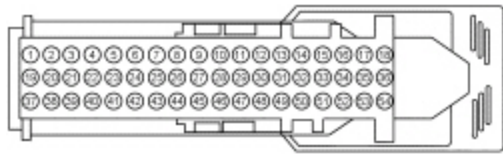
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to O4 .

O4 CHECK THE AUDIO CIRCUITS TO THE ACM FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 23	VME52 (BU)	—	Ground
C2383 Pin 24	RME52 (GY/OG)	—	Ground
C2383 Pin 25	VME53 (VT/GN)	—	Ground
C2383 Pin 26	RME53 (BN/WH)	—	Ground



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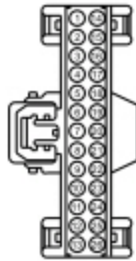
Are the resistances greater than 10,000 ohms?

Yes	GO to O5 .
No	REPAIR the circuit in question.

O5 CHECK THE CIRCUITS TO THE ACM FOR AN OPEN

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 23	VME52 (BU)	C240B Pin 12	VME52 (BU)
C2383 Pin 24	RME52 (GY/OG)	C240B Pin 25	RME52 (GY/OG)
C2383 Pin 25	VME53 (VT/GN)	C240B Pin 13	VME53 (VT/GN)
C2383 Pin 26	RME53 (BN/WH)	C240B Pin 26	RME53 (BN/WH)



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Are the resistances less than 3 ohms?

Yes	GO to O6 .
No	REPAIR the circuit in question.

O6 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM . REFER to Accessory Protocol Interface Module (APIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

O7 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ACM connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ACM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test P: The SYNC® System Audible Prompts Are Inoperative Or Do Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System for schematic and connector information.

Normal Operation and Fault Conditions

Refer to SYNC® System and Voice Recognition in Information and Entertainment System.

Possible Sources

- Customer's setting
- Wiring, terminals, or connectors
- APIM
- Audio DSP module

PINPOINT TEST P : THE SYNC® SYSTEM AUDIBLE PROMPTS ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY

P1 CHECK THE AUDIBLE PROMPT SETTING

- Operate the audio system in SYNC® mode.
- Verify the audible prompts are enabled. Refer to "SYNC® Voice Recognition Feature" in the Owner's Literature.
- Operate the VOICE button on the steering wheel controls and observe the SYNC® audible prompt.

Does the SYNC® system produce an audible prompt correctly?

Yes	The system is operating correctly at this time. The concern was caused by the customer's setting. INSTRUCT the customer in the correct operation of the audible prompt feature.
No	GO to P2 .

P2 CHECK FOR A VOLTAGE SIGNAL FROM THE APIM

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- Operate the audio system in SYNC® mode.
- While operating the VOICE button repeatedly, measure the **AC voltage** by back-probing audio DSP module connector [C3154A](#):

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 5	VME80 (VT/GN)	C3154A Pin 15	RME80 (BU/GY)

Is a fluctuating AC voltage present each time the VOICE switch is operated?

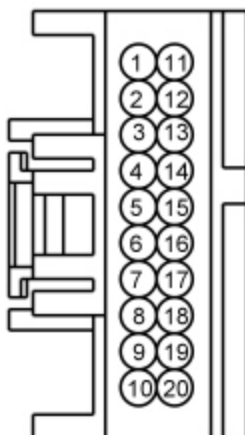
Yes	GO to P7 .
No	GO to P3 .

P3 CHECK THE AUDIBLE PROMPT CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Audio DSP Module [C3154A](#).
- Disconnect: APIM [C2383](#).
- Ignition ON.

- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 5	VME80 (VT/GN)	—	Ground
C3154A Pin 15	RME80 (BU/GY)	—	Ground



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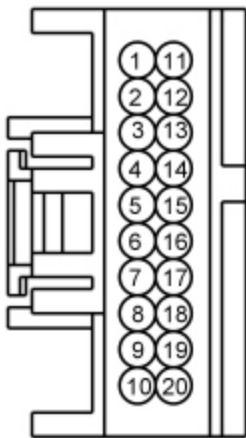
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to P4 .

P4 CHECK THE AUDIBLE PROMPT CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 5	VME80 (VT/GN)	—	Ground
C3154A Pin 15	RME80 (BU/GY)	—	Ground



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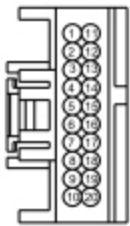
Are the resistances greater than 10,000 ohms?

Yes	GO to P5 .
No	REPAIR the circuit in question.

P5 CHECK THE AUDIBLE PROMPT CIRCUITS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154A Pin 5	VME80 (VT/GN)	C2383 Pin 3	VME80 (VT/GN)
C3154A Pin 15	RME80 (BU/GY)	C2383 Pin 4	RME80 (BU/GY)



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Are the resistances less than 3 ohms?

Yes	GO to P6 .
No	REPAIR the circuit in question.

P6 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.

- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

P7 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test Q: All SYNC® Services Features Are Inoperative Or Inaccurate

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to SYNC® Services in Information and Entertainment System.

Possible Sources

- Expired subscription
- APIM
- GPSM
- SYNC® Services server

PINPOINT TEST Q : ALL SYNC® SERVICES FEATURES ARE INOPERATIVE OR INACCURATE

Q1 VERIFY THE CUSTOMER'S SYNC® OWNER ACCOUNT AND SYNC® SERVICES REGISTRATION AND SUBSCRIPTION

- Verify that the customer has a SYNC® Owner Account on [SyncMyRide](#)
- Verify that the customer has registered for SYNC® Services and has an active subscription.

Does the customer have a SYNC® Owner Account and an active SYNC® Services subscription?

Yes	GO to Q2 .
No	If the customer does not have a SYNC® Owner Account, INFORM the customer that an account must be created in order to sign up for SYNC® Services. REFER to SyncMyRide . If the customer does not have an active SYNC® Services subscription, INFORM the customer that in order for SYNC® Services to function, the subscription needs to be activated. REFER to SyncMyRide .

Q2 VERIFY THE CUSTOMER'S REGISTERED PHONE NUMBER AND STATUS

- Verify that the mobile phone being used has the correct phone number registered with SYNC® Services.
- Verify that the mobile phone being used is selected as "Active" on [SyncMyRide](#).

Is the customer using the registered and active mobile phone associated with the SYNC® services account?

Yes	GO to Q3 .
No	INFORM the customer that the correct phone number must be registered on the SYNC® website and selected as "Active." REFER to SyncMyRide .

Q3 CHECK FOR APIM DTCS

- Ignition ON.
- Using a diagnostic scan tool, perform the APIM self-test.

Are any DTCs recorded in the APIM ?

Yes	REFER to the APIM DTC Chart in Information and Entertainment System.
No	GO to Q4 .

Q4 CHECK FOR GPSM DTCS

- Using a diagnostic scan tool, perform the GPSM self-test.

Are any DTCS recorded in the GPSM ?

Yes	REFER to the <u>GPSM</u> DTC Chart in Information and Entertainment System.
No	GO to Q5 .

Q5 VERIFY THE CUSTOMER'S PHONE CONNECTION TO SYNC® SERVICES

- Advise the customer to connect their phone to SYNC® Services. Caller ID blocking must be disabled on the customer's phone. Refer to the Owner's Literature.
- Operate a SYNC® Services feature.
- Verify that the phone call is placed and that the phone is connected to SYNC® Services.

Does the phone connect to SYNC® Services and operate correctly?

Yes	GO to Q6 .
No	GO to Pinpoint Test S .

Q6 OBSERVE THE SYNC® SERVICES SUBSCRIPTION COMMERCIAL

- Once the customer's phone is connected to SYNC® Services, observe the subscription commercial.

Does the SYNC® Services subscription commercial repeat with no selection option?

Yes	There is a potential SYNC® Services server issue, and the problem does not reside in the vehicle's audio system. CONTACT the Ford Technical Hotline for assistance.
No	GO to Q7 .

Q7 CHECK THE OPERATION OF THE DIRECTIONS FROM SYNC® SERVICES

- When prompted, select directions from SYNC® Services and observe the audible prompts from the audio system.

Is a single chime, audible prompt, then a short chime heard, indicating the navigation route is being sent to the vehicle?

Yes	SYNC® Services is operating correctly at this time.
No	If a single chime, audible prompt, then a repeating chime is heard, indicating the navigation route is being sent to the vehicle, GO to Q8 . If no chime or audible prompt is heard, indicating that the system was unable to locate the vehicle, there is a potential SYNC® Services server issue. The problem does not reside in the vehicle's audio system. CONTACT the Ford Technical Hotline for assistance.

Q8 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test R: Voice Recognition Is Inoperative Or Does Not Operate Correctly, Or During A Phone Call, Poor Quality Or No Outgoing Audio Is Heard On The Outside Device

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Voice Recognition in Information and Entertainment System.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B116A:01	Handset Microphone: General Electrical Failure	Set by the APIM when a short to voltage, short to ground, or open is detected on the microphone circuits.

Possible Sources

- Insufficiently quiet vehicle
- Inadequate phone reception
- Wiring, terminals, or connectors
- Microphone
- APIM

PINPOINT TEST R : VOICE RECOGNITION IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY, OR DURING A PHONE CALL, POOR QUALITY OR NO OUTGOING AUDIO IS HEARD ON THE OUTSIDE DEVICE

R1 CHECK THE AUDIBLE PROMPT FUNCTIONALITY

- Operate the audio system in radio tuner AM/FM mode.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.

Does the SYNC® system produce an audible prompt correctly, indicating voice recognition mode has been entered?

Yes	GO to R3 .
No	GO to R2 .

R2 CHECK THE OPERATION OF THE RH STEERING WHEEL SWITCHES

- Operate the audio system in AM/FM mode.
- Operate the VOLUME +/-, SEEK +/-, MEDIA, and PHONE steering wheel switches.

Do the VOLUME +/-, SEEK +/-, MEDIA, and PHONE steering wheel switches operate correctly?

Yes	GO to Pinpoint Test P .
No	GO to Pinpoint Test X to diagnose the RH steering wheel switches.

R3 CHECK THE OPERATION OF THE "PHONE" VOICE COMMAND

- Pair a Bluetooth phone with the SYNC® system. For information on the pairing process, refer to the Owner's Literature. If necessary, review the device compatibility list found on syncmyride.com.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.
- **NOTE:** *Make sure the interior of the vehicle is as quiet as possible. Excessive noise may prevent the system from correctly recognizing spoken commands.*

Clearly speak the word "phone."

Does the SYNC® system respond correctly to the "phone" voice command?

Yes	For a concern with the voice recognition being inoperative or not operating correctly, the system is operating correctly at this time. The concern may have been caused by an insufficiently quiet vehicle. Delete the just-paired Bluetooth phone from the SYNC® system. For information on deleting a Bluetooth device from the SYNC® system, refer to the Owner's Literature.
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	For a concern with poor quality or no outgoing audio being heard on the outside device during a phone call, GO to R4
No	GO to R5 .

R4 CHECK THE OUTGOING PHONE CALL QUALITY AND VOICE VOLUME

- Ensure that the Bluetooth phone paired in step 3 has adequate reception.
 - **NOTE:** *Make sure the interior of the vehicle is as quiet as possible. Excessive noise may prevent the call from having clear call quality and acceptable voice volume.*
- Using the SYNC® system, place a phone call. For information on making calls from the SYNC® system, refer to the Owner's Literature.
- Ensure that the called party's phone has adequate reception.
 - Instruct the called party to determine if the call quality is clear and of acceptable volume.
 - Delete the Bluetooth phone paired in step 3 from the SYNC® system. For information on deleting a Bluetooth device from the SYNC® system, refer to the Owner's Literature.

Is the outgoing call quality clear and of acceptable volume?

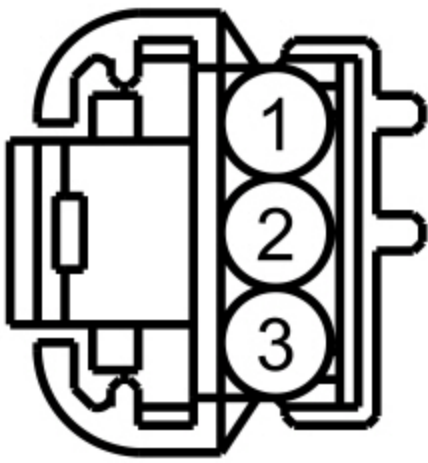
Yes	The system is operating correctly at this time. The concern may have been caused by poor phone reception or an insufficiently quiet vehicle.
No	GO to R5 .

R5 CHECK THE MICROPHONE AND MICROPHONE CIRCUITS FOR CORRECT VOLTAGE DROP

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- Ignition ON.
- While back-probing microphone [C9221](#), measure and record the **DC voltage**.

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C9221 Pin 3	DMM13 (BK)



Is the voltage between 2.7 and 6.2 volts?

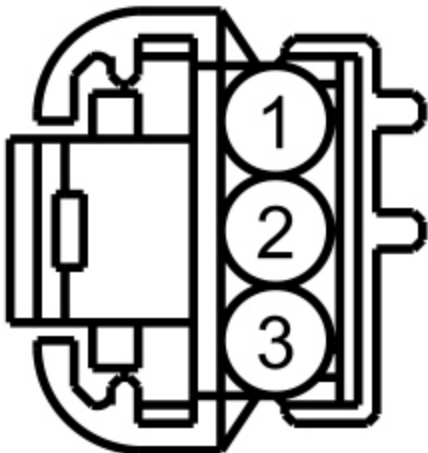
Yes	If diagnosing an intermittent concern, GO to R6 . If not diagnosing an intermittent concern, GO to R13
No	GO to R7 .

R6 CHECK THE MICROPHONE CIRCUITS FOR INTERMITTENT CONCERNS

NOTICE: This pinpoint test step directs testing circuits using a back-probe method. Use the special back-probe tool POM6411. Do not force test leads or other probes into connectors. Adequate care must be exercised to avoid connector terminal damage, while ensuring that good electrical contact is made with the circuit or terminal. Failure to follow these instructions may cause damage to wiring, terminals, or connectors, and may cause subsequent electrical faults.

- While back-probing microphone [C9221](#), measure the **DC voltage** while agitating the microphone harness back and forth and up and down (wiggle testing).

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C9221 Pin 3	DMM13 (BK)



- Review the recorded results from step 5.

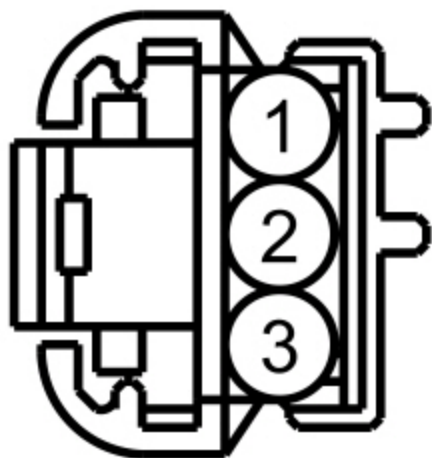
Does the voltage fluctuate more than .5 volts from the recorded result in step 5?

Yes	REPAIR the circuit(s), connector, or terminal(s) in question. For corrosion, clean pins or install a new connector or terminals. For damaged or bent pins, install new terminals or pins.
No	GO to R12 .

R7 CHECK THE MICROPHONE CIRCUITS FOR CORRECT VOLTAGE

- Ignition OFF.
- Disconnect: Microphone [C9221](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C9221 Pin 3	DMM13 (BK)



Is the voltage between 7.9 and 9.0 volts?

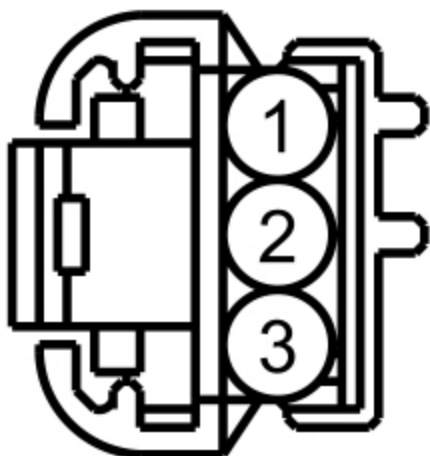
Yes	GO to R12 .
No	GO to R8 .

R8 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: APIM [C2383](#).
- Ignition ON.

- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	—	Ground
C9221 Pin 3	DMM13 (BK)	—	Ground



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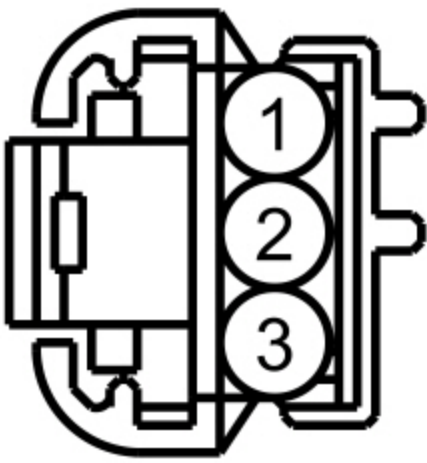
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to R9 .

R9 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	—	Ground
C9221 Pin 3	DMM13 (BK)	—	Ground



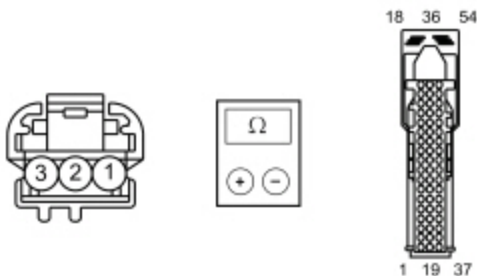
Are the resistances greater than 10,000 ohms?

Yes	GO to R10 .
No	REPAIR the circuit in question.

R10 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C2383 Pin 12	VMM13 (YE/GN)
C9221 Pin 3	DMM13 (BK)	C2383 Pin 13	DMM13 (BK)



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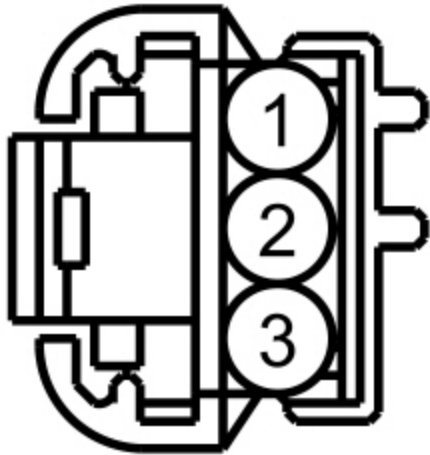
Are the resistances less than 3 ohms?

Yes	GO to R11 .
No	REPAIR the circuit in question.

R11 CHECK THE MICROPHONE CIRCUITS TO THE APIM FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C9221 Pin 2	VMM13 (YE/GN)	C9221 Pin 3	DMM13 (BK)



Is the resistance greater than 10,000 ohms?

Yes	GO to R13 .
No	REPAIR the circuit in question.

R12 ISOLATE THE MICROPHONE

- Install a new microphone. Refer to Microphone.
- If necessary, connect: ~~APIM~~ [C2383](#).
- Operate the audio system in AM/FM mode.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.
- Speak a command to test the voice recognition for normal operation.

Is the voice recognition functioning correctly?

Yes	The system is operating correctly at this time. The concern was caused by an inoperative microphone.
No	GO to R13 .

R13 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the ~~APIM~~ connector.
- Repair:

- Corrosion (clean module pins or install new connectors or terminals)
- Damaged or bent pins (install new terminals or pins)
- Pushed-out pins (install new pins as necessary)

- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test S: All Media Hub Functions Are Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Media Hub, Composite Audio-Video Input Mode, SD Card Mode, and **USB** Mode in Information and Entertainment System.

The media hub requires a power and ground circuit from the **APIM** to operate. Loss of power or ground disables all media hub functionality.

Possible Sources

- Device conflict
- Wiring, terminals, or connectors
- Media hub
- **APIM**

PINPOINT TEST S : ALL MEDIA HUB FUNCTIONS ARE INOPERATIVE

NOTE: Before carrying out this pinpoint test, make sure the MP3 device is operating correctly.

S1 CHECK FOR APIM BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **APIM** self-test.

Are any low or high voltage DTCs recorded in the **APIM** ?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
------------	--

No	GO to S2 .
-----------	----------------------------

S2 RESET THE APIM

- Remove any devices connected to the media hub.
- Carry out an **APIM** power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Test the system for normal operation.

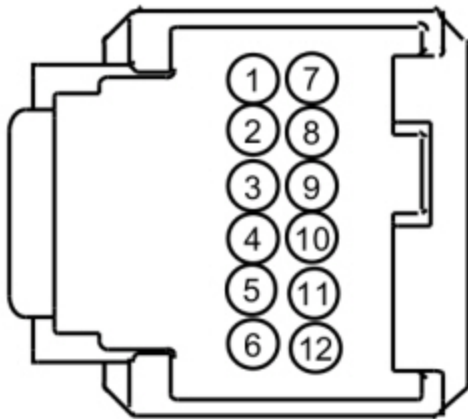
Does the system operate correctly?

Yes	The system is operating correctly at this time. The concern was corrected by performing the APIM power reset.
No	GO to S3 .

S3 CHECK THE MEDIA HUB VOLTAGE SUPPLY CIRCUIT

- Ignition OFF.
- Disconnect: Media Hub [C3618](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3618 Pin 1	LME89 (YE/BU)	—	Ground



-

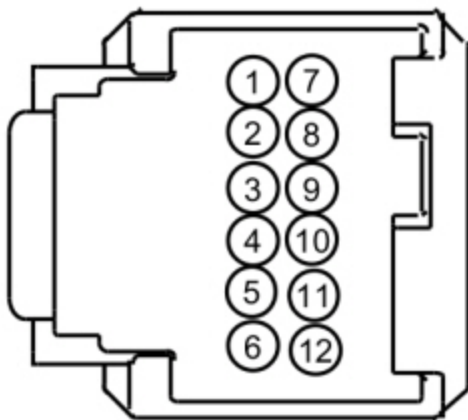
Is the voltage greater than 11 volts?

Yes	GO to S4 .
No	GO to S5 .

S4 CHECK THE MEDIA HUB GROUND CIRCUIT

- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3618 Pin 1	LME89 (YE/BU)	C3618 Pin 10	RME89 (WH/BU)



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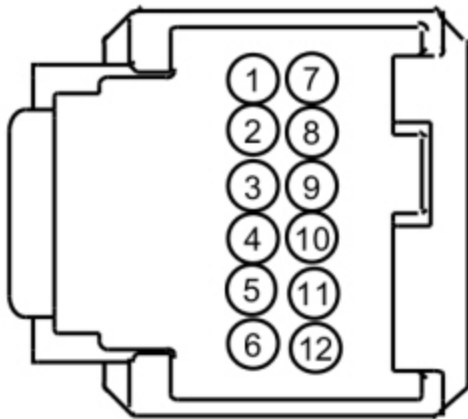
Is the voltage greater than 11 volts?

Yes	Install a new media hub. REFER to Media Hub.
No	GO to S5 .

S5 CHECK THE MEDIA HUB VOLTAGE SUPPLY CIRCUIT AND GROUND CIRCUIT FOR A SHORT TOGETHER

- Ignition OFF.
- Disconnect: APIM [C2383](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3618 Pin 1	LME89 (YE/BU)	C3618 Pin 10	RME89 (WH/BU)



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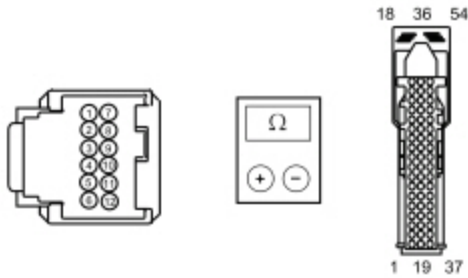
Is the resistance greater than 10,000 ohms?

Yes	GO to S6 .
No	Repair the circuit in question.

S6 CHECK THE MEDIA HUB VOLTAGE SUPPLY CIRCUIT AND GROUND CIRCUIT FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3618 Pin 1	LME89 (YE/BU)	C2383 Pin 8	LME89 (YE/BU)
C3618 Pin 10	RME89 (WH/BU)	C2383 Pin 38	RME89 (WH/BU)



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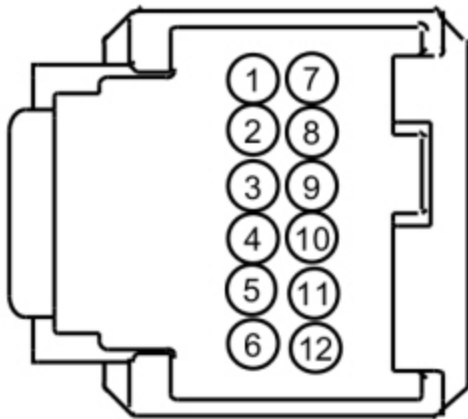
Are the resistances less than 3 ohms?

Yes	GO to S7 .
No	Repair the circuit in question.

S7 CHECK THE MEDIA HUB VOLTAGE SUPPLY CIRCUIT FOR A SHORT TO GROUND

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3618 Pin 1	LME89 (YE/BU)	—	Ground



-

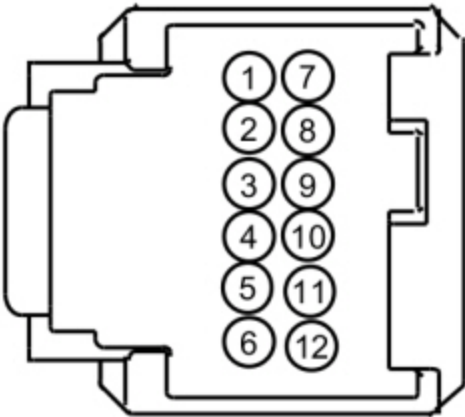
Is the resistance greater than 10,000 ohms?

Yes	GO to S8 .
No	REPAIR the circuit.

S8 CHECK THE MEDIA HUB GROUND CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3618 Pin 10	RME89 (WH/BU)	—	Ground



•

Is any voltage present?

Yes	REPAIR the circuit.

No	GO to S9 .
----	----------------------------

S9 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test T: The **USB** Port(s) Is/Are Inoperative Or Do/Does Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Media Hub and **USB** Mode in Information and Entertainment System.

Although there are two **USB** ports, there is only one **USB** cable connecting the media hub to the **APIM** . The **USB** cable is used for data transfer, and does not supply power. The Multi-Media Interface Tester 105-00120 is powered by the **USB** port.

USB and SD card data is transmitted to the **APIM** through the **USB** cable.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1252:11	USB Port: Circuit Short to Ground	Set by the APIM when a short to ground or over-current condition is detected in the USB cable. This can be caused by a fault in the USB cable, USB port, or customer's USB or SD card device.
B1252:13	USB Port: Circuit Open	Set by the APIM when an open is detected in the USB cable.

Possible Sources

- Device conflict
- USB cable
- Media hub
- APIM

PINPOINT TEST T : THE USB PORT(S) IS/ARE INOPERATIVE OR DO/DOES NOT OPERATE CORRECTLY

T1 CHECK THE USB PORT FOR POWER OUTPUT	
• Connect: Multi-Media Interface Tester 105-00120.	
Does the Multi-Media Interface Tester 105-00120 power on from both USB ports?	
Yes	GO to T2 .
No	If the Multi-Media Interface Tester 105-00120 powers on from only one USB port, INSTALL a new media hub. REFER to Media Hub. If the Multi-Media Interface Tester 105-00120 does not power on from either USB port, GO to Pinpoint Test R.
T2 RESET THE APIM	
• Remove any devices connected to the media hub. • Carry out an APIM power reset by disconnecting the battery for 1 minute, then reconnecting it. • Wait at least 2 minutes for the APIM to re-initialize. • Connect: Multi-Media Interface Tester 105-00120. • Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using each USB port.	
Is the audio output OK for each USB port?	
Yes	The system is operating correctly at this time. The concern was corrected by performing the APIM power reset.
No	If the audio output is not OK from one USB port, INSTALL a new media hub. REFER to Media Hub. If the audio output is not OK from both USB ports, GO to T3.
T3 CHECK THE SD CARD SLOT FUNCTIONALITY	

- Using a known good SD card, attempt to play an audio file using the SD card slot.

Is the audio output OK for the SD card slot?

Yes	INSTALL a new media hub. REFER to Media Hub.
No	GO to T4 .

T4 CHECK THE APIM FOR DTC B1252:11 OR DTC B1252:13

- Using a diagnostic scan tool, clear the **APIM** DTCs.
- Perform the **APIM** self-test.

Is DTC B1252:11 or DTC B1252:13 present?

Yes	GO to T5 .
No	GO to T7 .

T5 INSPECT THE USB CABLE, MEDIA HUB, AND APIM

- Ignition OFF.
- Disconnect: **USB** cable at the media hub.
- Disconnect: **USB** cable at the **APIM** .
- Inspect the **USB** cable ends, media hub **USB** connector port, and **APIM USB** connector port for damage. Remove any debris.

Is any damage present?

Yes	For the USB cable, INSTALL a new USB cable. REFER to Universal Serial Bus (USB) Cable and Port. For the media hub, INSTALL a new media hub. REFER to Media Hub. For the APIM , INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	GO to T6 .

T6 CHECK EACH USB PORT FOR FUNCTIONALITY

- Connect: **USB** cable at the media hub.
- Connect: **USB** cable at the **APIM** .
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using each **USB** port.

Is the audio output OK for each **USB** port?

Yes	The system is operating correctly at this time. The concern was caused by the USB connector not being seated correctly.
No	GO to T7 .

T7 ISOLATE THE USB CABLE

- INSTALL a new **USB** cable. REFER to Universal Serial Bus (USB) Cable and Port.
- Connect: Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using each **USB** port.

Is the audio output OK for each **USB port?**

Yes	The system is operating correctly at this time. The concern was caused by an inoperative USB cable.
No	GO to T8 .

T8 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test U: The SD Card Slot Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Media Hub and SD Card Mode in Information and Entertainment System.

SD card data is transmitted to the **APIM** through the **USB** cable.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1252:11	USB Port: Circuit Short to Ground	Set by the APIM when a short to ground or over-current condition is detected in the USB cable. This can be caused by a fault in the USB cable, USB port, or customer's USB or SD card device.
B1252:13	USB Port: Circuit Open	Set by the APIM when an open is detected in the USB cable.

Possible Sources

- Device conflict
- **USB** cable
- Media hub
- **APIM**

PINPOINT TEST U : THE SD CARD SLOT IS INOPERATIVE

U1 RESET THE APIM	
• Remove any devices connected to the media hub. • Carry out an APIM power reset by disconnecting the battery for 1 minute, then reconnecting it. • Wait at least 2 minutes for the APIM to re-initialize. • Using a known good SD card, attempt to play an audio file using the SD card slot.	
Is the audio output OK for the SD card slot?	
Yes	The system is operating correctly at this time. The concern was corrected by performing the APIM power reset.
No	GO to U2 .

U2 CHECK THE USB PORTS FOR FUNCTIONALITY	
• Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using each USB port.	
Is the audio output OK for each USB port?	

Yes	Install a new media hub. REFER to Media Hub.
No	If the audio output is not OK from one <u>USB</u> port, INSTALL a new media hub. REFER to Media Hub. If the audio output is not OK from both <u>USB</u> ports, GO to U3 .

U3 CHECK THE APIM FOR DTC B1252:11 OR DTC B1252:13

- Using a diagnostic scan tool, clear the APIM DTCs.
- Perform the APIM self-test.

Is DTC B1252:11 or DTC B1252:13 present?

Yes	GO to U4 .
No	Isolate the <u>USB</u> cable. INSTALL a new <u>USB</u> cable. Universal Serial Bus (USB) Cable and Port. Retest the system for normal operation. If the concern is still present, GO to U6 .

U4 INSPECT USB CABLE, MEDIA HUB, AND APIM

- Ignition OFF.
- Disconnect: USB cable at media hub.
- Disconnect: USB cable at APIM .
- Inspect the USB cable ends, media hub USB connector port, and APIM USB connector port for damage and remove any debris.

Is any damage present?

Yes	For the <u>USB</u> cable, INSTALL a new <u>USB</u> cable. REFER to Universal Serial Bus (USB) Cable and Port. For the media hub, INSTALL a new media hub. REFER to Media Hub. For the <u>APIM</u> , INSTALL a new <u>APIM</u> . Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	GO to U5 .

U5 RECHECK FOR APIM DTC B1252:11 OR DTC B1252:13

- Using a diagnostic scan tool, clear the APIM DTCs.
- Perform the APIM self-test.

Is DTC B1252:11 or DTC B1252:13 present?

Yes	GO to U6 .
No	Isolate the <u>USB</u> cable. INSTALL a new <u>USB</u> cable. REFER to Universal Serial Bus (USB) Cable and Port. Retest the system for normal operation. If the concern is still present, GO to U6 .

U6 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test V: The RCA Jacks Are Inoperative Or Have Poor Sound Quality

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Media Hub and Composite Audio-Video Input Mode in Information and Entertainment System.

The data transmitted through the RCA cables to the media hub is sent to the APIM through dedicated audio and video circuits.

Possible Sources

- Device conflict
- Wiring, terminals, or connectors
- Media hub
- APIM

PINPOINT TEST V : THE RCA JACKS ARE INOPERATIVE OR HAVE POOR SOUND QUALITY

V1 RESET THE APIM

- Remove any devices connected to the media hub.

- Carry out an **APIM** power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Using the Multi-Media Interface Tester 105-00120 and RCA Adapter 105-00121, attempt to play audio content using the RCA jacks.

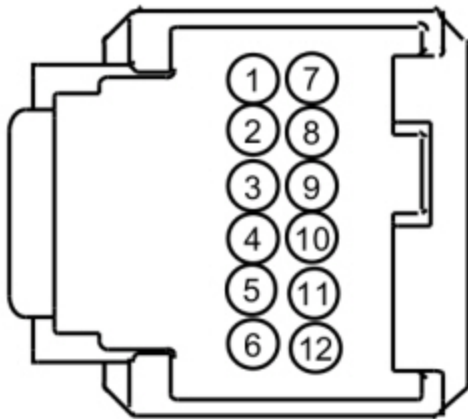
Is the audio output OK for the RCA jacks?

Yes	The system is operating correctly at this time. The concern was corrected by performing the APIM power reset.
No	GO to V2 .

V2 CHECK THE RCA CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Media Hub [C3618](#).
- Disconnect: **APIM** [C2383](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2362 Pin 2	DME88	—	Ground
C2362 Pin 3	VME88 (BU/WH)	—	Ground
C2362 Pin 4	DME45	—	Ground
C2362 Pin 5	VME45 (BU)	—	Ground
C2362 Pin 6	RME45 (YE/GN)	—	Ground
C2362 Pin 9	RME88 (GN/BU)	—	Ground
C2362 Pin 11	VME46 (BU/GN)	—	Ground
C2362 Pin 12	RME46 (WH/GN)	—	Ground



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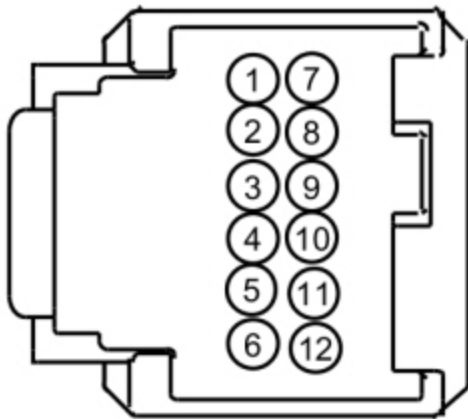
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to V3 .

V3 CHECK THE RCA CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2362 Pin 2	DME88	—	Ground
C2362 Pin 3	VME88 (BU/WH)	—	Ground
C2362 Pin 4	DME45	—	Ground
C2362 Pin 5	VME45 (BU)	—	Ground
C2362 Pin 6	RME45 (YE/GN)	—	Ground
C2362 Pin 9	RME88 (GN/BU)	—	Ground
C2362 Pin 11	VME46 (BU/GN)	—	Ground
C2362 Pin 12	RME46 (WH/GN)	—	Ground



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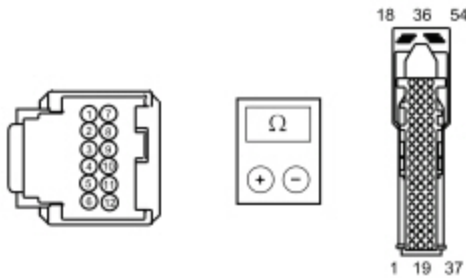
Are the resistances greater than 10,000 ohms?

Yes	REPAIR the circuit in question.
No	GO to V4 .

V4 CHECK THE RCA CIRCUITS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2362 Pin 2	DME88	C2383 Pin 36	DME88
C2362 Pin 3	VME88 (BU/WH)	C2383 Pin 34	VME88 (BU/WH)
C2362 Pin 4	DME45	C2383 Pin 9	DME45
C2362 Pin 5	VME45 (BU)	C2383 Pin 10	VME45 (BU)
C2362 Pin 6	RME45 (YE/GN)	C2383 Pin 11	RME45 (YE/GN)
C2362 Pin 9	RME88 (GN/BU)	C2383 Pin 35	RME88 (GN/BU)
C2362 Pin 11	VME46 (BU/GN)	C2383 Pin 28	VME46 (BU/GN)
C2362 Pin 12	RME46 (WH/GN)	C2383 Pin 29	RME46 (WH/GN)



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Are the resistances less than 3 ohms?

Yes	INSTALL a new media hub. REFER to Media Hub. Retest the system for normal operation. If the concern is still present, GO to V5 .
No	REPAIR the circuit in question.

V5 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM.
------------	---

	Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test W: Unable To Pair Bluetooth Device

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to Bluetooth Mode in Information and Entertainment System.

Possible Sources

- Incompatible Bluetooth device
- Customer error
- Customer's Bluetooth device
- APIM

PINPOINT TEST W : UNABLE TO PAIR BLUETOOTH DEVICE

W1 CHECK THE BLUETOOTH CONNECTION	
NOTE: <i>Performing a Master Reset returns all preference settings to the factory defaults, erases all the phone book and call histories, and deletes any devices paired with the SYNC® system.</i>	
• Connect: Multi-Media Interface Tester 105-00120. • Using the Multi-Media Interface Tester 105-00120, connect to the SYNC® system using Bluetooth. Follow the tool instructions. • Using a diagnostic scan tool, view the APIM PIDs. • Monitor the Bluetooth device paired PID (BT_PAIR) and the Bluetooth device connected PID (BT_CONN).	
Do the PIDs both read "Yes"?	
Yes	The SYNC® system is operating correctly at this time. CARRY OUT a SYNC® system Master Reset. REFER to the Owner's Literature. REVIEW the pairing process with the customer. If the customer's device still does not pair, the fault is with the customer's device.
No	GO to W2 .
W2 CHECK FOR CORRECT APIM OPERATION	

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)

- Damaged or bent pins (install new terminals or pins)
- Pushed-out pins (install new pins as necessary)

- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test X: The RH Steering Wheel Controls Are Inoperative Or Do Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell 130 , Audio System/Navigation for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Steering Wheel Switch Functions and Steering Wheel Switches in Information and Entertainment System.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1380:09	Steering Wheel Right Switch Pack: Component Failure	Set by the SCCM when the RH steering wheel switch is stuck.
B1380:11	Steering Wheel Right Switch Pack: Circuit Short To Ground	Set by the SCCM when a short to ground is detected in the RH steering wheel switch circuits.
B1380:17	Steering Wheel Right Switch Pack: Circuit Voltage Above Threshold	Set by the SCCM when a short to voltage is detected in the RH steering wheel switch circuits.

Possible Sources

- Wiring, terminals, or connectors
- Clockspring
- Steering wheel switch
- SCCM

PINPOINT TEST X : THE RH STEERING WHEEL CONTROLS ARE INOPERATIVE OR DO NOT OPERATE CORRECTLY

X1 CHECK THE RH STEERING WHEEL SWITCH PIDS

- Ignition ON.
- Monitor the steering wheel controls PIDs while pressing each **RH** steering wheel switch.
- Using a diagnostic scan tool, view the **SCCM** PIDs.
 - SW_VOLDOWN
 - SW_VOLUP
 - SW_SEEKMINUS
 - SW_SEEKPLUS
 - SW_MEDIA
 - SW_VOICE
 - SW_PHONE

Do the PID values agree with the switch positions?

Yes	GO to X10 .
No	If one or more, but not all PID values are incorrect or always read one particular switch position, INSTALL a new RH steering wheel switch. REFER to Section 211-05. If all PID values are incorrect or always read one particular switch position, GO to X2 .

X2 CHECK FOR IPC DTC U0212:00

- Ignition ON.
- Using a diagnostic scan tool, clear the DTCs from the **IPC**.
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, perform the **IPC** self-test.

Is DTC U0212:00 set in the IPC ?

Yes	REFER to Section 413-01.
No	GO to X3 .

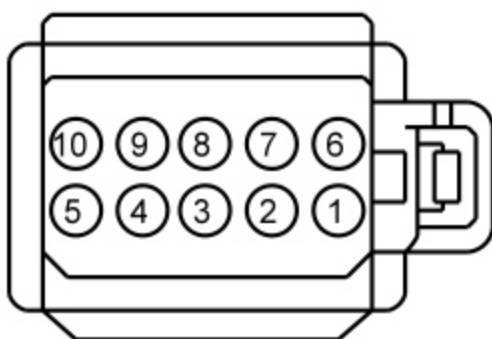
X3 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPrING AND STEERING WHEEL CONTROLS FOR A SHORT TO GROUND

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.
- Disconnect: **RH** Steering Wheel Switch [C2999](#).
- Disconnect: Clockspring [C218B](#).

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	—	Ground
C2999 Pin 4	RME14	—	Ground
C2999 Pin 6	VME54	—	Ground
C2999 Pin 7	RME54	—	Ground



-

Are the resistances greater than 10,000 ohms?

Yes	GO to X4 .
No	INSTALL a new steering wheel. REFER to Section 211-04.

X4 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C218B Pin 14	VME14
C2999 Pin 4	RME14	C218B Pin 6	RME14
C2999 Pin 6	VME54	C218B Pin 16	VME54
C2999 Pin 7	RME54	C218B Pin 8	RME54



- N0148938

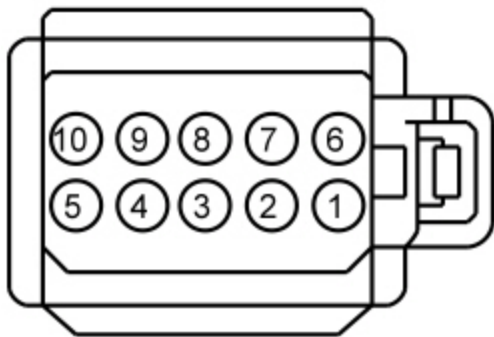
Are the resistances less than 3 ohms?

Yes	GO to X5 .
No	INSTALL a new steering wheel. REFER to Section 211-04.

X5 CHECK THE CIRCUITRY BETWEEN THE CLOCKSPRING AND STEERING WHEEL CONTROLS FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C2999 Pin 4	RME14
C2999 Pin 6	VME54	C2999 Pin 7	RME54



-

Are the resistances greater than 10,000 ohms?

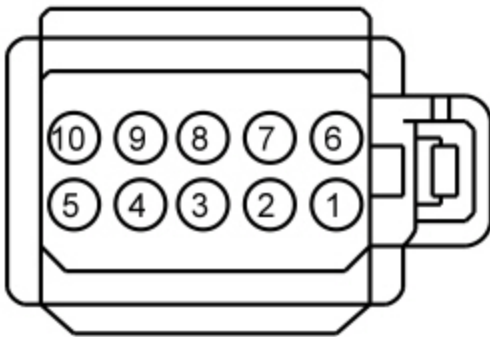
Yes	GO to X6 .
No	INSTALL a new steering wheel. REFER to Section 211-04.

X6 CHECK THE CLOCKSPRING FOR A SHORT TO GROUND

- Connect: Clockspring [C218B](#).

- Disconnect: Clockspring [C218C](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	—	Ground
C2999 Pin 4	RME14	—	Ground
C2999 Pin 6	VME54	—	Ground
C2999 Pin 7	RME54	—	Ground



•

Are the resistances greater than 10,000 ohms?

Yes	GO to X7 .
No	INSTALL a new clockspring. REFER to Section 501-20B.

X7 CHECK THE CLOCKSPRING FOR AN OPEN

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C218C Pin 6	—
C2999 Pin 4	RME14	C218C Pin 5	—
C2999 Pin 6	VME54	C218C Pin 10	—
C2999 Pin 7	RME54	C218C Pin 9	—



- N0148946

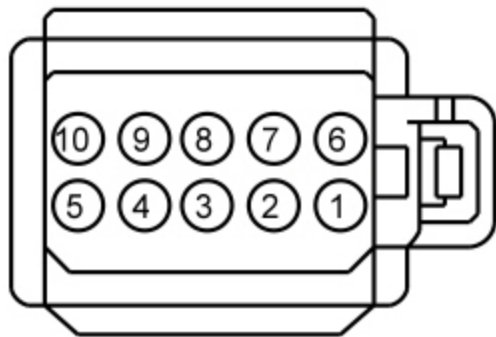
Are the resistances less than 3 ohms?

Yes	GO to X8 .
No	INSTALL a new clockspring. REFER to Section 501-20B.

X8 CHECK THE CLOCKSPRING FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2999 Pin 2	VME14	C2999 Pin 4	RME14
C2999 Pin 6	VME54	C2999 Pin 7	RME54



-

Are the resistances greater than 10,000 ohms?

Yes	GO to X9 .
No	INSTALL a new clockspring. REFER to Section 501-20B.

X9 CHECK FOR VOLTAGE AT THE SCCM

- Ignition ON.

- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2414D Pin 6	—	C2414D Pin 5	—
C2414D Pin 10	—	C2414D Pin 9	—



-

Is approximately 5 volts present?

Yes	INSTALL a new RH steering wheel switch. REFER to Section 211-05.
No	GO to X10 .

X10 CHECK FOR CORRECT SCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **SCCM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **SCCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new SCCM . REFER to Section 211-05.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test Y: The Audio System Does Not Operate Correctly From The FCIM

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Refer to Accessory Delay Feature, Audio Extended Play, and FCIM in Information and Entertainment System.

Possible Sources

- Network communication concern
- FCIM

PINPOINT TEST Y : THE AUDIO SYSTEM DOES NOT OPERATE CORRECTLY FROM THE FCIM

Y1 CHECK THE FDIM OPERATION

- Verify the FDIM screen operation.

Does the FDIM operate correctly?

Yes	GO to Y2 .
No	GO to Pinpoint Test XA .

Y2 VERIFY THAT THE ACM , APIM , AND FCIM PASS THE NETWORK TEST

- Using a diagnostic scan tool, perform the network test.

Did the ACM , APIM , and FCIM pass the network test?

Yes	GO to Y3 .
No	If the ACM failed the network test, REFER to Section 418-00 The ACM Does Not Respond To The Scan Tool. If the APIM failed the network test, REFER to Section 418-00 The APIM Does Not Respond To The Scan Tool. If the FCIM failed the network test, REFER to Section 418-00 The FCIM Does Not Respond To The Scan Tool.

Y3 RESET THE AUDIO SYSTEM MODULES AND RECHECK THE FCIM OPERATION

- Disconnect the battery for 5 minutes, then reconnect it.
- Turn the infotainment system on.
- Test the FCIM for normal operation.

Does the FCIM operate correctly?

Yes	The FCIM is operating correctly at this time. The concern was corrected by resetting the audio system modules.
No	GO to Y4 .

Y4 RETRIEVE THE ACM , APIM , AND FCIM DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - FCIM

Are any DTCS present in the ACM , APIM , or FCIM ?

Yes	For ACM DTCS, REFER to the ACM DTC Chart in Information and Entertainment System. For APIM DTCS, REFER to the APIM DTC Chart in Information and Entertainment System. For FCIM DTCS, REFER to the FCIM DTC Chart in Information and Entertainment System.
No	GO to Y5 .

Y5 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **FCIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test XA: The **EDIM Is Inoperative Or Does Not Operate Correctly**

Diagnostic Overview

Normal Operation and Fault Conditions

Information and Entertainment System.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B108E:01	Display: General Electrical Failure	Sets in the APIM when an error is detected from the FDIM .
U0162:00	Lost Communication With Navigation Display Module: No Sub Type Information	Set by the APIM when an internal connection concern or internal LCD panel concern is detected in the FDIM .

Possible Sources

- Communication network concern
- Uncalibrated touchscreen
- Configuration concern
- Object pressed against the FDIM
- FDIM
- APIM

PINPOINT TEST XA : THE FDIM IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY

XA1 REBOOT THE APIM AND RECHECK THE SYSTEM OPERATION

- Ignition ON.
- Simultaneously press and hold the SEEK UP and POWER buttons on the FCIM for 10 seconds. As the APIM reboot begins, the touchscreen turns black before the startup sequence begins.
- After the reboot is complete, operate the system to verify if the concern is still present.

Is the concern still present?

Yes	If the FDIM is blank or black, GO to XA4 . If the image shown on the FDIM is incorrect, the image color is incorrect, or the image is distorted, flickering, or unclear, GO to XA12 . If the FDIM is frozen, GO to XA2 . If the FDIM is unresponsive to touch, GO to XA2 . If the FDIM is slow to respond, GO to XA10 . For all other concerns, GO to XA4 .
No	The system is operating correctly at this time. VERIFY the customer specific concern by duplicating the steps that were taken when the original concern was generated.

XA2 CHECK FOR THE AUDIO SYSTEM VOLUME POP-UP ON THE FDIM WHILE ADJUSTING THE AUDIO SYSTEM VOLUME

- Operate the audio system in AM/EM mode.
- Observe the FDIM while adjusting the audio system volume up and down.

Does the FDIM display the audio system volume pop-up when the volume is adjusted up and down?

Yes	If the volume pop-up is displayed and the audio system output volume responds, GO to XA10 . If the volume pop-up is displayed but no sound is heard, REFER to the Symptom Charts in this section to diagnose the inoperative audio system feature(s).
No	GO to XA3 .

XA3 ENSURE THE TOUCHSCREEN IS CALIBRATED

- Calibrate the touchscreen. Refer to Touchscreen Calibration.
- Verify the touchscreen operation.

Does the FDIM touchscreen respond correctly?

Yes	The system is operating correctly at this time. The concern was a result of the touchscreen not being calibrated.
No	GO to XA13 .

XA4 VERIFY THAT THE MODULES PASS THE NETWORK TEST

- **NOTE:** A concern with the CAN may cause the FDIM touchscreen to be blank.

Using the diagnostic scan tool, perform the network test.

Do all modules pass the network test?

Yes	GO to XA5 .
No	Refer to Section 418-00.

XA5 VERIFY THE APIM IGNITION STATE PID

- Using a diagnostic scan tool, view the APIM PIDs .
- Monitor the Ignition State PID (KEYPOS).
- Ignition OFF.
- Ignition ON.

Does the PID value match the actual switch positions?

Yes	GO to XA6 .
No	DIAGNOSE the ignition switch input. Refer to Section 211-05.

XA6 CHECK FOR DTC U0162:00 SET IN THE APIM

- Using a diagnostic scan tool, perform the APIM self-test.

Is DTC U0162:00 set in the APIM ?

Yes	GO to XA7 .
No	GO to XA8 .

XA7 CLEAR THE APIM DIAGNOSTIC TROUBLE CODES (DTCS) AND RECHECK FOR DTC U0162:00

- Using a diagnostic scan tool, clear the **APIM** Diagnostic Trouble Codes (DTCs) and repeat the **APIM** self-test.

Is DTC U0162:00 still set in the APIM ?

Yes	GO to XA14 .
No	GO to XA8 .

XA8 CHECK FOR DTC B108E:01 SET IN THE APIM

- Using a diagnostic scan tool, perform the **APIM** self-test.

Is DTC B108E:01 set in the APIM ?

Yes	GO to XA9 .
No	GO to XA13 .

XA9 CLEAR THE APIM DIAGNOSTIC TROUBLE CODES (DTCS) AND RECHECK FOR DTC B108E:01

- Using a diagnostic scan tool, clear the **APIM** DTCs and repeat the **APIM** self-test.

Is DTC B108E:01 set in the APIM ?

Yes	GO to XA13 .
No	GO to XA14 .

XA10 INSPECT THE DISPLAY FOR CRACKS

- Ignition OFF.
- Remove the **FCIM** . REFER to Front Controls Interface Module (FCIM).
- NOTE:** *Minor cracks on the display can cause touch concerns.*

Carefully inspect the **FDIM** screen for cracks. The RGB Pixel Test and Display Test Pattern functions available through **APIM** Bezel Diagnostics can be used to assist in the inspection. REFER to Bezel Diagnostics.

Are any cracks present on the FDIM display?

Yes	CHECK OASIS for any applicable service articles: TSB , General Service Bulletin (GSB), SSM or ESA . If a service article exists for this concern, DISCONTINUE this test and FOLLOW the
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service article instructions. If no service articles address this concern, INSTALL a new FDIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.

No GO to [XA11](#).

XA11 CONFIRM THE FDIM TOUCH RESPONSE WITH THE FCIM REMOVED

NOTE: The FDIM can only sense one touch command at a time. If any objects are pressed against the touchscreen (including wiring or components behind the trim panel surrounding the FDIM), the FDIM can appear to be inoperative.

- Connect the FCIM electrical connectors, but do not install it.
- Ignition ON.
- Access the Bezel Diagnostics. Refer to Bezel Diagnostics.
- **NOTE:** The touchscreen is a resistive touch panel. This means that a finer touch point result in a more responsive touch panel. For example, the tip of a finger nail result in better response than the broad surface of the thumb.

After initiating the Touchscreen Activation Test, touch several points around the display and observe the display touch response. The most recent touch point is indicated by a blue box and prior touch points are indicated by white boxes.

Does the FDIM respond correctly to each touch point?

Yes If the FDIM responds correctly, the system is operating correctly at this time. REINSTALL the FCIM and ensure the FCIM opening is properly aligned with the FDIM and that no objects are in contact with the FDIM touchscreen. VERIFY the customer specific concern by duplicating the steps that were taken when the original concern was generated.

No GO to [XA14](#).

XA12 INSPECT THE FDIM IMAGE CLARITY

- Access Bezel Diagnostics. Refer to Bezel Diagnostics.
- While using the RGB Pixel Test and Display Test Pattern functions, inspect the display image quality.

Is the image quality concern present when using the RGB Pixel Test and Display Test Pattern functions?

Yes GO to [XA13](#).

No The system is operating correctly at this time. VERIFY the customer specific concern by duplicating the steps that were taken when the original concern was generated.

XA13 CHECK THE FDIM TO APIM CONNECTION

- Ignition OFF.
- Remove the FDIM and separate the FDIM and APIM . REFER to Front Display Interface Module (FDIM).
- **NOTICE:** Be careful not to damage the APIM and FDIM when inspecting and cleaning the connection.

Inspect and clean the connection between the **FDIM** and **APIM** using compressed air.

- Install the **FDIM** and **APIM** . Make sure the **APIM** is properly fastened to the **FDIM** and all connectors seat and latch correctly.
- Ignition ON.
- TEST the system for normal operation.

Is the concern still present?

Yes	CHECK OASIS for any applicable service articles: TSB , General Service Bulletin (GSB), SSM or ESA . If a service article exists for this concern, DISCONTINUE this test and FOLLOW the service article instructions. If no service articles address this concern, INSTALL a new FDIM . Refer to the appropriate Removal and Installation procedure in Section 415-00. If the concern is still present after FDIM replacement, GO to XA14 .
No	The system is operating correctly at this time. The DTC may have been set due to an intermittent fault in the connection between the FDIM and APIM .

XA14 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable service articles: TSB , General Service Bulletin (GSB), SSM or ESA . If a service article exists for this concern, DISCONTINUE this test and FOLLOW the service article instructions. If no service articles address this concern, INSTALL a new APIM . Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AA: Navigation Application Is Inoperative Or Unresponsive (If Equipped)

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Normal Operation and Fault Conditions

Removal of the navigation secure digital (SD) card results in the loss of navigation while the card is removed. This is normal operation.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
B1215:00	External Memory Card: No Sub Type Information	Set by the APIM when the navigation secure digital (SD) card is missing.

Possible Sources

- Media hub
- Navigation map secure digital (SD) card not present, not fully seated, or damaged
- Communications network concern
- GPSM
- APIM

PINPOINT TEST AA : NAVIGATION APPLICATION IS INOPERATIVE OR UNRESPONSIVE (IF EQUIPPED)

AA1 DETERMINE IF THE VEHICLE IS EQUIPPED WITH A FACTORY-INSTALLED NAVIGATION SYSTEM	
• Ensure that the correct VIN was entered on the technician service publication website. • Hover the mouse cursor over or click on the QASIS tab toward the top LH corner of the screen. • Select "HVBoM (Historical Vehicle Bill of Material)" • Locate the Limit to Base: text entry box toward the center of the screen. • Enter the navigation data secure digital (SD) card base part number "19H449" (do not include quotation marks). • Press Enter on the keyboard or click on "View BoM (Build of Material)". • If the vehicle is equipped with a factory-installed navigation system, a row will appear indicating "CARD - MEMORY" in the DESCRIPTION column. • If the vehicle is not equipped with a factory-installed navigation system, a row will not appear and the DESCRIPTION column will be blank.	
Does the technician service publication website indicate the vehicle is equipped with a factory-installed navigation system?	
Yes	GO to AA2 .
No	Ford Motor Company does not support the addition of a navigation system to vehicles not equipped with a factory-installed navigation system.

AA2 VERIFY THE APIM PASSES THE NETWORK TEST

- Using the diagnostic scan tool, perform the network test.

Did the APIM pass the network test?

Yes	GO to AA3 .
No	REFER to Section 418-00. See The APIM Does Not Respond to the Diagnostic Scan Tool.

AA3 CHECK THE USB PORTS FOR POWER OUTPUT

- Connect Multi-Media Interface Tester 105-00120.

Does the Multi-Media Interface Tester 105-00120 power on from both USB ports?

Yes	GO to AA4 .
No	If the Multi-Media Interface Tester 105-00120 powers on from only one USB port, INSTALL a new media hub. REFER to Media Hub. If the Multi-Media Interface Tester 105-00120 does not power on from either USB port, GO to Pinpoint Test S .

AA4 RESET THE APIM AND VERIFY OPERATION OF THE USB CABLE

- Remove any devices connected to the media hub.
- Carry out an APIM power reset by disconnecting the battery for 1 minute, then reconnecting it.
- Wait at least 2 minutes for the APIM to re-initialize.
- Connect Multi-Media Interface Tester 105-00120.
- Using the Multi-Media Interface Tester 105-00120, attempt to play an audio file using each USB port.

Is the audio output OK for each USB port?

Yes	GO to AA5 .
No	If the audio output is not OK from one USB port, INSTALL a new media hub. REFER to Media Hub. If the audio output is not OK from from both USB ports, GO to Pinpoint Test T .

AA5 CHECK FOR A NAVIGATION MAP DATA SECURE DIGITAL (SD) CARD FAULT MESSAGE

- Ignition ON.
- Observe the FDIM display.

Does the FDIM display "SD CARD FAULT?"

Yes	GO to AA6 .
No	GO to AA9 .

AA6 VERIFY THE NAVIGATION MAP DATA SECURE DIGITAL (SD) CARD IS PRESENT

- Check for the presence of a navigation map data secure digital (SD) card in the media hub secure digital (SD) slot.

Is the navigation map data secure digital (SD) card present and fully seated?

Yes	GO to AA7 .
No	If a non-Ford supplied secure digital (SD) card is present, EXPLAIN to the customer that only a Ford supplied map data card will work with the navigation system. INSTALL the navigation map data secure digital (SD) card. TEST the system for normal operation.

AA7 INSPECT THE NAVIGATION MAP DATA SECURE DIGITAL (SD) CARD

NOTE: The media hub secure digital (SD) card slot is spring loaded. To remove the secure digital (SD) card, push the card inward and release it. Do not attempt to forcefully pull the secure digital (SD) card from the secure digital (SD) card slot. This may cause damage to the secure digital (SD) card pins.

- Remove the navigation map data secure digital (SD) card.
- Inspect the navigation map data secure digital (SD) card case and pins for damage.

Is any damage present?

Yes	INSTALL a new navigation map data secure digital (SD) card.
No	If a known good map data secure digital (SD) card is available, GO to AA8 . If a known good map data secure digital (SD) card is NOT available, GO to AA9 .

AA8 VERIFY NAVIGATION SYSTEM OPERATION WITH KNOWN GOOD MAP DATA SECURE DIGITAL (SD) CARD

- Insert a known good navigation map data secure digital (SD) card with the correct software version level.
- Confirm the navigation system operates correctly.

Does the navigation system operate correctly with the known good navigation map data secure digital (SD) card?

Yes	INSTALL a new navigation map data secure digital (SD) card.
No	GO to AA9 .

AA9 CHECK FOR DIAGNOSTIC SCAN TOOL COMMUNICATION WITH THE GPSM

- Ignition ON.
- Using a diagnostic scan tool, perform the network test.

Does the diagnostic scan tool communicate with the GPSM ?

Yes	GO to AA10 .
No	REFER to Section 418-00 The <u>GPSM</u> Does Not Respond To The Scan Tool.

AA10 RETRIEVE THE DTCS FROM THE APIM AND GPSM

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - APIM
 - GPSM

Are any DTCS present in the APIM or GPSM ?

Yes	For <u>APIM</u> DTCS, REFER to the <u>APIM</u> DTC Chart in Information and Entertainment System For <u>GPSM</u> DTCS, REFER to the <u>GPSM</u> DTC Chart in Information and Entertainment System
No	GO to AA11 .

AA11 REPROGRAM THE APIM

- Perform the APIM Standard Programming. REFER to Accessory Protocol Interface Module (APIM) Standard Programming
- Override the Service Pack installation and perform a Full Flash/Force CIP .
- Once the APIM programming is complete and the system has re-initialized, confirm the navigation system operates correctly.

Does the navigation system operate correctly?

Yes	The navigation system is operating correctly at this time. The concern was resolved by reprogramming the <u>APIM</u> .
No	GO to AA12 .

AA12 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect and inspect the GPSM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the GPSM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK <u>QASIS</u> for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new <u>GPSM</u> .
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REFER to Global Positioning System Module (GPSM).

If the concern is still present, GO to [AA13](#).

No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.
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AA13 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AB: The Compass Is Inoperative

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Possible Sources

- Network communication concern
- **APIM**
- **GPSM**

PINPOINT TEST AB : THE COMPASS IS INOPERATIVE

AB1 VERIFY THAT THE APIM AND GPSM PASS THE NETWORK TEST

- Using a diagnostic scan tool, perform the network test.

Did the **APIM** and **GPSM** pass the network test?

Yes	GO to AB2 .
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No	If the APIM failed the network test, REFER to Section 418-00 The APIM Does Not Respond To The Scan Tool. If the GPSM failed the network test, REFER to Section 418-00 The GPSM Does Not Respond To The Scan Tool.
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AB2 RETRIEVE THE APIM AND GPSM DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - APIM**
 - GPSM**

Are any DTCs present in the **APIM** or **GPSM** ?

Yes	For APIM DTCs, REFER to the APIM DTC Chart in Information and Entertainment System For GPSM DTCs, REFER to the GPSM DTC Chart in Information and Entertainment System
No	GO to AB3 .

AB3 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect and inspect the **GPSM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **GPSM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AC: DTC U0100:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0100:00	Lost Communication with ECM/PCM "A": No Sub Type Information	Set by the <u>GPSM</u> when network messages are missing from the <u>PCM</u> for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AC : DTC U0100:00

AC1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AC2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AC2 RECHECK FOR DTC U0100:00

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

Is DTC U0100:00 set in the GPSM ?

Yes	GO to AC3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AC3 CHECK THE PCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the PCM self-test.

Are any low or high voltage DTCs recorded in the PCM ?

Yes	REFER to <u>PCM</u> DTC Chart in Section 303-14.
No	GO to AC4 .

AC4 CHECK THE GPSM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **GPSM** self-test.

Are any low or high voltage DTCS recorded in the **GPSM ?**

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AC5 .

AC5 CHECK FOR CORRECT PCM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **PCM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **PCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new PCM . REFER to Section 303-14.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AD: DTC U0140:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0140:00	Lost Communication With Body Control Module: No Sub Type Information	Set by the APIM and GPSM when network messages are missing from the BCM for more than 5 seconds with the ignition in RUN.

AD1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AD2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AD2 RECHECK FOR DTC U0140:00

- Ignition ON.
- Using a diagnostic scan tool, clear all **CMDTCs** .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all **CMDTCs** .

Is DTC U0140:00 set in the **APIM or **GPSM** ?**

Yes	GO to AD3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AD3 CHECK THE BCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **BCM** self-test.

Are any low or high voltage DTCS recorded in the **BCM ?**

Yes	REFER to BCM DTC Chart in Section 419-10.
No	GO to AD4 .

AD4 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - **APIM**
 - **GPSM**

Are low or high voltage DTCS recorded in the **APIM or **GPSM** ?**

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AD5 .

AD5 CHECK FOR DTC U0140:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0140:00 set in more than one audio system module?

Yes	GO to AD6 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AD6 CHECK FOR CORRECT BCM OPERATION

- Ignition OFF.
- Disconnect and inspect all the BCM connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the BCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. .
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AE: DTC U0151:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0151:00	Lost Communication With Restraints Control Module: No Sub Type Information	Set by the APIM when the airbag deployment notification message from the RCM is missing for more than 5 seconds with the ignition in the RUN position.

PINPOINT TEST AE : DTC U0151:00

AE1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AE2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AE2 RECHECK FOR DTC U0151:00

- Ignition ON.
- Using a diagnostic scan tool, clear all **CMDTCs** .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all **CMDTCs** .

Is DTC U0151:00 set in the **APIM** ?

Yes	GO to AE3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AE3 CHECK THE RCM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **RCM** self-test.

Are any low or high voltage DTCs recorded in the **RCM** ?

Yes	For DTC U3003:16, REFER to RCM DTC Chart in Section 501-20B. For DTC U3003:16, REFER to RCM DTC Chart in Section 501-20B.
No	GO to AE4 .

AE4 CHECK THE APIM FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the **APIM** self-test.

Are any low or high voltage DTCS recorded in the **APIM** ?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AE5 .

AE5 CHECK FOR CORRECT RCM OPERATION

- Ignition OFF.
- Disconnect and inspect all the **RCM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **RCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new RCM . REFER to Section 501-20B.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AF: DTC U0155:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type	Set by the ACM , APIM , audio DSP module, FCIM , and GPSM when network messages are missing from the IPC for

DTC	Description	Fault Trigger Conditions
	Information	more than 5 seconds with the ignition in RUN.

PINPOINT TEST AF : DTC U0155:00

AF1 VERIFY THE CUSTOMER'S CONCERN	
- Ignition ON. - Verify if there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to AF2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AF2 RECHECK FOR DTC U0155:00	
- Ignition ON. - Using a diagnostic scan tool, clear all CMDTCs . - Ignition OFF. - Ignition ON. - Wait at least 10 seconds. - Using a diagnostic scan tool, retrieve all CMDTCs .	
Is DTC U0155:00 set in the ACM , APIM , audio DSP module, ECIM , or GPSM ?	
Yes	GO to AF3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AF3 CHECK THE IPC FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS	
Using a diagnostic scan tool, perform the IPC self-test.	
Are any low or high voltage DTCs recorded in the IPC ?	
Yes	REFER to IPC DTC Chart in Section 413-01.
No	GO to AF4 .

AF4 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS	
- Using a diagnostic scan tool, perform the self-test for the following modules: ACM	

- APIM
- Audio DSP module
- FCIM
- GPSM

Are low or high voltage DTCs recorded in the ACM , APIM , audio DSP module, FCIM , or GPSM ?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AF5 .

AF5 CHECK FOR DTC U0155:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0155:00 set in more than one audio system module?

Yes	GO to AF6 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AF6 CHECK FOR CORRECT IPC OPERATION

- Ignition OFF.
- Disconnect and inspect the IPC connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the IPC connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new IPC . REFER to Section 413-01.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AG: DTC U016A:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U016A:00	Lost Communication With Global Positioning System Module: No Sub Type Information	Set by the <u>APIM</u> when network messages are missing from the <u>GPSM</u> for more than 5 seconds with the ignition in the RUN position.

PINPOINT TEST AG : DTC U016A:00

AG1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AG2 .
No	The system is operating correctly at this time. The DTC may have been set due to an intermittent fault in the network wiring. CHECK the <u>MS-CAN</u> wiring between the <u>APIM</u> and the <u>GPSM</u> for a fault.

AG2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Carry out the network test using the scan tool.

Does the GPSM pass the network test?

Yes	GO to AG3 .
No	REFER to Section 418-00.

AG3 CHECK FOR ANY RECENT PMI CHANGES

NOTE: *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Check the vehicle service history for recent service actions related to the GPSM .

Have there been any recent service actions with either module?

Yes	CARRY out a manual PMI on the GPSM using As-Built data. REFER to the diagnostic scan tool instructions.
No	GO to AG4 .

AG4 RETRIEVE THE RECORDED DTCS FROM THE GPSM

- Clear the GPSM DTCs.
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Repeat the GPSM self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	REFER to the GPSM DTC chart in this section.
No	GO to AG5 .

AG5 RETRIEVE THE RECORDED DTCS FROM THE APIM

- Repeat the APIM self-test.

Is DTC U3003:16 or DTC U3003:17 recorded?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN
No	GO to AG6 .

AG6 RECHECK THE APIM DTCS

- Perform the APIM self-test.

Is DTC U016A:00 still present?

Yes	GO to AG7 .
No	The system is operating correctly at this time. The DTC may have been set due to an intermittent fault in the network wiring.

AG7 CHECK FOR DTC U016A:00 SET IN OTHER MODULES

- Clear all DTCs.
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.

- Repeat the **APIM** self-test.

Is DTC U016A:00 set in any other module?

Yes	GO to AG8 .
No	GO to AG9

AG8 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect the **GPSM** connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the **GPSM** connector and make sure it seats correctly.

Is the concern still present?

Yes	INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

AG9 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect the **APIM** connector.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the **APIM** connector and make sure it seats correctly.
- Wait 2 minutes for the **APIM** to re-initialize.

Is the concern still present?

Yes	INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test AH: DTC U0184:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0184:00	Lost Communication With Radio: No Sub Type Information	Set by the APIM and audio DSP module when network messages are missing from the ACM for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AH : DTC U0184:00

AH1 VERIFY THE CUSTOMER'S CONCERN	
- Ignition ON. - Verify if there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to AH2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
AH2 RECHECK FOR DTC U0184:00	
- Ignition ON. - Using a diagnostic scan tool, clear all CMDTCs . - Ignition OFF. - Ignition ON. - Wait at least 10 seconds. - Using a diagnostic scan tool, retrieve all CMDTCs .	
Is DTC U0184:00 set in the APIM and audio DSP module?	
Yes	GO to AH3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AH3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - Audio DSP module

Are low or high voltage DTCS recorded in the ~~ACM~~ , ~~APIM~~ , or audio ~~DSP~~ module?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AH4 .

AH4 CHECK FOR DTC U0184:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0184:00 set in more than one audio system module?

Yes	GO to AH5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AH5 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AI: DTC U0238:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0238:00	Lost Communication With Digital Audio Control Module "D" : No Sub Type Information	Set by the ACM and APIM when network messages are missing from the audio DSP module for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AI : DTC U0238:00

AI1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AI2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AI2 RECHECK FOR DTC U0238:00

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

Is DTC U0238:00 set in the ACM and APIM ?

Yes	GO to AI3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AI3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - Audio DSP module

Are low or high voltage DTCs recorded in the ACM , APIM , or audio DSP module?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AI4 .

AI4 CHECK FOR DTC U0238:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0238:00 set in more than one audio system module?

Yes	GO to AI5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AI5 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio DSP module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio DSP module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AJ: DTC U0253:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0253:00	Lost Communication With Accessory Protocol Interface Module: No Sub Type Information	Set by the ACM and audio DSP module when network messages are missing from the APIM for more than 5 seconds with the ignition in RUN.

PINPOINT TEST AJ : DTC U0253:00

AJ1 VERIFY THE CUSTOMER'S CONCERN

- Ignition ON.
- Verify if there is an observable symptom present.

Is an observable symptom present?

Yes	GO to AJ2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AJ2 RECHECK FOR DTC U0253:00

- Ignition ON.
- Using a diagnostic scan tool, clear all CMDTCs .
- Ignition OFF.
- Ignition ON.
- Wait at least 10 seconds.
- Using a diagnostic scan tool, retrieve all CMDTCs .

Is DTC U0253:00 set in the ACM and audio DSP module?

Yes	GO to AJ3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AJ3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - Audio DSP module

Are low or high voltage DTCs recorded in the ACM , APIM , or audio DSP module?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AJ4 .

AJ4 CHECK FOR DTC U0253:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0253:00 set in more than one audio system module?

Yes	GO to AJ5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AJ5 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AK: DTC U0256:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0256:00	Lost Communication With Front Controls Interface Module "A": No Sub Type Information	Set by the ACM APIM , and audio DSP module when network messages are missing from the FCIM for more than 5 seconds with the ignition in the RUN position.

PINPOINT TEST AK : DTC U0256:00

AK1 VERIFY THE CUSTOMER'S CONCERN	
- Ignition ON. - Verify if there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to AK2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
AK2 RECHECK FOR DTC U0256:00	
- Ignition ON. - Using a diagnostic scan tool, clear all CMDTCs . - Ignition OFF. - Ignition ON. - Wait at least 10 seconds. - Using a diagnostic scan tool, retrieve all CMDTCs .	
Is DTC U0256:00 set in the ACM , APIM , or audio DSP module?	
Yes	GO to AK3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AK3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - Audio DSP module
 - FCIM

Are low or high voltage DTCs recorded in the ACM , APIM , audio DSP module, or FCIM ?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AK4 .

AK4 CHECK FOR DTC U0256:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0256:00 set in more than one audio system module?

Yes	GO to AK5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AK5 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the FCIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the FCIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AL: DTC U0257:00

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U0257:00	Lost Communication With Front Controls / Display Interface Module: No Sub Type Information	NOTE: The <i>FDIM</i> is a non-networked module attached to the <i>APIM</i>. The <i>APIM</i> acts as the gateway module for messages sent over the <i>I-CAN</i> between the <i>FDIM</i>, <i>ACM</i>, and <i>FCIM</i>. Set by the <i>ACM</i>, audio <i>DSP</i> module, and <i>FCIM</i> when <i>I-CAN</i> network messages are missing from the <i>APIM</i> for more than 5 seconds with the ignition in RUN. This DTC may be set in conjunction with DTC U0253:00.

PINPOINT TEST AL : DTC U0257:00

AL1 VERIFY THE CUSTOMER'S CONCERN	
- Ignition ON. - Verify if there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to AL2 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AL2 RECHECK FOR DTC U0257:00	
- Ignition ON. - Using a diagnostic scan tool, clear all <i>CMDTCs</i>. - Ignition OFF. - Ignition ON. - Wait at least 10 seconds. - Using a diagnostic scan tool, retrieve all <i>CMDTCs</i>.	

Is DTC U0257:00 set in the ACM , audio DSP module, or FCIM ?

Yes	GO to AL3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AL3 CHECK FOR BATTERY VOLTAGE OUT-OF-RANGE DTCS

- Using a diagnostic scan tool, perform the self-test for the following modules:
 - ACM
 - APIM
 - Audio DSP module
 - FCIM

Are low or high voltage DTCS recorded in the ACM , APIM , audio DSP module, or FCIM ?

Yes	For DTC U3003:16, GO to Pinpoint Test AM . For DTC U3003:17, GO to Pinpoint Test AN .
No	GO to AL4 .

AL4 CHECK FOR DTC U0257:00 SET IN MULTIPLE MODULES

- Review the recorded results from the self-test.

Is DTC U0257:00 set in more than one audio system module?

Yes	GO to AL5 .
No	If there is an observable symptom, GO to Symptom Chart — General Audio System to diagnose the observed symptom. If there is no observable symptom, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

AL5 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the APIM to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AM: DTC U3003:16

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Wiring Diagrams Cell [130](#) , Audio System/Navigation for schematic and connector information.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	Set by the ACM , APIM , audio DSP module, FCIM , and GPSM when the supply voltage falls below 10 volts for at least 10 seconds during normal operation or for more than 250 milliseconds during the self-test.

PINPOINT TEST AM : DTC U3003:16

AM1 RECHECK FOR DTC U3003:16	
- Ignition ON. - Using a diagnostic scan tool, clear the DTCs for the module in question. - Wait at least 15 seconds. - Using a diagnostic scan tool, repeat the self-test for the module in question.	
Is DTC U3003:16 still present?	
Yes	GO to AM2 .
No	The system is operating correctly at this time. The DTC may have been set due to a previous low battery voltage condition.
AM2 CHECK FOR CHARGING SYSTEM DTCS IN THE PCM	
Using a diagnostic scan tool, retrieve all CMDTCs .	

Are any voltage-related DTCs set in the PCM ?

Yes	REFER to PCM DTC Chart in Section 303-14.
No	GO to AM3 .

AM3 CHECK THE BATTERY CONDITION AND STATE OF CHARGE

- Check the battery condition and verify that the battery is fully charged. REFER to Battery Condition Test in Section 414-01.

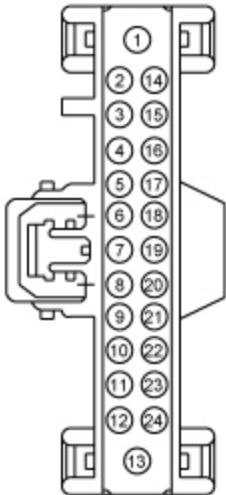
Is the battery OK and fully charged?

Yes	GO to AM4 .
No	REFER to the Symptom Chart in Section 414-00.

AM4 CHECK THE MODULE VOLTAGE SUPPLY

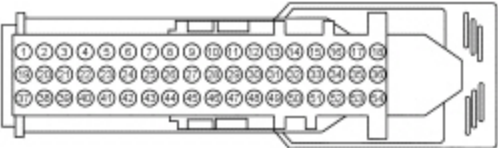
- Ignition OFF.
- Measure and record the voltage at the battery.
- Disconnect the audio system module setting DTC U3003:16.
- Ignition ON.
- Measure the voltage between the module setting DTC U3003:16 and ground.
- For the ~~ACM~~ , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240A Pin 1	SBP29 (GY/RD)	—	Ground



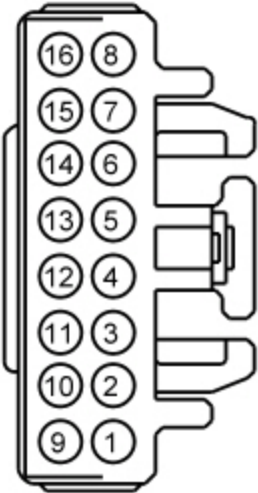
- For the **APIM** , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 1	SBP02 (YE/RD)	—	Ground



-
- For the audio **DSP** module, measure the **voltage** between:

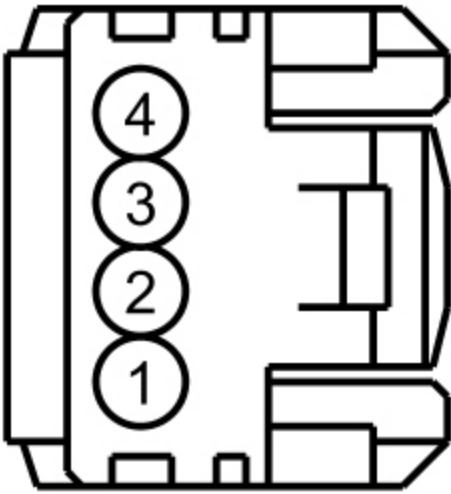
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154C Pin 5	SBP19 (BU/RD)	—	Ground
C3154C Pin 6	SBP19 (BU/RD)	—	Ground



-
- For the **FCIM** , measure the **voltage** between:

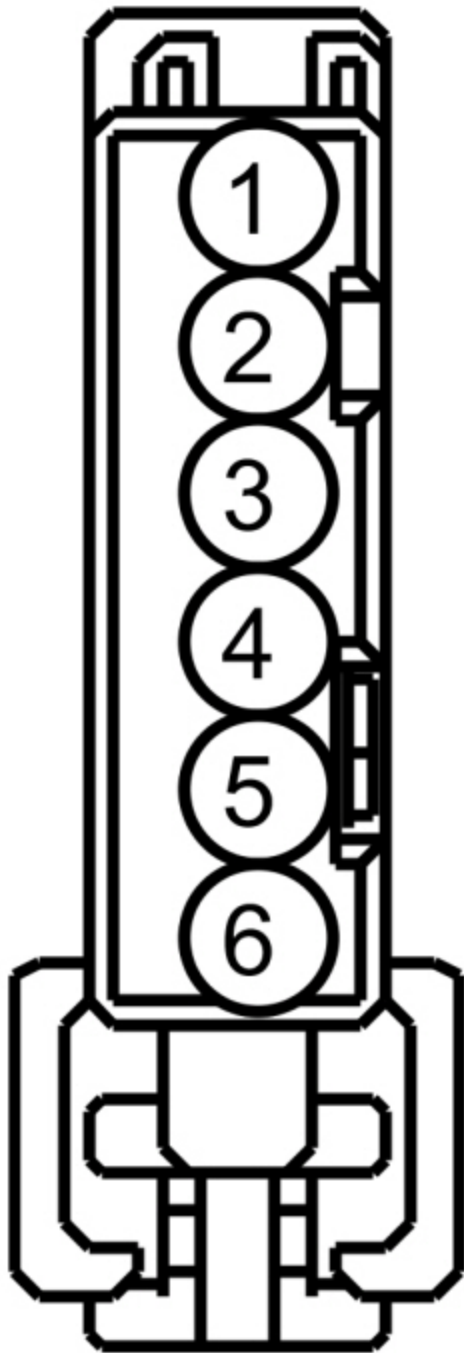
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit

C2402 Pin 1	SBP09 (RD)	—	Ground
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- For the GPSM , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2398 Pin 1	SBP09 (RD)	—	Ground



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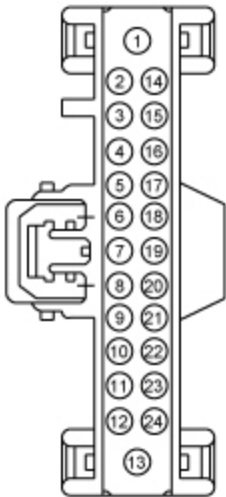
Is the voltage within 0.2 volts of the recorded battery voltage?

Yes	GO to AM5 .
No	REPAIR the circuit(s) for high resistance.

AM5 CHECK THE SUSPECT MODULE GROUND CIRCUIT FOR CONTINUITY

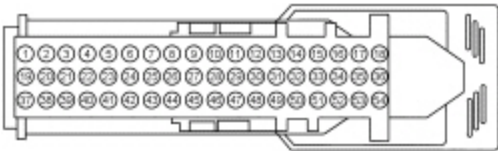
- Measure the voltage at the suspect module.
- For the ~~ACM~~ , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C240A Pin 1	SBP29 (GY/RD)	C240A Pin 13	GD114 (BK/BU)



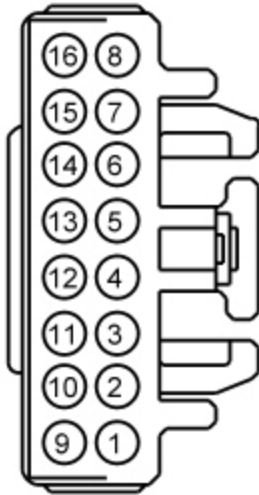
-
- For the **APIM** , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2383 Pin 1	SBP02 (YE/RD)	C2383 Pin 37	GD114 (BK/BU)



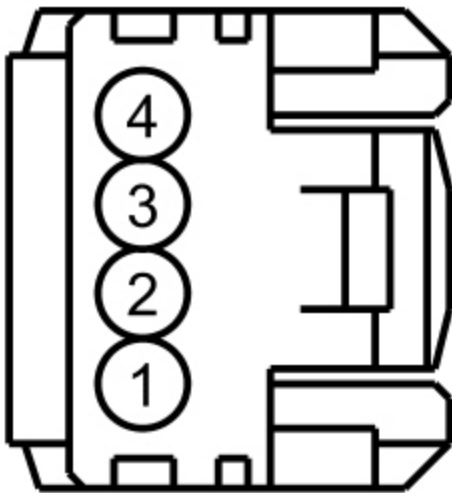
-
- For the audio **DSP** module, measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C3154C Pin 5	SBP19 (BU/RD)	C3154C Pin 15	GD148 (BK/YE)
C3154C Pin 6	SBP19 (BU/RD)	C3154C Pin 16	GD148 (BK/YE)



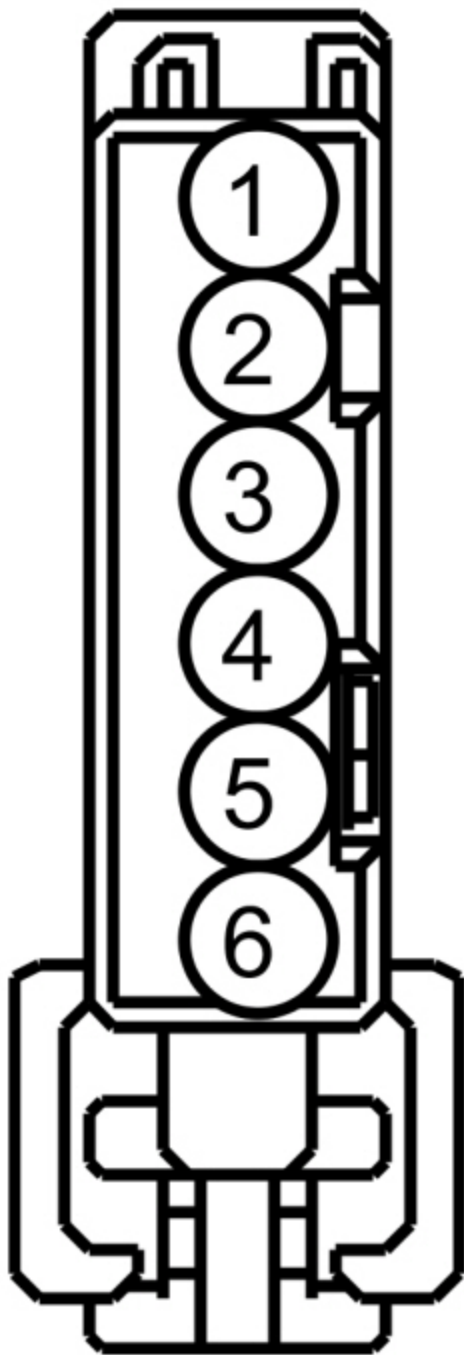
-
- For the **ECIM** , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2402 Pin 1	SBP09 (RD)	C2402 Pin 4	GD133 (BK)



-
- For the **GPSM** , measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2398 Pin 1	SBP09 (RD)	C2398 Pin 6	GD114 (BK/BU)



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Is the suspect module voltage greater than 11 volts?

Yes	For the <u>ACM</u> , GO to AM6 . For the <u>APIM</u> , GO to AM7 . For the audio <u>DSP</u> module, GO to AM8 . For the <u>FCIM</u> , GO to AM9 . For the <u>GPSM</u> , GO to AM10 .
No	REPAIR the circuit for high resistance.

- Ignition OFF.
- Disconnect and inspect all the **ACM** connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **ACM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AM7 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AM8 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)

- Damaged or bent pins (install new terminals or pins)
- Pushed-out pins (install new pins as necessary)
- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AM9 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **FCIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AM10 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect and inspect the **GPSM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **GPSM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AN: DTC U3003:17

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

DTC Fault Trigger Conditions

DTC	Description	Fault Trigger Conditions
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	Set by the ACM , APIM , audio DSP module, ECIM , and GPSM when the supply voltage is more than 15.8 volts for more than 250 milliseconds during normal operation or the self-test.

PINPOINT TEST AN : DTC U3003:17

AN1 RECHECK FOR DTC U3003:17	
• Turn the engine off. • Ignition ON. • Using a diagnostic scan tool, clear the DTC for the module in question. • Wait at least 10 seconds. • Using a diagnostic scan tool, perform the self-test for the module in question.	
Is DTC U3003:17 present?	
Yes	GO to AN2 .
No	The system is operating correctly at this time. The DTC may have been set previously during battery charging or while jump starting the vehicle.
AN2 CHECK FOR HIGH VOLTAGE DTCS SET IN OTHER MODULES	
• Ignition ON. • Using a diagnostic scan tool, retrieve all CMDTCs .	

- Wait at least 10 seconds.
- Using a diagnostic scan tool, perform the self-test for the module in question.

Are high voltage DTCs such as B1676, P0563, U3003:17 set in more than one module?

Yes	REFER to Section 414-01 to diagnose an overcharging condition.
No	GO to AN3 .

AN3 CHECK THE BATTERY VOLTAGE

- Turn off all interior lights, exterior lights, and accessories.
- Start and run the engine at approximately 2,000 RPM for 3 minutes while monitoring the battery voltage.

Does the battery voltage rise to 15.5 volts or higher?

Yes	REFER to Section 414-01 to diagnose an overcharging condition.
No	For the ACM , GO to AN4 . For the APIM , GO to AN5 . For the audio DSP module, GO to AN6 . For the FCIM , GO to AN7 For the GPSM , GO to AN8 .

AN4 CHECK FOR CORRECT ACM OPERATION

- Ignition OFF.
- Disconnect and inspect all the ~~ACM~~ connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the ~~ACM~~ connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new ACM . REFER to Audio Control Module (ACM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AN5 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.

- Disconnect and inspect the **APIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **APIM** connector. Make sure it seats and latches correctly.
- Wait at least 2 minutes for the **APIM** to re-initialize.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AN6 CHECK FOR CORRECT AUDIO DSP MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all the audio **DSP** module connectors.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the audio **DSP** module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new audio DSP module. REFER to Audio Digital Signal Processing (DSP) Module — Front Captain's Chair or Audio Digital Signal Processing (DSP) Module — Split Bench.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AN7 CHECK FOR CORRECT FCIM OPERATION

- Ignition OFF.
- Disconnect and inspect the **FCIM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)

- Damaged or bent pins (install new terminals or pins)
- Pushed-out pins (install new pins as necessary)

- Reconnect the **FCIM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new FCIM . REFER to Front Controls Interface Module (FCIM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

AN8 CHECK FOR CORRECT GPSM OPERATION

- Ignition OFF.
- Disconnect and inspect the **GPSM** connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the **GPSM** connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK QASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new GPSM . REFER to Global Positioning System Module (GPSM).
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test AO: Voice Recognition Does Not Operate Correctly To Voice Navigate To A Destination

Diagnostic Overview

Refer to Wiring Diagrams Cell [130](#) for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Information and Entertainment System, Navigation.

Visual Inspection and Diagnostic Pre-checks

- Verify the key is in the ON position (navigation function may be limited in accessory mode on some vehicles).

- Verify the vehicle is located in a clear, open area to ensure the satellite information can be received by the vehicle.
- Verify the navigation Secure Digital (SD) card is present and fully seated in the Secure Digital (SD) card slot.
- Run the vehicle for at least one minute before testing the system in order for the navigation and voice recognition application to load properly.

PINPOINT TEST AO : VOICE RECOGNITION DOES NOT OPERATE CORRECTLY TO VOICE NAVIGATE TO A DESTINATION

AO1 DETERMINE IF THE VEHICLE IS EQUIPPED WITH A FACTORY-INSTALLED NAVIGATION SYSTEM

- Make sure the correct VIN was entered on the technician service publication website.
- Hover the mouse cursor over or click on the OASIS tab toward the top LH corner of the screen.
- Select HVBoM (Historical Vehicle Bill of Material).
- Locate the Limit to Base: text entry box toward the center of the screen.
- Enter the navigation map data Secure Digital (SD) card base part number "19H449" (do not include quotation marks).
- Press Enter on the keyboard or click on "View BoM (Build of Material)".
 - If the vehicle is equipped with a factory-installed navigation system, a row will appear indicating "CARD-MEMORY" in the DESCRIPTION column.
 - If the vehicle is **not** equipped with a factory-installed navigation system, a row will **not** appear and the DESCRIPTION column will be blank.

Does the technician service publication website indicate the vehicle is equipped with a factory-installed navigation system?

Yes	GO to AO2 .
No	Ford Motor Company does not support the addition of a navigation system to vehicles not equipped with a factory-installed navigation system.

AO2 CHECK THE FDIM TOUCHSCREEN RH UPPER TAB

- Ignition ON.
- Check the navigation status of the FDIM touchscreen RH upper tab with the vehicle running for at least one minute.

Does the FDIM touchscreen RH upper tab read “NAVIGATION”?

Yes	GO to AO4 .
No	GO to AO3 .

AO3 REPROGRAM THE APIM

- Perform the APIM Standard Programming. Refer to Accessory Protocol Interface Module (APIM) Standard Programming.
- Override the Service Pack Installation and perform a Full Flash/Force CIP .

- Once the **APIM** programming is complete and the system has re-initialized, confirm that the **RH** upper tab of the **EDIM** touchscreen displays “NAVIGATION” and the navigation system operates correctly.

Does the **RH upper tab of the **EDIM** touchscreen display “NAVIGATION” and the navigation system operate correctly?**

Yes	The concern was resolved by reflashing the APIM . TEST the voice recognition system for correct operation.
No	GO to AO5 .

AO4 VERIFY THAT NO ROUTE PREFERENCE, NAVIGATION, TRAFFIC OR AVOID AREA RESTRICTIONS ARE SELECTED

- Verify that Route Preferences are not set to avoid freeways, toll roads, ferries and car trains. Refer to the Owner's Literature.
- Verify that Traffic Preferences are set properly. Refer to the Owner's Literature.
- Change preferences, as necessary, and operate the Voice Navigation system.

Does the Voice Navigation system operate as expected?

Yes	If the concern is not duplicated, the system is operating correctly at this time. ADVISE the customer of the correct operation of the route preference, navigation, traffic and avoid area restrictions.
No	GO to AO5 .

AO5 CHECK THE AUDIBLE PROMPT FUNCTIONALITY

- Operate the audio system in AM/FM mode.
- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.

Does the SYNC® system produce an audible prompt correctly indicating voice recognition mode has been entered?

Yes	GO to AO6 .
No	GO to Pinpoint Test P .

AO6 CHECK THE VOICE RECOGNITION OPERATION

- Operate the VOICE steering wheel switch and listen for the SYNC® audible prompt.
- NOTE:** Make sure the interior of the vehicle is as quiet as possible. Excessive noise may prevent the system from correctly recognizing spoken commands.

Clearly speak the word "Navigation".

Does the SYNC® system respond correctly to the "Navigation" voice command?

Yes	The voice recognition of the navigation system destination concern may have been caused by an insufficiently quiet vehicle or a destination not being present in the map database. GO to A07 .
No	GO to Pinpoint Test R .

A07 CHECK OPERATION OF NAVIGATION SYSTEM WITH MANUAL DESTINATION ENTRY

NOTE: Dragging your finger and pinching two fingers will not scroll to a location or zoom in or out as in other devices on the market. Holding your finger on a location off-center will cause the map to scroll to that location. Refer to the Owner's Literature for more information about navigating the map screen.

- Using the map feature, touch the screen at the requested destination.
- Verify there is a road present on the map at the requested location and verify the address of the destination is correct.

Does the navigation system properly navigate to the destination requested?

Yes	The information on the navigation secure digital (SD) card may not include the correct map data, or is in a different format than what is expected. The system is operating correctly at this time. NOTIFY the customer that there may be an updated map upgrade available for purchase.
No	GO to Pinpoint Test AA .

Pinpoint Test XB: The FDIM Changes Brightness

Diagnostic Overview

Normal Operation and Fault Conditions

The FDIM display brightness can be automatically or manually adjusted. Automatic adjustment relies on the illumination dimming level message sent from the BCM to the APIM .

Possible Sources

- Customer setting
- Communication network concern
- Instrument panel backlighting system concern
- FDIM
- APIM

PINPOINT TEST XB : THE FDIM CHANGES BRIGHTNESS

XB1 VERIFY THE DISPLAY BRIGHTNESS CONCERN IS NOT CAUSED BY SYSTEM SETTINGS

- Using the touchscreen, review the display settings under Settings>Display.
- Test the system to verify if the concern is caused by a display setting such as Mode being set to "Auto" and Auto Dim turned ON or OFF.

Was the concern caused by a setting within the Display Settings menu?

Yes	REVIEW the display brightness settings and system operation with the customer.
No	GO to XB2 .

XB2 REBOOT THE APIM AND RECHECK THE SYSTEM OPERATION

- Ignition ON.
- Simultaneously press and hold the SEEK UP and POWER buttons on the **FCIM** for 10 seconds. As the **APIM** reboot begins, the touchscreen turns black before the startup sequence begins.
- After the reboot is complete, operate the system to verify if the concern is still present.

Is the concern still present?

Yes	GO to XB3 .
No	The system is operating correctly at this time. VERIFY the customer's specific concern by duplicating the conditions present when the original concern was generated.

XB3 VERIFY OPERATION OF THE NETWORKED INSTRUMENT PANEL BACKLIGHTING SOURCES

- Illuminate the ambient light sensor to simulate daytime lighting conditions.
- While adjusting the instrument panel backlighting intensity up and down, inspect the **FCIM** and **IPC** .
- Cover the ambient light sensor to simulate nighttime lighting conditions.
- While adjusting the instrument panel backlighting intensity up and down, inspect the **FCIM** and **IPC** .

Does the instrument panel backlighting operate correctly?

Yes	GO to XB4 .
No	Refer to Section 413-00.

XB4 VERIFY THAT THE APIM AND BCM PASS THE NETWORK TEST

- Using a diagnostic scan tool, perform the network test.

Did the **APIM and **BCM** pass the network test?**

Yes	GO to XB5 .
No	Refer to Section 418-00.

XB5 CHECK THE APIM CMDTCS

- Using a diagnostic scan tool, retrieve all **CMDTCS** .

Are any **CMDTCS present in the **APIM** ?**

Yes	REFER to the APIM DTC Chart In this section.
No	GO to XB6 .

XB6 TEMPORARILY REMOVE THE POWER FROM THE APIM AND RECHECK THE SYSTEM OPERATION

- Ignition OFF.
- Disconnect: Negative Battery Cable from the battery post.
- Wait for 1 minute.
- Connect: Negative Battery Cable to the battery post.
- Ignition ON.
- Test the system to verify the concern is still present.

Is the concern still present?

Yes	GO to XB7 .
No	The system is operating correctly at this time. VERIFY the customer's specific concern by duplicating the steps that were taken when the original concern was generated.

XB7 CHECK FOR CORRECT APIM OPERATION

- Ignition OFF.
- Disconnect and inspect the APIM connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the APIM connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable service articles: <u>TSB</u> , General Service Bulletin (GSB), <u>SSM</u> or <u>PSA</u> . If a service article exists for this concern, DISCONTINUE this test and FOLLOW the service article instructions. If no service articles address this concern, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00.
No	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.